

Rivington Street Studio Limited

Coventry City Council

Stoke Park Secondary School Expansion

SPS-RSS-00-ZZ-SH- A-6500

Stage 4 Specification

For Tender

T01

10-10-2025

New Year 7 teaching block with general and specialist classrooms, a central heart space, external areas, and site reconfiguration, together with conversion of the Youth Centre into a Drama block.

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B10

Whole project

Clauses

100 Whole project

1. **Description:** Single Storey Classroom Building and refurbishment of existing Youth Centre into Drama Building
2. **Early-stage**
 - 2.1. **Client brief:** Year 7 classrooms and drama block with landscaping
3. **Requirements**
 - 3.1. **Performance requirements**
 - 3.1.1. **Structural performance strategy:** Refer to Structural Engineers documents and drawings
 - 3.1.2. **Fire performance strategy:** Refer to Stage 4 Fire Engineers Report
 - 3.1.3. **Energy performance strategy:** Refer to M&E Specification
 - 3.1.4. **Fabric Thermal Performance:** Walls U value - 0.13w/m2k
Floor U Value – 0. 15w/m2k
Roof U Value – 0. 15w/m2k
Window U Value – 1.4w /m2k (& g-value of 0.4)
Door U value 1.8 w/m2k
Air Permeability 3.0
 - 3.1.5. **Acoustic performance strategy:** Refer to Stage 4 Acoustic Design Review
 - 3.1.6. **Environmental performance strategy:** Refer to M&E Specification

Ω End of Section

F10 Brick/ block walling

Types of walling

110 Clay bricks New Building

1. Bricks: To BS EN 771-1.
 - 1.1. Manufacturer: [Ibstock Brick Ltd](#)
 - 1.1.1. Contact details
 - 1.1.1.1. Address: Leicester Road
Ibstock
Leicestershire
LE67 6HS
 - 1.1.1.2. Telephone: [+44 \(0\)1530 261999](tel:+44(0)1530261999)
 - 1.1.1.3. Web: <https://ibstock.co.uk/ibstock-brick>
 - 1.1.1.4. Email: enquiries@ibstock.co.uk
 - 1.1.2. Product reference: [Arden Red- Clay Facing Brick](#)
 - 1.2. Standard: To BS EN 771-1.
 - 1.3. Brick description: Facing brick.
 - 1.4. Work size (length x width x height): 215 x 102 x 65 mm.
 - 1.5. Tolerances
 - 1.5.1. Mean value: T2.
 - 1.5.2. Range: R1.
 - 1.6. Compressive strength
 - 1.6.1. Mean compressive strength (minimum): 15 N/mm².
 - 1.7. Durability designation: F2.
 - 1.8. Density
 - 1.8.1. Gross dry: 1840 kg/m³.
 - 1.8.2. Tolerance: T2, R1.
 - 1.9. Water absorption: 13%.
 - 1.10. Configuration: Single frog.
 - 1.11. Thermal conductivity: 1.0 W/mK.
 - 1.12. Water vapour permeability: 5/10 µ.
 - 1.13. Bond strength: 0.15 N/mm².
 - 1.14. Reaction to fire: Class A1 to BS EN 13501-1.
 - 1.15. Type: Stock.
 - 1.16. Weight: 2.4 kg.
 - 1.17. Active soluble salts: S2.
 - 1.18. Mortar: As section Z21.
 - Mortar: As section Z21.
 - Standard: BS EN 998-2.
 - Manufacturer: CPI Euromix or similar, www.cpieuromix.com
 - Colour: Allow for coloured mortar, colour reference tbc
 - 1.19. Bond: Stretcher
 - 1.20. Joints: Bucket handle
 - 1.21. Specials: Pistol bricks at brickwork support

355 Aggregate concrete blocks

1. Blocks: To BS EN 771-3.
 - 1.1. Manufacturer: [Lignacite Ltd](#)
 - 1.1.1. Contact details
 - 1.1.1.1. Address: Norfolk House
High Street
Brandon
Suffolk
IP27 0AX
 - 1.1.1.2. Telephone: [+44 \(0\)1842 810678](tel:+44(0)1842810678)
 - 1.1.1.3. Web: <https://lignacite.co.uk/>
 - 1.1.1.4. Email: info@lignacite.co.uk
 - 1.1.2. Product reference: Lignacite SP 100mm or 140mm 7.3kn concrete blocks
 - 1.2. Standard: To BS EN 771-3.
 - 1.3. Block description:
 - 1.4. Appearance: Medium grey in colour with a fine textured surface.
 - 1.5. Configuration: Group 1.
 - 1.6. Compressive strength
 - 1.6.1. Mean value: 7.3 N/mm².
 - 1.6.2. Category: M4.
 - 1.7. Thermal conductivity: 0.79 W/m·K at 3% moisture content.
 - 1.8. Thermal resistance: 0.177 m²K/W at 3% moisture content.
 - 1.9. Work sizes (length x width x height): 440 x 140 x 215 mm.
 - 1.10. Tolerance category: D1.
 - 1.11. Density
 - 1.11.1. Net dry density: 1450 kg/m³.
 - 1.12. Reaction to fire: Class A1.
 - 1.13. Water absorption by capillarity: 5/15 µ.
 - 1.14. Moisture movement: <0.7 mm/m.
 - 1.15. Weighted sound reduction index, *R_w*: No finish: 49 dB; paint finish: 50 dB.
 - 1.16. Fire resistance: Non-loadbearing wall (criteria E1): 4 hours; loadbearing wall (criteria RE1): 3 hours.
 - 1.17. Third-party certification: BSI Kitemark approved KM 09694.
 - 1.18. Unit weight: 19.2 kg.

Testing

410 Compressive strength of mortar for each walling type

1. Testing authority:
2. Test method: To BS EN 1015-11.
3. Preliminary tests procedure: As follows:
 - 3.1. Specimens
 - 3.1.1. Number of specimens: Six.
 - 3.1.2. Type: 40 x 40 x 160 mm prism.
 - 3.1.3. Preparation: At least six weeks before walling commences.
 - 3.2. Specimen testing: Half of specimens at seven days. Remainder at 28 days.

- 3.2.1. Retarded mixes: Extend curing periods to include retardation period.
- 3.3. Response to result: If mean compressive strength at 28 days is not within the range given below repeat tests with more suitable sand or next higher
- 4. Site tests procedure: As follows.
 - 4.1. Number of specimens: Six per 150m² of walling or per storey whichever the more frequent.
 - 4.2. Specimen types: As preliminary test, but prepared during construction.
 - 4.3. Specimen testing: Half of specimens at seven days. Remainder at 28 days.
 - 4.3.1. Retarded mixes: Extend curing periods to include retardation period.
- 5. Required test mean compressive strength at 28 days (N/mm²): To be within the following range:
 - 5.1. Walling type:
 - 5.1.1. Preliminary tests minimum (N/mm²):
 - 5.1.2. Preliminary tests maximum (N/mm²):
 - 5.1.3. Site tests minimum (N/mm²):
 - 5.1.4. Site tests maximum (N/mm²):
- 6. Results: Submit.

Workmanship generally

430 Conditioning of clay bricks and blocks

- 1. Bricks and blocks delivered warm from manufacturing process: Do not use until cold.
- 2. Absorbent bricks in warm weather: Wet to reduce suction. Do not soak.

440 Conditioning of concrete bricks/ blocks

- 1. Autoclaved concrete bricks/ blocks delivered warm from manufacturing process: Do not use.
- 2. Age of nonautoclaved concrete bricks/ blocks: Do not use until at least four weeks old.
- 3. Avoidance of suction in concrete bricks/ blocks: Do not wet.
 - 3.1. Use of water retaining mortar admixture: Submit details.

460 Mortar designations

- 1. Mix proportions: For a specified designation select a mix from the following:
 - 1.1. Designation (i) (BS EN 998-2 M12 equivalent)
 - 1.1.1. 1:0-¼:3 (Portland cement:lime:sand with or without air entraining additive).
 - 1.1.2. 1:3 (Portland cement:sand and air entraining additive).
 - 1.2. Designation (ii) (BS EN 998-2 class M6 equivalent)
 - 1.2.1. 1:½:4-5 (Portland cement:lime:sand with or without air entraining additive).
 - 1.2.2. 1:3 (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).
 - 1.2.3. 1:2½-3½ (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).
 - 1.2.4. 1:3-4 (Portland cement:sand and air entraining additive).
 - 1.3. Designation (iii) (BS EN 998-2 class M4 equivalent)
 - 1.3.1. 1:1:5-6 (Portland cement:lime:sand with or without air entraining additive).
 - 1.3.2. 1:3½-4 (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).
 - 1.3.3. 1:4-5 (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).
 - 1.3.4. 1:5-6 (Portland cement:sand and air entraining additive).

1.4. Designation (iv) (BS EN 998-2 class M2 equivalent)

- 1.4.1. 1:2:8-9 (Portland cement:lime:sand with or without air entraining additive).
- 1.4.2. 1:4½ (masonry cement:sand containing Portland cement and lime in approximate ratio 1:1, and an air entraining additive).
- 1.4.3. 1:5½-6½ (masonry cement:sand containing Portland cement and inorganic materials other than lime and air entraining additive).
- 1.4.4. 1:7-8 (Portland cement:sand and air entraining additive).

2. **Batching:** Mix proportions by volume.

3. **Mortar type:** Continuous throughout any one type of masonry work.

500 Laying generally

- 1. **Mortar joints:** Fill vertical joints. Lay bricks, solid and cellular blocks on a full bed.
- 2. **AAC block thin mortar adhesive and gypsum block adhesive joints:** Fill vertical joints. Lay blocks on a full bed.
- 3. **Clay block joints**
 - 3.1. **Thin-layer mortar:** Lay blocks on a full bed.
 - 3.2. **Interlocking perpend:** Butted.
- 4. **Bond where not specified:** Half-lap stretcher.
- 5. **Vertical joints in brick and concrete block facework:** Even widths. Plumb at every fifth cross joint.

520 Accuracy

- 1. **Courses:** Level and true to line.
- 2. **Faces, angles and features:** Plumb.
- 3. **Permissible deviations**
 - 3.1. **Position in plan of any point in relation to the specified building reference line and/ or point at the same level:** ± 10 mm.
 - 3.2. **Straightness in any 5 m length:** ± 5 mm.
 - 3.3. **Verticality up to 3 m height:** ± 10 mm.
 - 3.4. **Verticality up to 7 m height:** ± 14 mm.
 - 3.5. **Overall thickness of walls:** ± 10 mm.
 - 3.6. **Level of bed joints up to 5 m (brick masonry):** ± 11 mm.
 - 3.7. **Level of bed joints up to 5 m (block masonry):** ± 13 mm.

560 Coursing brickwork

- 1. **Gauge:** Four brick courses including bed joints to 300 mm.

580 Laying frogged bricks

- 1. **Single frogged bricks:** Frog uppermost.
- 2. **Double frogged bricks:** Larger frog uppermost.
- 3. **Frog cavity:** Fill with mortar.

595 Lintels

- 1. **Bearing:** Ensure full length masonry units occur immediately under lintel ends.

635 Jointing

- 1. **Profile:** Consistent in appearance.

645 Accessible joints not exposed to view

1. Jointing: Struck flush as work proceeds.

665 Pointing

1. Description:
2. Joint preparation: Remove debris. Dampen surface.
3. Mortar: As section Z21.
 - 3.1. Standard:
 - 3.2. Mix:
 - 3.3. Additional requirements:
4. Profile:

671 Fire-stopping

1. Avoidance of fire and smoke penetration: Fit tightly between cavity barriers and masonry. Leave no gaps.

690 Adverse weather

1. General: Do not use frozen materials or lay on frozen surfaces.
2. Air temperature requirements: Do not lay bricks/ blocks:
 - 2.1. In cement-gauged mortars when at or below 3°C and falling or unless it is at least 1°C and rising.
 - 2.2. In hydraulic lime:sand mortars when at or below 5°C and falling or below 3°C and rising, or as manufacturer's/ supplier's recommendations.
 - 2.3. In thin-layer mortars when outside the limits set by the mortar manufacturer.
3. Temperature of walling during curing: Above freezing until hardened.
4. Newly erected walling: Protect at all times from:
 - 4.1. Rain and snow.
 - 4.2. Drying out too rapidly in hot conditions and in drying winds.

Additional requirements for facework

710 The term facework

1. Definition: Applicable in this specification to brick/ block walling finished fair.
 - 1.1. Painted facework: The only requirement to be waived is that relating to colour.

730 Brick/ Concrete block samples

1. General: Before placing orders with suppliers submit for approval of appearance labelled samples of the following:
2. Selection of samples: Representative of the range in variation of appearance.

740 Finished masonry work reference panels

1. General: Before proceeding to construct the following walling types, construct panels as specified. Give notice when panels are dry.
2. Selection of masonry units: Reasonably representative of the average quality of the whole order to be delivered
3. Panel types
 - 3.1. Walling type: F10/110

- 3.1.1. Location:
- 3.1.2. Size: 1200 x 1200
- 3.1.3. Other requirements: Colour mortar options to match existing

750 Colour consistency of masonry units

- 1. **Colour range:** Submit proposals of methods taken to ensure that units are of consistent and even appearance within deliveries.
- 2. **Conformity:** Check each delivery for consistency of appearance with previous deliveries and with approved reference panels; do not use if variation is excessive.
- 3. Facing bricks should be blended on site from a minimum of three packs to ensure an even distribution of colour and texture variation.
- 4. **Finished work:** Free from patches, horizontal stripes and racking back marks.

760 Appearance

- 1. **Brick/ block selection:** Do not use units with damaged faces or arrises.
- 2. **Cut masonry units:** Where cut faces or edges are exposed cut with table masonry saw.
- 3. **Quality control:** Lay masonry units to match relevant reference panels.
 - 3.1. **Setting out:** To produce satisfactory junctions and joints with built-in features and components.
 - 3.2. **Coursing:** Evenly spaced using gauge rods.
- 4. **Lifts:** Complete in one operation.
- 5. **Methods of protecting facework:** Submit proposals.

780 Ground level

- 1. **Commencement of facework:** Not less than 150 mm below finished level of adjoining ground or external works level.

800 Toothed bond

- 1. **New and existing facework in same plane:** Bond together at every course to achieve continuity.

830 Cleanliness

- 1. **Facework:** Keep clean.
- 2. **Mortar on facework:** Allow to dry before removing with stiff bristled brush.
- 3. **Removal of marks and stains:** Rubbing not permitted.

Ω End of Section

F30

Accessories/ sundry items for brick/ block/ stone walling

Cavities

110 Concrete fill to base of cavity

1. Concrete generally: To [BS EN 206](#) and [BS 8500-2](#).
2. Concrete type:
 - 2.1. Workability: High.
3. Extent: Maintain 75 mm between top of fill and external ground level and a minimum of 225 mm between top of fill and ground-level dpc.
4. Placement: Compact to eliminate voids.

120 Cleanliness

1. Cavity base and faces, ties, insulation and exposed dpcs: Free from mortar and debris.

132 Perpend joint plastics weepholes

1. Manufacturer: [Rytons Building Products Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: Design House
Orion Way
Kettering Business Park
Kettering
Northamptonshire
NN15 6NL
 - 1.1.2. Telephone: [+44 \(0\)1536 511874](tel:+44(0)1536511874)
 - 1.1.3. Web: <https://www.vents.co.uk>
 - 1.1.4. Email: admin@rytons.com
 - 1.2. Product reference: RYTWEEP - Rytons Rytweep
2. Colour: Clear.
3. Accessories:
4. Locations: Through outer leaf immediately above base of cavity, at cavity trays, stepped dpcs and external openings. 75 mm above top of cavity fill at base of cavity.
5. Provision: At not greater than 1000 mm centres and not less than two over each opening.

160 Airbricks

1. Manufacturer: [Cordek Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: Spring Copse Business Park
Slinfold
West Sussex
RH13 0SZ
 - 1.1.2. Telephone: [+44 \(0\)1403 799600](tel:+44(0)1403799600)
 - 1.1.3. Web: www.cordek.com
 - 1.1.4. Email: info@cordek.com
 - 1.2. Product reference: [Air Brick \(CGV-014\)](#)
2. Work sizes: 210 x 70 x 57 mm.
3. Material

- 3.1. Colour: Red
- 4. Weight: 0.125 kg.
- 5. Free airflow: 7600 mm².
- 6. Material: Polypropylene.

171 Ventilation gratings

- 1. Manufacturer: [Cordek Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: Spring Copse Business Park
Slinfold
West Sussex
RH13 0SZ
 - 1.1.2. Telephone: [+44 \(0\)1403 799600](tel:+44(0)1403799600)
 - 1.1.3. Web: www.cordek.com
 - 1.1.4. Email: info@cordek.com
 - 1.2. Product reference: Adjustable Cavity Vent
- 2. Apertures: Louvre.
- 3. Material: Injection moulded polypropylene.
- 4. Colour: red
- 5. Accessories: Extensions as required to meet levels,

180 Cavity closers

- 1. Manufacturer: [ARC Building Solutions Limited](#)
 - 1.1. Contact details
 - 1.1.1. Address: Unit J Gildersome Spur,
Leeds
LS27 7JZ
 - 1.1.2. Telephone: [+44 \(0\)113 2529428](tel:+44(0)1132529428)
 - 1.1.3. Web: www.arcbuildingsolutions.co.uk
 - 1.1.4. Email: sales@arcbuildingsolutions.co.uk
 - 1.2. Product reference: Eco-Closer Plus
- 2. Materials: Rock fibre mineral wool.
- 3. ThermalConductivity: 0.035 W/mK.
- 4. Width: To suit cavity
- 5. Flange: 20 mm flange.
- 6. Refer to 18, 30, 31 & 37 series drawings for locations

Reinforcing/ fixing accessories

220 Cavity wall ties

- 1. Manufacturer: [Leviat](#)
 - 1.1. Contact details
 - 1.1.1. Address: President Way
President Park,
Sheffield
South Yorkshire
S4 7UR
 - 1.1.2. Telephone: [+44 \(0\) 114 275 5224](tel:+44(0)1142755224)

- 1.1.3. Web: www.leviat.com
- 1.1.4. Email: info.uk@leviat.com
- 1.2. Product reference: [Ancon 25/14 Brick Ties](#)
- 2. Tie type: Type 1
- 3. Length: to suit cavity
- 4. Movement: Rotation through 90° and can be easily moved to the required position.

225 Fixing ties in masonry cavity walls

- 1. Embedment in mortar beds (minimum): 50 mm.
- 2. Placement: Sloping slightly downwards towards outer leaf, without bending. Drip centred in the cavity and pointing downwards.
- 3. Spacing: Staggered in alternate courses.
 - 3.1. Horizontal centres:
 - 3.2. Vertical centres:
- 4. Provision of additional ties: Within 225 mm of reveals of unbonded openings and at the vertical reveals of unsupported masonry.
 - 4.1. Spacing:

282 General-purpose slip ties

- 1. Manufacturer: [Leviat](#)
 - 1.1. Contact details
 - 1.1.1. Address: President Way
President Park,
Sheffield
South Yorkshire
S4 7UR
 - 1.1.2. Telephone: [+44 \(0\) 114 275 5224](tel:+44(0)1142755224)
 - 1.1.3. Web: www.leviat.com
 - 1.1.4. Email: info.uk@leviat.com
 - 1.2. Product reference: [Ancon PPS Movement Tie - 2D Detail](#)
- 2. Material: Austenitic stainless steel.
- 3. Length: 225 mm.

Flexible damp-proof courses/ cavity trays

330 Physical insertion vertical damp-proof course systems

- 1. Description:
- 2. Manufacturer: [Visqueen](#)
 - 2.1. Contact details
 - 2.1.1. Address: Visqueen
Heanor Gate Industrial Estate
Derbyshire
Heanor
Derbyshire
United Kingdom
DE75 7RG
 - 2.1.2. Telephone: [+44 \(0\) 333 202 6800](tel:+44(0)3332026800)
 - 2.1.3. Web: www.visqueen.com

2.1.4. Email: enquiries@visqueen.com

2.2. Product reference: [Visqueen Zedex CPT High Performance DPC](#)

3. Material: Co-polymer thermoplastic.
4. Third party certification: British Board of Agrément certified, 94/3059 / CE Mark EN 14909.
5. Colour: Black.
6. Accessories:
7. Length: 20 000 mm.
8. Mass: 750 g/m².
9. Joint strength: 220 N.
10. TensileStrength: 14 Mpa.
11. Elongation: 500%.
12. Reaction to fire: F Class.
13. Tensile strength MD: 17 Mpa.
14. Resistance to static loading - 20kg load: <20 kg.
15. Application temperature:
16. Verification:

345 Polypropylene (PP) damp-proof courses and cavity trays Type A

1. Manufacturer: [Visqueen](#)

1.1. Contact details

1.1.1. Address: Visqueen
Heanor Gate Industrial Estate
Derbyshire
Heanor
Derbyshire
United Kingdom
DE75 7RG

1.1.2. Telephone: [+44 \(0\) 333 202 6800](tel:+44(0)3332026800)

1.1.3. Web: www.visqueen.com

1.1.4. Email: enquiries@visqueen.com

1.2. Product reference: [Visqueen Zedex CPT High Performance DPC](#)

2. Material: Co-polymer thermoplastic.
3. Third party certification: British Board of Agrément certified, 94/3059 / CE Mark EN 14909.
4. Colour: Black.
5. Accessories:
6. Length: 20 000 mm.
7. Mass: 750 g/m².
8. Joint strength: 220 N.
9. TensileStrength: 14 Mpa.
10. Elongation: 500%.
11. Reaction to fire: F Class.
12. Tensile strength MD: 17 Mpa.
13. Resistance to static loading - 20kg load: <20 kg.
14. Application temperature:
15. Verification:

385A Preformed cavity trays

1. Description:
2. Manufacturer: [Visqueen](#)
 - 2.1. Contact details
 - 2.1.1. Address: Visqueen
Heanor Gate Industrial Estate
Derbyshire
Heanor
Derbyshire
United Kingdom
DE75 7RG
 - 2.1.2. Telephone: [+44 \(0\) 333 202 6800](tel:+441202333202)
 - 2.1.3. Web: www.visqueen.com
 - 2.1.4. Email: enquiries@visqueen.com
3. Product reference: Visqueen Ultimate Preformed Units
4. Use/location: All cavity wall constructions with a masonry outer leaf and a structural framing system or masonry inner leaf where a cavity is formed requiring a three-dimensional linear run of cavity tray formed on site to prevent water ingress to the internal leaf. Preformed unit types as shown on construction drawings.
5. Preparation/Installation: Ensure Visqueen Ultimate Preformed Units are installed following good practice set out in BS 8215:1991 and PD 6697:2019. All preformed units must be sandwiched between even beds of wet mortar in the inner and outer leaf. Alternatively, the preformed units can be surface fixed (face-fixed) to the inner leaf using Visqueen HP Tanking Primer, Visqueen Zedex Jointing Tape, Visqueen Zedex Fixing Strip and appropriate fixings for the substrate. All unsupported preformed unit lap joints to be fully supported with Visqueen DPC Joint Supports. All preformed unit lap joints to be overlapped by 100mm and sealed with Visqueen Zedex Jointing Tape applied within the overlap joint. Complete all work strictly in accordance with relevant Visqueen product technical datasheets.

390 Junction cloaks/ stop ends for site-formed dpcs/ cavity trays

1. Three dimensional changes in shape: Form to provide a free draining and watertight installation. Seal laps.
2. Alternative use of preformed junction cloaks/ stop ends: Submit proposals.

Installation of dpcs/ cavity trays

415 Installation of horizontal dpcs

1. Placement: In continuous lengths on full even bed of fresh mortar, with 100 mm laps at joints and full laps at angles.
2. Width: At least full width of leaf unless otherwise specified. Edges of dpc not covered with mortar or projecting into cavity.
3. Overlying construction: Immediately cover with full even bed of mortar to receive next masonry course.
4. Overall finished joint thickness: As close to normal as practicable.

425 Installation of ground-level dpcs

1. Joint with damp-proof membrane: Continuous and effectively sealed.

435 Installation of stepped dpcs in external walls

1. External walls on sloping ground: Install dpcs not less than 150 mm above adjoining finished ground level.

445 Installation of sill dpcs

1. **Form and placement:** In one piece and turned up at back when sill is in contact with inner leaf.

455 Installation of coping/ capping dpcs

1. **Placement:** Bed in one operation to ensure maximum bond between masonry units, mortar and dpc.
2. **Dpcs crossing cavity:** Provide rigid support to prevent sagging.

475 Installation of site-formed cavity trays

1. **Requirements to prevent downward ingress of water**
 - 1.1. **Profiles:** To match those shown on drawings. Firmly secured.
 - 1.2. **Joint treatment:** Use continuous length wherever possible, otherwise lap at least 100 mm and seal to produce a free draining and watertight installation.
 - 1.3. **Horizontal cavity trays:** Support using cavity closer.
 - 1.4. **Sloping cavity trays:** Prevent sagging.
 - 1.5. **Cleanliness:** Free from debris and mortar droppings.

485 Installation of cavity trays over openings and other cavity bridgings

1. **Length:** To extend not less than 150 mm beyond ends of lintels/ bridgings.

495 Installation of gas-resistant dpcs/ cavity trays

1. **Joint treatment:** Use continuous length wherever possible, otherwise lap at least 150 mm and seal to form a gas and watertight installation.
2. **Joint with damp-proof membrane:** Overlap dpc/ cavity tray not less than 150 mm.

515 Dpc/ cavity tray leading edge in facework – flush

1. **Treatment at face of masonry:** Finish flush and clear of mortar at the following locations:

525 Dpc/ cavity tray leading edge in facework – set back

1. **Treatment at face of masonry:** Set back 5 mm from face of wall with recessed mortar joint to expose edge at the following locations:

535 Dpc/ cavity tray leading edge in facework – projecting

1. **Treatment at face of masonry:** Projecting 5 mm from face of wall at the following locations:

560 Installation of vertical dpcs

1. **Form:** In one piece wherever possible.
 - 1.1. **Joints:** Upper part overlapping lower not less than 100 mm.

570 Installation of jamb dpcs at openings

1. **Joint with cavity tray/ lintel at head:** Full underlap.
2. **Joint with sill/ horizontal dpc at base:** Full overlap.
3. **Projection into cavity:** Not less than 25 mm.
4. **Relationship with frame:** In full contact.

Joins

610 Movement joints with sealant

1. Joint preparation and sealant application: As section Z22.
2. Filler:
 - 2.1. Thickness: To match design width of joint.
 - 2.2. Manufacturer: Grace Construction Ltd
 - 2.2.1. Product reference: Serviced Aerofil 1
 - 2.3. Placement: Build in as work proceeds with no projections into cavities and to correct depth to receive sealant system.
3. Sealant
 - 3.1. Manufacturer: Dow Corning
 - 3.1.1. Product reference: Dow Corning 797 One Part Silicone
 - 3.2. Colour: To match adjoining brickwork

630 Unexposed contraction joints

1. Formation: Close butt as work proceeds.

650 Pointing in flashings

1. Joint preparation: Free of debris and lightly wetted.
2. Pointing mortar: As for adjacent walling.
3. Placement: Fill joint and finish flush.

655 Pointing in asphalt skirtings

1. Joint preparation: Free of debris and lightly wetted.
2. Pointing mortar: 1:4 cement: sand incorporating a bonding agent.
 - 2.1. Colour: Match adjacent work.
3. Placement: Fill joint and finish flush.

660 Pinning up to soffits

1. Top joint of loadbearing walls: Fill and consolidate with mortar.

Proprietary sills/ lintels/ copings/ dressings

720 Precast concrete thresholds to doors

1. Standard: To BS 5642-1.
2. Material: Precast concrete
3. Manufacturer: Contractor's choice
 - 3.1. Product reference: Contractor's choice
4. Dimensions: As shown on drawings 30 series.
5. Mortar for bedding/ jointing: Cement-gauged, as section Z21.
 - 5.1. Standard: To BS EN 998-2
6. Joints: Flush
7. Bedding one-piece sills: Leave bed joints open except under end bearings and masonry mullions. On completion, point to match adjacent work.

Miscellaneous items

840 Openings for frames

1. **Formation:** Use accurate, rigid templates to required size.

850 Wall plates

1. **Placement:** On full bed of mortar to correct horizontal level.

Ω End of Section

G10

Structural steel framing

General requirements/ information

112 Contractor's design

1. Description:
2. Design concept:
3. Design responsibility: Design and detail members and joints to suit the conceptual layout detailed in the contract drawings.
 - 3.1. Other responsibilities:
4. Structural requirements
 - 4.1. Generally: As section B50.
 - 4.1.1. Modifications:
 - 4.2. Design: Complete in accordance with the designated code of practice to satisfy specified performance criteria.
 - 4.3. Connections:
 - 4.4. Fixings to foundations and walls:
 - 4.5. Limitations on steelwork layout:
 - 4.6. Additional requirements:
5. Design and production information:

115 Design constraints – general

1. Members forming bracing systems or girders of lattice construction: Unless detailed or instructed otherwise, position so that their lines of action intersect at a point.
2. Bolts
 - 2.1. Diameter (minimum):
 - 2.2. Number per connection (minimum): Two, unless otherwise indicated.
 - 2.3. Other requirements:
3. Punching of bolt holes:
4. Welds:
5. Other constraints:

120 Drawings and calculations

1. Information required:
2. Requirement: Before preparing detailed fabrication drawings, submit:
 - 2.1. General arrangement drawings with individual steel members clearly identified.
 - 2.2. Calculations for major connections.

123 Drawings and calculations prepared by Contractor

1. Information required:
2. General arrangement drawings: Submit before preparing calculations. Clearly identify:
 - 2.1. Individual steel members.
 - 2.2. Conflicts with other work.
 - 2.3. Proposed changes to contract drawings.
3. Member and joint calculations: Submit before preparing fabrication drawings.

125 Specification standard

1. **Standard:** Comply with latest edition of National Structural Steelwork Specification (NSSS).
 - 1.1. **Additional requirements:**
 - 1.2. **Document availability:** For the duration of the work, at fabrication shop and on site.
2. **References to Engineer in NSSS:** For the purpose of this contract, interpret such references as being to
- 2.1. **Exceptions:**

Frame systems

150 Light steel wall framing systems Type A

1. **Manufacturer:** [Metsec \(voestalpine Metsec\)](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** Broadwell Road
Oldbury
West Midlands
B69 4HF
 - 1.1.2. **Telephone:** +44 (0) 121 601 6000
 - 1.1.3. **Web:** <https://www.metsec.com>
 - 1.1.4. **Email:** david.atkinson@voestalpine.com
2. **System performance:** Structural design supervision and checking. Design of framing systems generally.
3. **Framing members:** Cold-formed galvanized steel sections.
4. **Connections**
 - 4.1. **Connection fasteners:** Self drilling tek screws.
5. **Anchors to concrete and masonry**
 - 5.1. **Type of anchor:** By structural engineers design.
6. **Movement joints:** Deflection allowance at the top of the wall.
7. **Insulation:**
8. **Sheathing board:** British Gypsum Glasroc X - 12 mm.
9. **Framing:** Metsec SFS 180 mm stud depth.
10. **Inner layer:** British Gypsum Wall Board, 12.5 mm.
11. **Face layer:** British Gypsum Wall Board, 12.5 mm.
12. **Fasteners:** Drywall screws.
13. **System accessories:** Flatstrap / partition brace.
14. **Internal face:** 60/ 60 minutes (insulation/ integrity).
15. **External face:** 60/ 60 minutes (insulation/ integrity).
16. **Fire performance:** 60 minutes (boundary wall); 60 minutes (non-boundary wall).
17. **Thermal performance (U-value maximum):** Variable.
18. **Air tightness:** Variable.
19. **Dimensional constraints:** Variable.
20. **Product reference:** [Metsec SFS Infill Wall with British Gypsum Glasroc X sheathing board, British Gypsum Gyproc plasterboard, Rockwool Duoslab insulation \(DXXX-X1-N180-B20 - 180 mm stud\)](#)

Cold-formed materials - Not Used

Fabrication

180 Notification of commencement

1. Notice: Give notice before fabrication is due to start.
 - 1.1. Period of notice (minimum):

190 Marking

1. Identifying and recording materials and components: Submit details of proposed methods.
2. Location of marks
 - 2.1. Generally: Visible for checking after erection.
 - 2.2. Weathering steel: On surfaces not exposed to open view in the completed work.
3. Steel to be blast cleaned, pickled, metal sprayed or galvanized: Marked so that subsequent treatment cannot obliterate the marking.

Welding

255 Site welding

1. Usage: Permitted only where indicated on drawings.
2. Working conditions: Suitable and safe. Do not weld when surfaces are wet or when ambient temperature is below 0°C.

270 Additional welds

1. Welds (including tack welds) not indicated on drawings: Not permitted without approval.

Bolt assemblies

370 Galvanized coating to bolt assemblies

1. Standard: To BS 7371-6.
2. Galvanizing: Applied by fastener manufacturer. Passivated and lubricated if no additional coatings are specified. Nuts tapped after galvanizing.
3. Use/location:

Erection

405 Outline method of erection

1. Documentation:

410 Pre-erection checks

1. Scope: At least 7 days before proposed erection start date, check the following:
 - 1.1. Foundations and other structures to which steelwork will be attached: Accuracy of setting out.
 - 1.2. Holding down bolts: Position, protruding length, slackness and condition.
2. Inaccuracies and defects: Report without delay.
3. Permission to commence erection: Obtain.

420 Setting out

1. **Permissible deviations:** In addition to the requirements of the NSSS, add permissible deviations for different types of dimension and locations, as necessary.

Testing

465 Testing

1. **Testing:** Arrange the following tests. Prepare test pieces as necessary.
2. **Test:**
 - 2.1. **Testing authority:**
 - 2.2. **Frequency/ Number:**
 - 2.3. **Level of acceptability:**
 - 2.4. **Other requirements:**
3. **Test and examination results:** Submit copies immediately they are available.

475 Products

1. **Steel:** Submit test certificates.

Protective coatings

521 Alternative manufacturers

1. **Short list of manufacturers:** Obtain coating materials from one only of the following:
2. **Selected manufacturer:** Submit details before ordering materials.

Protective coating systems - Not Used

Preparation for painting

710 Offsite preparation and painting

1. **Working area:** Covered and properly lit, heated and ventilated.
2. **Sequence of working:** Select from the following and submit proposals:
 - 2.1. **Fabricate, blast clean, prime.**
 - 2.2. **Blast clean, fabricate, remove flash rust with a light overall sweep blast, prime.**
 - 2.3. **Blast clean, apply weldable prefabrication primer, fabricate, prime.**
3. **Prefabrication primer (option 3):** Type recommended by manufacturer of post fabrication primer.
 - 3.1. **Thickness of post fabrication primer coat:** May be reduced if and as recommended by manufacturer.
4. **Surfaces inaccessible after assembly:** Apply full treatment and coating system including, if necessary, local application of site coatings.

Painting

810 Environmental conditions

1. **General requirements prior to starting coating work**
 - 1.1. **Surfaces:** Unaffected by moisture or frost.
 - 1.2. **Steel temperature:** At least 3°C above dew point, with conditions stable or improving, and not high enough to cause blistering or wrinkling of the coating.
 - 1.3. **Relative humidity:** Below 85%.

825 Stripe coat

1. External angles, nuts, bolt heads, rough weld seams, and areas difficult to coat: Apply an additional stripe coat of

Ω End of Section

H72

Aluminium strip/ sheet fully supported roof and wall coverings/ flashings

Types of aluminium work

420 Cover flashings

1. Description: Copings to brick parapet
2. Manufacturer: Alumasc Exterior Building Products Ltd or Equal & Approved
3. Aluminium: Aluminium sheet to BSEN515:1993
 - 3.1. Finish: PPC
 - 3.2. Thickness: 3mm
4. Dimensions
 - 4.1. Lengths: Not more than 2 m, with end to end joints lapped not less than 100 mm.
 - 4.2. Cover to roofing upstand: Not less than 75 mm, with bottom edge welted 15 mm.
5. Fixing: As recommended by manufacturer

421 Cover flashings

1. Description: Flashings to roof lanterns and roof edges
2. Manufacturer: Alumasc Exterior Building Products Ltd or Equal & Approved
3. Aluminium: Aluminium sheet to BSEN515:1993
 - 3.1. Finish: PPC
 - 3.2. Thickness: 2mm
4. Dimensions
 - 4.1. Lengths: Not more than 2 m, with end to end joints lapped not less than 100 mm.
 - 4.2. Cover to roofing upstand: Not less than 75 mm, with bottom edge welted 15 mm.
5. Fixing: Metal Framing to subcontractor design
6. Insulation: Min 50mm mineral wool insulation under steels

422 Aluminium canopies

1. Description: Aluminium canopy with integral gutter, RWP and soffit panels with recessed joints
2. Manufacturer: Alumasc Exterior Building Products Ltd or Equal & Approved
3. Aluminium: Aluminium sheet to BSEN515:1993
 - 3.1. Finish: PPC
 - 3.2. Thickness: 3mm
4. Dimensions
 - 4.1. Lengths: Not more than 2 m, with end to end joints lapped not less than 100 mm.
 - 4.2. Cover to roofing upstand: Not less than 75 mm, with bottom edge welted 15 mm.
5. Fixing: Supported by primary steel frame to structural engineers detail, plywood lining with membranes by aluminium steel subcontractor, panels fixings including soffit recessed panel joint to metal framing to subcontractor design

423 Door Canopy

1. Manufacturer: [Alumasc Water Management Solutions](#)
2. Contact details
Rivington Street Studio Limited
10-10-2025

- 2.1. Address: Station Road
Burton Latimer
Kettering
Northamptonshire
NN15 5JP
- 2.2. Telephone: +44 (0)1536 383810
- 2.3. Web: <https://www.alumascwms.co.uk/>
- 2.4. Email: marketing@alumascwms.co.uk
3. Product reference: Alumasc Skyline Door Canopy
4. Shelter description: Door Canopy.
5. Dimensions
 - 5.1. Plan size: 2500mm wide x 900mm depth
 - 5.2. Height: 90mm
6. Frame
 - 6.1. Material: Aluminium.
 - 6.2. Finish: Powder coated.
 - 6.3. Colour: RAL colour to match windows TBC
7. Roof covering: Aluminium.
8. Weather resistance: Weathered System.
9. Fixings: Aluminium Bracketry.
10. Other requirements: Fixing requirements back to primary steel TBC with Structural Engineer

424 Flashings to cladding

1. Description: Flashing to base and head of rainscreen
2. Manufacturer: Alumasc Exterior Building Products Ltd or Equal & Approved
3. Aluminium: Aluminium sheet to BSEN515:1993
 - 3.1. Finish: PPC
 - 3.2. Thickness: 2mm
4. Dimensions
 - 4.1. Lengths: Not more than 2 m, with end to end joints lapped not less than 100 mm.
 - 4.2. Cover to roofing upstand: Not less than 75 mm, with bottom edge weltd 15 mm.
5. Fixing: As recommended by manufacturer
6. Other Requirements: Continuous along length of building perimeter and to form window cills

440 Aluminium flashings Type A

1. Manufacturer: Contractor's choice
2. Product reference: [Aluminium Roof Edge Drip Trim](#)
3. Material: Aluminium sheet to BSEN515:1993
4. Item: Roof edge drip trim.
5. Form of delivery: Preformed item.
6. Size
 - 6.1. Thickness (minimum): 2mm.
7. Finish: Polyester powder-coated.
8. Colour: RAL colour to match other roof copings
9. Texture: Plain.

10. Depth: 150 mm.

470 Aluminium Flashing to roof upstand

1. **Description:** Folded aluminium fascia panel
2. **Aluminium:** Aluminium sheet to BSEN515:1993
 - 2.1. **Finish:** PPC to match copings
 - 2.2. **Thickness:** 3mm
3. **Dimensions**
 - 3.1. **Lengths:** Not more than 2 m.
4. **Fixing:** Allow for movement fixed to aluminium angles Butt-jointed panels with 5 mm gap Behind gap, fix black EPDM strip

880 Leftover material

1. **Spare covering material:** Clean, adhesive-free off-cuts to be removed from site and taken to an Altro Recofloor collection points for recycling.

Contact Altro for further details of this scheme and the Recowall recycling scheme. Altro Limited, tel: +44(0)1462 707600, fax: +44 (0)1462 707515, email: enquiries@altro.com, website www.altro.com

880 Leftover material

1. **Spare covering material:** Clean, adhesive-free off-cuts to be removed from site and taken to an Altro Recofloor collection points for recycling.

Contact Altro for further details of this scheme and the Recowall recycling scheme. Altro Limited, tel: +44(0)1462 707600, fax: +44 (0)1462 707515, email: enquiries@altro.com, website www.altro.com

General requirements/ preparatory work

510 Workmanship generally

1. **Standard:** Generally to BS EN 14783, BS EN 507, CP 143-15 and latest edition of the FTMRC publication 'UK guide to good practice in fully supported metal roofing and cladding'.
2. **Fabrication and fixing:** To provide a secure, free draining and completely weathertight installation.
3. **Operatives:** Trained in the application of aluminium coverings/ flashings. Submit records of experience on request.
4. **Measuring, marking, cutting and forming:** Prior to assembly wherever possible.
5. **Marking out:** With pencil, chalk or crayon. Do not use scribes or other sharp instruments without approval.
6. **Folding:** With mechanical or manual presses to give straight, regular and tight bends, leaving panels free from ripples, kinks, buckling and cracks. Use hand tools only for folding details that cannot be pressed.
7. **Surface protection:** Fully coat surfaces to be embedded in concrete or mortar with high build bitumen-based paint, after folding.
8. **Sharp metal edges:** Fold under or remove as work proceeds.
9. **Joints:** Do not use sealants to attain waterproofing.
10. **Finished aluminium work:** Fully supported, adequately fixed to resist wind uplift and able to accommodate thermal movement without distortion or stress.
 - 10.1. **Protection:** Prevent staining, discolouration and damage by subsequent works.

515 Welding

1. In situ welding: Not permitted.

516 Welding

1. In situ welding: Permitted subject to completion of a 'hot work permit' form and compliance with its requirements.

520 Aluminium strip/ sheet

1. Standard:
 - 1.1. Stamped or labelled with alloy designation, temper, finish and thickness.
2. Manufacturer:
 - 2.1. Product reference:

530 Integrity of aluminium

1. Requirement: Design coverings/ flashings and methods of attachment to prevent loss of weathertightness and permanent deformation due to wind pressure or suction.
2. Structural requirements
 - 2.1. Wind loads:

535 Integrity of aluminium

1. Requirement: Design coverings/ flashings and methods of attachment to prevent loss of weathertightness and permanent deformation due to wind pressure or suction.
2. Wind loads:
3. Structural requirements
 - 3.1. Generally: As section B50.
 - 3.1.1. Modifications:
4. Design: Complete in accordance with the designated code of practice to satisfy specified performance criteria.

550 Lightning protection

1. Aluminium coverings: Attach the following to a lightning protection system
2. Electrical continuity: Provide between aluminium strips/ sheets via wetting of joints.

555 Layout

1. Setting out of longitudinal and cross joints: Submit proposals.

560 Control samples

1. General: Complete areas of finished work and obtain approval of appearance before proceeding:
2. Size:
3. Location:

610 Suitability of substrates

1. Condition: Dry and free of dust, debris, grease and other deleterious matter.

630 Plywood overlay

1. **Standard:** Manufactured to an approved national standard and to BS EN 636, section 8 (plywood for use in humid conditions).
 - 1.1. **Sheet size:** 2400 or 1200 x 1200 mm and 6 mm thick.
2. **Laying:** Parallel to perimeter edges with cross joints staggered and a 0.5 to 1 mm gap between sheets.
3. **Fixing:** With 25 mm annular ringed shank aluminium or galvanized steel nails, at 300 mm grid centres over the area of each sheet and at 150 mm centres along edges, set in 10 mm from perimeter edges and in pairs across joints.
 - 3.1. **Nail heads:** Set flush with or just below surface.

640 Timber for use with aluminium work

1. **Quality:** Planed, free from wane, splits, pitch pockets, decay and insect attack (ambrosia beetle excepted).
2. **Moisture content:** Not more than 22% at time of fixing and covering.
3. **Preservative treatment:** Organic solvent as section Z12, and Wood Protection Association Commodity Specification C8.

Fixing

790 Screw fixing into joints/ chases

1. **Joint/ Chase:** Rake out to a depth of not less than 25 mm.
2. **Aluminium:** Fold into joint/ chase and up back face.
3. **Fixing:** Into back face with sherardized, zinc plated steel or aluminium screws, washers and plastics plugs at not more than 450 mm centres, at every change of direction, and with at least two fixings for each piece of aluminium.
4. **Sealant:**
5. **Application:** As section Z22.

Jointing

810 Forming details

1. **Folds and welts:** Form without thinning or splitting the strip/ sheet.
2. **Thermal movement:** Form details with appropriate allowance for movement, without impairment of security at full expansion or contraction.

Ω End of Section

H92 Rainscreen cladding

Clauses

Read in conjunction with Preliminaries and general conditions

Tendering

10 Information to be provided with tender

1. Submit the following cladding particulars
 - 1.1. Typical plan, section and elevation drawings at suitable scales.
 - 1.2. Typical detailed drawings at large scales, including all interfaces.
 - 1.3. Technical information and certification demonstrating compliance with specification of proposed incorporated products and finishes, including all products.
 - 1.4. Certification, reports and calculations demonstrating compliance with specification of proposed cladding.
 - 1.5. Proposals for connections to and support from the primary support structure.
 - 1.6. Proposals for primary support structure additional to that shown on preliminary design drawings.
 - 1.7. Schedule of builders' work, special provisions and special attendance by others.
 - 1.8. Examples of standard documentation from which project quality plan will be prepared.
 - 1.9. Preliminary fabrication and installation method statements and programme.
 - 1.10. Proposals for replacing damaged or failed products.
 - 1.11. Areas of non-compliance with specification.

Type(s) of rainscreen cladding

120 Fibre-reinforced cement boards to Ground Floor

1. Rainscreen cladding system
 - 1.1. Manufacturer: [EQUITONE by Etex](#)
 - 1.1.1. Contact details
 - 1.1.1.1. Address: Etex (Exteriors) UK Ltd
Wellington Road
Burton upon Trent
DE14 2AP
 - 1.1.1.2. Telephone: [+44 \(0\)1283 501 555](tel:+44(0)1283501555)
 - 1.1.1.3. Web: <https://www.equitone.com/en-gb/>
 - 1.1.1.4. Email: infouk@etexgroup.com
 - 1.1.2. Product reference: [EQUITONE \[Natura\] - Fibre Cement Rainscreen Cladding \(Charcoal N073\)](#)
 - 1.2. Standard: To EN 12467, EN 15804.
 - 1.3. Fire rating: To EN 13501-1, A2-s1,d0.
 - 1.4. Weather resistance: To BS EN 12467, category A.
 - 1.5. Bending strength: To BS EN 12467, Class 4.
 - 1.6. Sheet size
 - 1.6.1. Nominal thickness: 12 mm
 - 1.6.2. Length: 2500 mm.

- 1.6.3. Width: 1250 mm.
- 1.7. Edges: Fitted with a decorative trim profile.
- 1.8. Colour: Charcoal N073 TBC as part of planning condition
- 1.9. Fixing: Rivet fixed to match board colour
- 1.10. Surface texture: Silk smooth.
- 1.11. Joint size: 8mm.
- 1.12. Durability classification: To EN 12467, Category A.
- 1.13. Apparent density: To EN 12467, 1650 kg/m³.
- 1.14. BendingStrength: To BS EN 12467, 24 N/mm² (perpendicular); 18.5 N/mm² (parallel).
- 1.15. WaterImpermeability: To EN 12467, pass.
- 1.16. Freeze-thaw resistance: To EN 12467, pass.
- 1.17. Movement: 0.01 mm/Mk.
- 1.18. ThermalConductivity: 0.6 W/Mk.
- 1.19. Insulation: As P17/217B
- 1.20. Other requirements: Black insect mesh to all ventilation gaps
- 1.21. Support framing: To subcontractor design. Note: framing members black spray painted prior to cladding install where visible through gaps in panels

121 Fibre-reinforced cement boards to Roof Areas

- 1. Rainscreen cladding system
 - 1.1. Manufacturer: [EQUITONE by Etex](#)
 - 1.1.1. Contact details
 - 1.1.1.1. Address: Etex (Exteriors) UK Ltd
Wellington Road
Burton upon Trent
DE14 2AP
 - 1.1.1.2. Telephone: [+44 \(0\)1283 501 555](tel:+44(0)1283501555)
 - 1.1.1.3. Web: <https://www.equitone.com/en-gb/>
 - 1.1.1.4. Email: infouk@etexgroup.com
 - 1.1.2. Product reference: [EQUITONE \[Natura\] - Fibre Cement Rainscreen Cladding \(Charcoal N073\)](#)
 - 1.2. Standard: To EN 12467, EN 15804.
 - 1.3. Fire rating: To EN 13501-1, A2-s1,d0.
 - 1.4. Weather resistance: To BS EN 12467, category A.
 - 1.5. Bending strength: To BS EN 12467, Class 4.
 - 1.6. Sheet size
 - 1.6.1. Nominal thickness: 8 mm.
 - 1.6.2. Length: 2500 mm.
 - 1.6.3. Width: 1250 mm.
 - 1.7. Edges: Fitted with a decorative trim profile.
 - 1.8. Colour: Charcoal N073 TBC as part of planning condition
 - 1.9. Fixing: Rivet fixed to match board colour
 - 1.10. Surface texture: Silk smooth.
 - 1.11. Joint size: 8mm.
 - 1.12. Durability classification: To EN 12467, Category A.
 - 1.13. Apparent density: To EN 12467, 1650 kg/m³.

- 1.14. **BendingStrength:** To BS EN 12467, 24 N/mm² (perpendicular); 18.5 N/mm² (parallel).
- 1.15. **WaterImpermeability:** To EN 12467, pass.
- 1.16. **Freeze-thaw resistance:** To EN 12467, pass.
- 1.17. **Movement:** 0.01 mm/Mk.
- 1.18. **ThermalConductivity:** 0.6 W/Mk.
- 1.19. **Insulation:** Phenolic insulation as H92/776
- 1.20. **Other requirements:** Black insect mesh to all ventilation gaps
- 1.21. **Support framing:** To subcontractor design. Note: framing members black spray painted prior to cladding install where visible through gaps in panels

General requirements/ preparatory work

210 Rainscreen cladding system design

- 1. **Rainscreen cladding system and associated features:** Complete detailed design in accordance with this specification and the preliminary design drawings and submit before commencement of fabrication.
- 2. **Standard:** In accordance with the [CWCT Standard for systemised building envelopes](#).
 - 2.1. **Reference information:**
- 3. **Appearance and fit:** Ensure that position and alignment of parts and features are as shown on preliminary design drawings. Accommodate deviations in the primary support structure.
- 4. **Related works:** Coordinate in the detailed design.
- 5. **Detailed design:** Complete and submit proposals for approval before fabrication.
- 6. **Submission of alternative proposals:** Preliminary design drawings indicate intent. Other reasonable proposals will be considered.

230 Information to be provided during detailed design

- 1. **Submit the following cladding particulars**
 - 1.1. A schedule of detailed drawings and dates for submission for comment.
 - 1.2. A schedule of loads that will be transmitted from the rainscreen cladding to the structure.
 - 1.3. Proposed fixing details and systems relevant to the structural design and construction with methods of adjustment and tolerances.
 - 1.4. A schedule of fabrication tolerances/ size tolerances.
 - 1.5. A detailed testing programme in compliance with the main contract master programme.
 - 1.6. A detailed fabrication and installation programme in compliance with the main contract master programme.
 - 1.7. Proposals to support outstanding applications for Building Regulations consents or relaxations.

232 Quality management

- 1. **Required documentation:** Project specific method statement in accordance with [CWCT Standard for systemised building envelopes](#) and guidance in [CWCT Technical Note 53](#).
- 2. **Content:** In accordance with [BS EN ISO 9001](#), and including the following:
 - 2.1. Name of the quality manager.
 - 2.2. Quality assessment procedures.
 - 2.3. Inspection procedures to be adopted in checking the work.
 - 2.4. Stages at which check lists will be used, and samples of the lists.
 - 2.5. List of work procedures on the correct use of materials or components, both off site and on site.

- 2.6. List of product information with latest revisions.
 - 2.7. Subcontractors involved in the work.
 - 2.8. Subcontractors' quality plans.
 - 2.9. Storage, handling, transport and protection procedures.
 - 2.10. Procedure for registering and reporting non-compliances.
 - 2.11. Maintenance procedures and calibration records.
 - 2.12. Certification that completed work complies with specification.
 - 2.13. Check list register to ensure that all items have been inspected and non-compliances discharged.
3. **Timing:** Submit during detailed design.

235 Information to be provided before commencement of testing or manufacture of rainscreen cladding system

1. **Submit the following cladding particulars**
 - 1.1. Detailed drawings to fully describe fabrication and installation.
 - 1.2. Detailed calculations to prove compliance with design/ performance requirements.
 - 1.3. Project-specific fabrication, handling and installation method statements.
 - 1.4. Certification for incorporated components manufactured by others confirming their suitability for proposed locations in the rainscreen cladding.
 - 1.5. Recommendations for spare parts for future repairs or replacements.
 - 1.6. Recommendations for safe dismantling and recycling or disposal of products.

240 Product samples

1. **General:** Before commencing detailed design, submit labelled samples of: cladding in chosen colour.

250 Samples of fixings

1. **General:** During detailed design, submit labelled samples of each type of fixing, together with manufacturers' recommended torque figures.

Design/performance requirements

310 CWCT 'Standard for systemised building envelopes'

1. **General:** Unless specified or agreed otherwise, comply with:
 - 1.1. Part 2 – Loads, fixings and movement.
 - 1.2. Part 3 – Air, water and wind resistance.
 - 1.3. Part 4 – Operable components, additional elements and means of access.
 - 1.4. Part 5 – Thermal, moisture and acoustic performance.
 - 1.5. Part 6 – Fire performance.
 - 1.6. Part 7 – Robustness, durability, tolerances and workmanship.
2. **Project performance requirements specified in this subsection:** Read in conjunction with [CWCT](#) performance requirements.

335 Integrity

1. **Requirement:** The rainscreen cladding must resist wind loads, dead loads and design live loads, and accommodate deflections and movements without damage.

350 Deflection under wind load

1. **Requirement:** Requirement: For listed components, at positive and negative applications of the design wind pressure, normal deflections are not to exceed: Refer Structural Engineer calculations.

360 Wind resistance – cyclic loading

1. **Requirement:** No reduction in the integrity of the rainscreen cladding must occur when subjected to the test sequence given in [CWCT Standard for systemised building envelopes](#), clause 8.14.6.
2. **Test method:** As clause 665.

380 General movement

1. **Requirement:** Rainscreen cladding must accommodate anticipated building movements as follows:
.....

385 Thermal movement

1. **Service temperature range:** To [CWCT Standard for systemised building envelopes](#), clause 2.7.2, Tables 2.2 and 2.3.

430 Thermal performance

1. **Method of calculation:** Weighted U-value.
2. **U-value of rainscreen wall (average):** Average U-value of rainscreen wall: 0.15 W/m²K

450 Air and vapour control layer

1. **Condensation risk within rainscreen wall:** Determine using the method described in [BS 5250](#). Where required, provide a suitable air and vapour control layer to ensure that damage and nuisance from condensation is reduced.

470 Fire performance of rainscreen cladding

1. **Reaction to fire classification**
 - 1.1. **External surfaces:** To [BS EN 13501-1](#), Class A1
 - 1.2. **Internal (cavity) surfaces:** To [BS EN 13501-1](#), Class A1

495 Durability

1. **Design life of the rainscreen cladding system:** Not less than 40 years.
2. **Secondary components:** Submit details together with required maintenance regime, replacement periods and methods of replacement.

Testing

510 Comparison (type) testing

1. **Commencement of fabrication and installation of rainscreen cladding:** Not until test results and reports showing compliance with this specification have been submitted.

530 Testing authority

1. **Requirement:** Project testing must be carried out by a [United Kingdom Accreditation Service \(UKAS\)](#)-approved independent laboratory.

680 Site spray bar test

1. **Requirement:** To [CWCT Standard test methods for building envelopes](#), part 10.

685 Site hose test

1. Requirement: To [CWCT Standard test methods for building envelopes](#), part 9.
 - 1.1. Joints to be tested:

Products

730 Mechanical fixings – material requirements

1. **Stainless steel:** To the relevant part(s) of BS EN ISO 3506, Grade A2 generally, Grade A4 when used in severely corrosive environments.
2. **Carbon steel:** To [BS 4190](#) and suitable for galvanizing or other protective coating.
3. **Aluminium:** To BS EN 755.

732 Adhesives

1. **General:** Not degradable by moisture or water vapour.

735 Fixings and fasteners

1. **Type and use:** Reviewed and approved by manufacturers. Submit confirmatory information on request.
2. **Dimensions:** Not less than recommended by their manufacturers.
3. **Adjustment capability:** Sufficient in three dimensions to accommodate primary support structure and rainscreen cladding fabrication/ installation tolerances.

770 General sealants

1. **Selection:** In accordance with [BS 6213](#).
2. **Classification and requirements:** To [BS EN ISO 11600](#).
3. **Reaction to contact products and finishes:** Stable and compatible.

775 Thermal insulation

1. **Mineral Wool Slab Insulation :** As P10/217B

776 Phenolic foam boards

1. **Manufacturer:** [Unilin Insulation UK Limited](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** Park Road
Holmewood Enterprise Park
Chesterfield
Derbyshire
United Kingdom
S42 5UY
 - 1.1.2. **Telephone:** [+44 \(0\)371 2221033](tel:+44(0)3712221033)
 - 1.1.3. **Web:** <http://www.unilininsulation.co.uk>
 - 1.1.4. **Email:** info.ui@unilin.com
 - 1.2. **Product reference:** [Safe-R SR/FB Framing Board Insulation](#)
2. **Standard:** To BS EN 13166.
3. **Third-party certification:** BBA Certification No. 23/6884. BS EN ISO 9001 Quality Management System. BS EN ISO 14001 Environmental Management System. ISO OHSMS 45001 Health and Safety Management System.
4. **Fire performance:** Euroclass C s1-d0.

5. Thermal conductivity (maximum): 0.020–0.021 W/m·K.
6. Compressive strength (minimum) at 10% compression: 100 kPa.
7. Cross section: Rectangular.
8. Thickness: 120mm
9. Face size (length x width): 2400 x 1200 mm.
10. Recycled content: 0%.
11. Edges: Straight.
12. Facing: Low emissivity composite foil.

785 Breather membrane

1. Manufacturer: Submit proposals
2. Standard: To [BS EN 13859-2](#).
3. Continuity: No breaks. Minimize joints.
 - 3.1. Penetrations and abutments: Attach to breather membrane with tape. Achieve full bond.
 - 3.2. Laps: Not less than 150 mm, bond with tape. Achieve full bond.
4. Tape: As recommended by breather membrane manufacturer.
5. Repairs: Lapped patch of breather membrane material secured with continuous band of tape on edges.
6. Junctions at flashings, sills, gutters etc. Overlap and allow free drainage to exterior.

Finishes

830 Powder coating

1. Requirement: As section Z31.

840 Anodizing

1. Requirement: As section Z33.

Fabrication and installation

910 Generally

1. Electrolytic corrosion: Take necessary measures to prevent.
2. Identification of products: Mark or tag to facilitate identification during assembly, handling, storage and installation. Do not mark surfaces visible in the complete installation.

912 Metalwork

1. Requirement: As section Z11, unless specified otherwise in this section.

915 Timber treatment

1. Requirement: As section Z12, unless specified otherwise in this section.
2. Exposed cut and drilled surfaces:

922 Fixings/ adhesives application

1. Requirement: As section Z20, unless specified otherwise in this section.

925 Sealant application

1. Requirement: As section Z22, unless specified otherwise in this section.

930 Assembly

1. **Location:** Carry out as much assembly as possible in the workshop.
2. **Joints:** Other than movement joints and designed open joints, joints must be rigidly secured, reinforced where necessary and fixed with hairline abutments.
3. **Displacement of components in assembled units:** Submit proposals for reassembly on site.

970 Rainscreen cladding installation

1. **Tightening mechanical fasteners:** To manufacturer's recommended torque figures. Do not overtighten fasteners intended to permit differential movement.
2. **Protective coverings:** Remove only where necessary to facilitate installation and from surfaces that will be inaccessible on completion.

975 Welding

1. **In situ welding:** Not permitted.

980 Interfaces

1. **Installation:** Locate flashings, closers, etc. correctly, and neatly overlap cladding to form a weathertight junction.

985 Damage

1. **Repairs:** Do not repair cladding without approval.
 - 1.1. **Approval:** Will not be given where the proposed repair will impair performance or appearance.

995 Maintenance

1. **Maintenance manual:** Incorporate details within the building manual in accordance with [CWCT Standard for systemised building envelopes](#), clause 7.6.1.
 - 1.1. **Materials certification and test reports to be included:** As required

Ω End of Section

J30

Liquid-applied tanking/ damp-proofing

Types of tanking/ damp proofing

110 Polyurethane (PUR) waterproof coatings

1. Membrane system:
 - 1.1. Manufacturer: [Visqueen](#)
 - 1.1.1. Contact details
 - 1.1.1.1. Address: Visqueen
Heanor Gate Industrial Estate
Derbyshire
Heanor
Derbyshire
United Kingdom
DE75 7RG
 - 1.1.1.2. Telephone: [+44 \(0\) 333 202 6800](#)
 - 1.1.1.3. Web: [www.visqueen.com](#)
 - 1.1.1.4. Email: [enquiries@visqueen.com](#)
 - 1.1.2. Product reference: [Visqueen Axiom Guard PRO](#)
 - 1.2. Third-party certification: BBA certificate.
 - 1.3. Admixture: One-part polyurethane.
 - 1.4. Colour: Dark grey.
 - 1.5. Coverage: 8–10 m² per 15 kg drum (1.5–1.9 kg/m² application rate).
 - 1.6. Application temperature range: 5–30 °C.
 - 1.7. Curing time: 24 hours (approximately).
 - 1.8. Top coat: Axiom Guard PRO Topcoat.
 - 1.9. Matting: Not required.

Execution

205 Suitability of substrate

1. Substrates generally: Smooth, even textured, clean, dry and frost-free
2. Moisture content and stability of substrate: Must not impair integrity of finished tanking/ damp-proofing
3. Vertical and horizontal surfaces: Correctly prepared and free from irregularities
4. Curing period for concrete substrates (minimum): 24 hours approx

210 Coating application

1. Adjacent surfaces exposed to view in finished work: Protect.
2. Coatings
 - 2.1. Coverage: Coverage 8–10 m² per 15 kg drum (1.5–1.9 kg/m² application rate).
 - 2.2. Application generally: As manufacturers recommendations

Completion - Not Used

Ω End of Section

J40

Flexible sheet waterproofing/ damp-proofing

To be read with preliminaries/ general conditions. - Not Used

Types of tanking/ damp proofing

140 Plastics sheets Type A

1. Manufacturer: [Visqueen](#)
 - 1.1. Contact details
 - 1.1.1. Address: Visqueen
Heanor Gate Industrial Estate
Derbyshire
Heanor
Derbyshire
United Kingdom
DE75 7RG
 - 1.1.2. Telephone: +44 (0) 333 202 6800
 - 1.1.3. Web: www.visqueen.com
 - 1.1.4. Email: enquiries@visqueen.com
 - 1.2. Product reference: [Visqueen Radon R400 Membrane](#)
2. Purpose: Gas retardance.
3. Standard: CE Mark EN 13967:2017.
4. Performance characteristics
 - 4.1. Tensile strength (minimum): 15 N/mm².
 - 4.2. Tear resistance: 250 N.
 - 4.3. Elongation to break: 400%.
 - 4.4. Resistant to underground gases: Radon.
 - 4.5. Water vapour resistance (minimum): 941 MN·s/g.
 - 4.6. Fire performance: F class.
5. Third-party certification: British Board of Agrément (BBA) Certificate.
6. Form: Multi-folded roll.
7. Physical properties
 - 7.1. Dimensions
 - 7.1.1. Thickness (minimum): 0.4 mm.
 - 7.1.2. Width (minimum): 4 m.
 - 7.1.3. Roll length (minimum): 20 m.
8. Recycled content: None – virgin material.
9. Radon permeability: 8x10⁽⁻¹²⁾ m²/s.
10. Radon transmittance: 21x10⁽⁻⁹⁾ m/s.
11. Joint strength: 245 N.
12. Impact resistance: 200 mm.
13. Application temperature:

190 Bitumen sheets

1. Manufacturer: [Visqueen](#)

1.1. Contact details

- 1.1.1. Address: Visqueen
Heanor Gate Industrial Estate
Derbyshire
Heanor
Derbyshire
United Kingdom
DE75 7RG
- 1.1.2. Telephone: +44 (0) 333 202 6800
- 1.1.3. Web: www.visqueen.com
- 1.1.4. Email: enquiries@visqueen.com

1.2. Product reference: [Visqueen Gas Resistant Self Adhesive Membrane](#)

- 2. Standard: To BS 8102:2009.
- 3. Performance characteristics: Resistant to underground gases / Damp proof membrane.
- 4. Third-party certification: British Board of Agrément (BBA) certified.
- 5. Form: Flexible self-adhesive roll.
- 6. Material
 - 6.1. Carrier material:
 - 6.2. Bitumen modification: Adhesive-modified bitumen.
 - 6.3. Root inhibitor: Metal foil.
 - 6.4. Gas inhibitor: Aluminium foil.
- 7. Weight (minimum): 1.1 kg/m².
- 8. Thickness (minimum): 1 mm.
- 9. Surface finishes
 - 9.1. Top: Self adhesive bitume.
 - 9.2. Underside:
- 10. Accessories:
- 11. Length: 20 000 mm.
- 12. Width: 1000 mm.
- 13. Fire performance: F Class.
- 14. Application temperature:
- 15. Watertightness: 60 kPa.
- 16. Resistance to static loading - 20kg load: 20 kg.
- 17. Reaction to fire: E Class
- 18. Tensile elongation property: >130 %.

Workmanship

310 Workmanship generally

- 1. Condition of substrate
 - 1.1. Clean and even-textured, free from voids and sharp protrusions.
 - 1.2. Moisture content:
- 2. Air and surface temperature:
- 3. Condition of membrane at completion
 - 3.1. Neat, smooth and fully supported, dressed well into abutments and around intrusions.
 - 3.2. Completely impervious and continuous.

3.3. Undamaged. Prevent puncturing during following work.

4. Permanent overlying construction:

370 Preformed collars for pipes, ducts, cables, etc.

1. **Description::** A white surfaced, self adhesive strip, supplied in 300mm / 500mm x 10m rolls to ensure continuity of damp protection at complex junctions and stanchions.
2. **Installation:** Ensure surfaces are smooth, clean and dry. Prime the substrate to which the strip is to be adhered with Visqueen HP Tanking Primer and allow to dry. Do not apply primer to surface of membrane. Cut the strip to size ensuring overlap of minimum 150mm on to the membrane and the primed substrate. Progressively remove the release paper while pressing the strip firmly on to the surface. Work from the centre of the strip outwards to remove any trapped air. When the weather is cold, store strip in a warm, dry place until required. Installation is not recommended below 5C.
3. **Manufacturer:** Visqueen Building Products, Heanor Gate Industrial Estate, Derbyshire, DE75 7RG. Tel: 0845 302 4758 Email: enquires@visqueenbuilding.co.uk Web: www.visqueenbuilding.co.uk

3.1. **Product reference:** Visqueen Detailing Strip

4. **Other uses::** Where these pass through DPM and tanking membranes, junctions must be made completely watertight using Visqueen PFU-105 Preformed Pipe Cloak (Top Hat) Units fully bonded/ sealed to both pipes and sheeting. Finish the fix of these details using a stainless steel jubilee clip.
Where water proofing around circular piles the preformed pipe cloak to be bonded with Visqueen Zedex DPC Jointing Tape and sealed with Visqueen Detailing Strip. The unit to be secured with a jubilee clip.
NB Where non-standard Preformed Units are specified supply detailed isometric drawings fully dimensioned for each Preformed Unit required.

380 Protection of external vertical and horizontal tanking

1. **Manufacturer:** Visqueen Building Products, Maerdy Industrial Estate, Rhymney, Tradegar, NP22 5PY. Tel: 01685 804672 Fax: 01685 842580 Email: enquires@visqueenbuilding.co.uk Web: www.visqueenbuilding.co.uk
 - 1.1. **Product reference:** Visqueen TreadGUARD 1500, Visqueen Protec&Drain
2. **Use/Location::** Cover DPM/ tanking with Visqueen TreadGUARD 1500.
3. **Preparation/Installation::** Fix to Visqueen Gas Resistant Self Adhesive Membrane by means of Zedex DPC Jointing Tape, providing support as necessary to ensure Visqueen TreadGUARD 1500 or Protec&Drain remains fully in contact with tanking and does not move during backfilling operations.
All boards to be butt jointed and taped using Visqueen PRO Single Side Tape.

Ω End of Section

J41

Reinforced bitumen membrane roof coverings

Types of roof covering

Important Note

1. Please note that changes made to the content of this document, outside of the available choices may impact technical suitability and eligibility to meet Bauder Limited's requirements for guarantee. For additional items to be added, not already included, please contact your local Area Technical Manager.

Design Information and Supporting Documents

1. This specification is to be read in conjunction with the supporting Specification-Appendix & Torch Free Report (where applicable), Calculations (where available), Bauder Installation Guides and Bauder Standard Detail Drawings.

This specification has been produced based on the information supplied at the time of writing and is deemed to apply subject to the conditions outlined below, unless additional calculations proving otherwise have been completed and issued to you by Bauder Ltd or an approved supplier.

Windloads: Suitable for roofs where the design load does not exceed 3.2KN/m². Should the site be situated in a location subject to increased windloads or have a Design Windload Pressure suspected to be exceeding this, Bauder Ltd must be informed and a site specific windload calculation must be completed.

U-Values: U-Values quoted are based on the Bauder waterproofing system construction including insulation and underlying deck material only. This does not include the supporting structure and/or any other materials within the construction below the deck, unless Bauder has been advised otherwise prior to producing the calculations. Refer to the project specific U-Value Calculation for additional information.

Drainage: Where Bauder Ltd have produced supporting drainage calculations based on the data supplied, and the resulting calculation states that 1 drainage outlet will be sufficient, Bauder Ltd additionally recommends the use of overflows on all roofs and that there should always be at least 2 outlets and/or overflows per drainage area.

Fire Regulations (Fire Safety: Approved Document B 2019) - Compartmentation

1. Approved Document B Vol 1: Dwellings Section 5 Internal Fire Spread covers Compartmentation (Compartmentation is covered in Section 8 of Vol 2: Buildings other than dwellings and for Scotland, Wales and NI similar consideration is necessary for both Dwellings and Buildings other than Dwellings) and there is a consideration to be made if the roof goes over a compartment wall. For New Build we will expect the Principal Designer to establish Building Control requirements and notify accordingly before construction. For Refurbishment it is not always easy for the roofing team to establish if there is a compartment wall under the roof and what is its full orientation, we will rely on the specifier or client to tell Bauder Ltd if there is a known compartment wall under the existing roof. As the requirements have proven to be open to various interpretations across Building Control authorities, the Bauder advice is to consult with the Building Control officer for the building.

NBS Section J41 - Description of Works

1. Section J41 deals with the installation of the Bauder Waterproofing System, comprising coverings of multiple layers of reinforced bituminous membranes laid and jointed using self-adhesive and/or torch application as required. It includes where required, the air and vapour control layer, thermal insulation, underlayer and capping sheet membranes (root resistant for green roof systems) and presumes the deck substrate and roof falls as stated within the specification below. Accessories are included where relevant.

It is intended for use on projects where the detailed design is completed by the specifier (architect or landscape architect) with technical assistance from the manufacturer as required and should be read in conjunction with any project specific drawings provided.

'Safe2Torch' advice

1. *The application of torch-on materials to or in the vicinity of combustible deck materials does not conform to the recommendations of BS8217:2005, clause 7.3.2.1, paragraph 3, or the advice given in the 'Safe2Torch' document produced by the National Federation of Roofing Contractors. When encountering an area which contains combustible material a minimum 900mm deep zone of the flat area around the material and any detail flashing to the material itself there is a requirement for 'TORCH-FREE' detailing. In these instances an appropriate alternative Bauder self-adhesive membrane should be used as described in: 'TORCH-FREE' & 'Safe to Torch' DETAILING - ALTERNATIVE MEMBRANES AND APPLICATION. The 'TORCH-FREE' & 'SAFE TO TORCH' detailing and method of application will be described in the Additional Items section and the 'TORCH-FREE' & 'SAFE TO TORCH' DETAILING section of this specification and further detailed in the Bauder 'TORCH-FREE' & 'Safe to Torch' drawings.*

Please note, there could be 'TORCH-FREE' areas within the roof area(s), however there are either no roof plans available or the design is not sufficiently complete at this stage in the project to enable Bauder to complete a 'TORCH-FREE' roof plan. Once this plan becomes available and the design is sufficiently complete, please contact the Bauder Area Technical Manager (details at the end of the specification) and this can then be created for this area.

110A Main Roof Bitumen Membrane Roof on Tapered Rockwool insulation

1. Manufacturer: [Bauder Ltd](#)
2. Contact details
 - 2.1. Address: 70 Landseer Road
Ipswich
Suffolk
IP3 0DH
 - 2.2. Telephone: +44 (0)1473 257671
 - 2.3. Web: www.bauder.co.uk
 - 2.4. Email: info@bauder.co.uk
3. Product reference: [Bauder Total Warm Roof System - Self Adhered](#)
4. Substrate: New Plywood deck.

For the purpose of the tapered insulation scheme design, the roof deck is assumed to be level.

- Roof falls to be provided by the tapered insulation scheme.
- The design should take account of construction tolerances, permitted deviations and deflections under load, as per Item 4.4 of BS6229:2018.
- No deflections or back-falls.

5. Primer: **Bauder Activator-Primer (Canister), APR01-Black**, applied to the roof substrate and all upstands and skirtings. For application method and guidance information, refer clause 660B.
6. Air and vapour control layer
 - 6.1. Type: **BauderTEC KSD FBS**, 2.5 mm thick aluminium lined, elastomeric bitumen self-adhesive air and vapour control layer mica finish. Installation as clauses 670G, 710.
7. Insulation: BauderROCK Tapered Insulation
Thickness: Varies from 175mm to 355 to achieve u value of 0.15.
Facing: 300 g/m³ glass fibre fleece (Base Boards – raw).
Thermal conductivity (maximum): 0.039 W/mK.
Fire performance: A2-s1,d0.
Face Size (length x width): 1200 x 1000 mm.

Fixing: Activator-Primer (Canister) APR01 – Black. or PU Insulation Adhesive – Tin.

Angle Fillets: BauderROCK angle fillets insulated upstands.

Please note: where Bauder angle fillets are used they should be of the same Bauder insulation material.

Installation as clauses 680C and C & 775

8. **Insulating vertical upstands:** BauderROCK Insulation flatboard insulation as detailed. Support and installation to Bauder recommendations
9. **Waterproof covering**
 - 9.1. **Primer:** **Bauder Activator-Primer (Canister), APR01-Black**, applied to the upper surface of the Bauder insulation. For application method and guidance information, refer clause 660B.
 - 9.2. **First layer**
 - 9.2.1. **Type:** **BauderTEC KSA DUO**, 3 mm thick, 200g/m² glass-fibre reinforced, elastomeric self-adhesive bitumen underlayer.
Attachment: As clauses 710, 747A.
 - 9.3. **Top layer**
 - 9.3.1. **Type:** **Bauder K5K**, 5.2 mm thick, tensile strength 1000N/50mm, 250g/m² polyester reinforced, elastomeric bitumen torch applied capping sheet, Charcoal Grey finish.
 - 9.3.2. **Attachment:** As clauses 710, 750A.
10. **Flashings and detail work:** **Bauder K5K** capping sheet, mineral slate finish as per main field area. Install as clauses 773, 775 & 777.
11. **Surface protection:** Designated areas of maintenance access walkway defined using **Bauder capping sheet**, in a contrasting colour finish. Install as clause 863.
12. **System accessories:**
 - **Bauder Bitumen Compact Vertical Outlet DN 100** or **Bauder Bitumen Vertical Outlet DN 70 / DN 150**, (as appropriate) complete with pre-attached bitumen connection flange and dome grating. To be supplied and installed – quantity as required. When this outlet is used within a warm roof construction with insulation depth 60mm or greater, the **Bauder Extension Unit** (supplied separately) must be used. Installation as clause 784A.
 - **Bauder Bitumen Letterbox Parapet Outlet DN 205** including stainless-steel leaf guard, complete with pre-attached bitumen connection flange. To be supplied and installed – quantity as required. Installation as clause 784B.
 - **Bauder Parapet Emergency Overflow Stainless Steel DN 70** (where required)
Installation as clause 784G.
 - **Bauder GRP edge trim** (where required) to all perimeter kerbs – Profile: Client to confirm / Type 1 (44mm) / Type 2 (44mm) / Type 3 (70mm) / Type 4 (100mm) / Type 6 (150mm).
Colour: Client to confirm / Black / Grey. Installation, as clause 785.
 - **Bauder Lightning Conductor Clips** (where required) – Type 4 for PVC coated tapes and uncoated tapes. Installation, as clause 786.
13. **Flashings:** Capping as H72/420
14. **PV mounting system:** BauderSOLAR F XL PV mounting solution for framed solar modules. Loose laid and positioned on the Bauder capping sheet membrane, in accordance with the Bauder scheme provided and secured using the associated Bauder bituminous membrane restraint system to ensure no penetration of the waterproofing system.
15. **Additional requirements:** 210, 230, 515, 520, 560, 561, 562, 910, 940
16. **Guarantee information:** Refer to clause 950A

110B Roof Lanterns and Heart Space Bitumen Membrane roof

1. **Manufacturer:** [Bauder Ltd](#)
2. **Contact details**

- 2.1. Address: 70 Landseer Road
Ipswich
Suffolk
IP3 0DH
- 2.2. Telephone: +44 (0)1473 257671
- 2.3. Web: www.bauder.co.uk
- 2.4. Email: info@bauder.co.uk
3. Product reference: [Bauder Total Warm Roof System - Self Adhered](#)
4. Substrate: New Plywood deck.
 - Designed and constructed to provide a minimum finished slope of (1°) 1:60.The design should take account of construction tolerances, permitted deviations and deflections under load, as per Item 4.4 of BS6229:2018.
 - No deflections or back-falls.
 - Preparation: As clause 610A.
5. Primer: **Bauder Activator-Primer (Canister), APR01-Black**, applied to the roof substrate and all upstands and skirtings. For application method and guidance information, refer clause 660B.
6. Air and vapour control layer
 - 6.1. Type: **BauderTEC KSD FBS**, 2.5 mm thick aluminium lined, elastomeric bitumen self-adhesive air and vapour control layer mica finish. Installation as clauses 670G, 710.
7. Insulation: **BauderPIR FA-TE Flatboard**, aluminium foil faced, highly efficient rigid urethane insulation 140mm thick to achieve the required U value (refer Clause 230). Thicknesses greater than 160mm will be supplied in two or more layers of insulation board. This product has a zero ODP and a Green guide rating of 'A'. **BauderPIR T KL 50 angle fillets** 50 mm x 50 mm for use with insulated & un-insulated upstands. Installation as clauses 680D and 775.
8. Insulating vertical upstands: **BauderPIR FA-TE flatboard**, aluminium foil faced, zero ODP, highly efficient rigid urethane insulation.

The vertical upstand(s) should be insulated, typically an external wall to a conditioned/habited space will be insulated in its own right e.g. a cavity wall, but there is still a requirement for a nominal 30mm thickness of insulation on the external face of the wall to reduce thermal bridging at the roof/wall intersection.

This insulation will be a minimum of 300mm in height from the deck surface to the top of the upstand and will ideally be fitted before the flat insulation so that it is retained at the base. **Bauder Insulated Upstand Support Brackets** should be installed to secure the top and provide a hard-leading edge.

Rooflight upstands (e.g. Builder's Kerbs) that are not supplied by the rooflight supplier as part of the rooflight assembly must be insulated with 60mm thick **BauderPIR FA-TE flatboard** insulation to achieve a maximum U value of 0.35W/m²K.

Installation as clause 681B.
9. Waterproof covering
 - 9.1. Primer: **Bauder Activator-Primer (Canister), APR01-Black**, applied to the upper surface of the Bauder insulation. For application method and guidance information, refer clause 660B.
 - 9.2. First layer
 - 9.2.1. Type: **BauderTEC KSA DUO**, 3 mm thick, 200g/m² glass-fibre reinforced, elastomeric self-adhesive bitumen underlayer.Attachment: As clauses 710, 747A.
 - 9.3. Top layer
 - 9.3.1. Type: **Bauder K5K**, 5.2 mm thick, tensile strength 1000N/50mm, 250g/m² polyester reinforced, elastomeric bitumen torch applied capping sheet, Charcoal Grey finish.
 - 9.3.2. Attachment: As clauses 710, 750A.
10. Flashings and detail work: **Bauder K5K** capping sheet, mineral slate finish as per main field area. Install as clauses 773, 775 & 777.

11. **Surface protection:** Designated areas of maintenance access walkway defined using **Bauder capping sheet**, in a contrasting colour finish. Install as clause 863.
12. **System accessories:**
 - **Bauder Bitumen Compact Vertical Outlet DN 100** or **Bauder Bitumen Vertical Outlet DN 70 / DN 150**, (as appropriate) complete with pre-attached bitumen connection flange and dome grating. To be supplied and installed – quantity as required. When this outlet is used within a warm roof construction with insulation depth 60mm or greater, the **Bauder Extension Unit** (supplied separately) must be used. Installation as clause 784A.
 - **Bauder Bitumen Letterbox Parapet Outlet DN 205** including stainless-steel leaf guard, complete with pre-attached bitumen connection flange. To be supplied and installed – quantity as required. Installation as clause 784B.
 - **Bauder Parapet Emergency Overflow Stainless Steel DN 70** (where required) Installation as clause 784G.
 - **Bauder GRP edge trim** (where required) to all perimeter kerbs – Profile: Client to confirm / Type 1 (44mm) / Type 2 (44mm) / Type 3 (70mm) / Type 4 (100mm) / Type 6 (150mm). Colour: Client to confirm / Black / Grey. Installation, as clause 785.
 - **Bauder Lightning Conductor Clips** (where required) – Type 4 for PVC coated tapes and uncoated tapes. Installation, as clause 786.
13. **Flashings:** Capping as H72/421
14. **Additional requirements:** 210, 230, 515, 520, 560, 561, 562, 910, 940
15. **Guarantee information:** Refer to clause 950A

130 Canopy - Bauder Total Roof System

1. **Manufacturer:** [Bauder Ltd](#)
2. **Contact details**
 - 2.1. **Address:** 70 Landseer Road
Ipswich
Suffolk
IP3 0DH
 - 2.2. **Telephone:** [+44 \(0\)1473 257671](tel:+44(0)1473257671)
 - 2.3. **Web:** www.bauder.co.uk
 - 2.4. **Email:** info@bauder.co.uk
3. **Product reference:** [Bauder Total Roof System Reinforced Bitumen Membrane Cold Roof Covering System Self-Adhered \(with Torch-On Capping Sheet\)](#)
4. **Substrate:** New Plywood deck.
 - Designed and constructed to provide a minimum finished slope of (1°) 1:60. The design should take account of construction tolerances, permitted deviations and deflections under load, as per Item 4.4 of BS6229:2018.
 - No deflections or back-falls.
 - **Fire protection layer:** Supply and Install 6mm Magply Magnesium Oxy Sulphate Board directly over the deck to meet fire test classification B_{ROOF}(t4).
 - Preparation: As clause 610A.
5. **Prime:** **Bauder Activator-Primer (Canister), APR01-Black**, applied to the roof substrate and all upstands and skirtings. For application method and guidance information, refer clause 660B.
6. **Waterproof covering**
 - 6.1. **First layer**
 - 6.1.1. **Type:** **BauderTEC KSA DUO**, 3 mm thick, 200g/m² glass-fibre reinforced, elastomeric self-adhesive bitumen underlayer.
 - 6.1.2. **Attachment:** Attachment: As clauses 710, 747A.
 - 6.2. **Top layer**

- 6.2.1. **Type:** **Bauder K5K**, 5.2 mm thick, tensile strength 1000N/50mm, 250g/m² polyester reinforced, elastomeric bitumen torch applied capping sheet, Charcoal Grey finish.
- 6.2.2. **Attachment:** As clauses 710, 750A.
7. **Flashings and detail work:** **Bauder K5K capping sheet**, mineral slate finish as per main field area. Install as clauses 773, 775 & 777.
8. **Surface protection:** Designated areas of maintenance access walkway defined using Bauder capping sheet, in a contrasting colour finish. Install as clause 863.
9. **System accessories:**
- **Bauder Bitumen Compact Vertical Outlet DN 100** or **Bauder Bitumen Vertical Outlet DN 70 / DN 150**, (as appropriate) complete with pre-attached bitumen connection flange and dome grating. To be supplied and installed – quantity as required. When this outlet is used within a warm roof construction with insulation depth 60mm or greater, the **Bauder Extension Unit** (supplied separately) must be used. Installation as clause 784A.
 - **Bauder Bitumen Letterbox Parapet Outlet DN 205** including stainless-steel leaf guard, complete with pre-attached bitumen connection flange. To be supplied and installed – quantity as required. Installation as clause 784B.
 - **Bauder GRP edge trim** (where required) to all perimeter kerbs – Profile: Client to confirm / Type 1 (44mm) / Type 2 (44mm) / Type 3 (70mm) / Type 4 (100mm) / Type 6 (150mm). Colour: Client to confirm / Black / Grey. Installation, as clause 785.
 - **Bauder Lightning Conductor Clips** (where required) – Type 4 for PVC coated tapes and uncoated tapes. Installation, as clause 786.

B235955/1 JFK Special School Beckton ASD Extension

1.

Performance

210 Roof performance

1. **General:** Secure, free-draining and weathertight.

230 Thermal performance

1. **Requirement:** Determine type and thickness of insulation and integral or separate overlay to satisfy the following criteria:
- 1.1. **Thermal transmittance of roof (maximum):** 0.15W/m²K 140mm)
- U-Values quoted are based on the Bauder waterproofing system construction including insulation and underlying deck material only. This does not include the supporting structure and/or any other materials within the construction below the deck, unless Bauder has been advised otherwise prior to producing the calculations. Refer to the project specific U-Value Calculation for additional information.
- 1.2. **Finished surface:** Suitably even, stable and robust to receive roof covering.
- 1.3. **Insulation compliance:** To relevant harmonized European Standard, or Agrément-certified.
- 1.4. **IMPORTANT NOTE:** : Where Tapered Insulation is required, the thickness stated above is an approximate average thickness, please refer to the Bauder Tapered Scheme Drawings for the actual U-Value and min/max insulation thicknesses achieved.

240 Fire performance

1. **Classification:** BROOF(t4), in accordance with BS EN 13501-5

Products

330 Timber trims, etc.

1. **Quality:** Planed. Free from wane, pitch pockets, decay and insect attack (except ambrosia beetle damage).
2. **Moisture content at time of covering (maximum):** 22%.
3. **Preservative treatment:** Contract Administrator to advise. Please note organic solvent based timber preservatives are not permitted, as may affect the waterproofing.

331 Preformed metal hard edge insulation protection angles

1. **Material:** Galvanised mild steel
2. **Thickness:** 1mm
3. **Dimensions:** 50mm x 50mm
4. **Length:** 3m max

Execution generally

515 Adverse weather

1. **General:** Do not lay coverings in high winds, wet or damp conditions or in extremes of temperature unless effective temporary cover is provided over working area.
2. **Unfinished areas of roof:** Keep dry. Protect edges of laid membrane from wind action.

520 Incomplete work

1. **End of working day:** Provide temporary seal to prevent water infiltration.
2. **On resumption of work:** Cut away tail of membrane from completed area and remove from roof.

560 General Workmanship Requirements

1. Installation of the Bauder waterproofing system may only be carried out by trained and certified operatives approved by Bauder Ltd and who carry current ID badges. These should be available for inspection at all times.
2. Workmanship must comply with Codes of Practice BS 8217:2005 (or alternatively Bauder Ltd.'s specification where otherwise stated). Non-compliant workmanship will not be permitted, even if the system is watertight. The client will be told that all such faults must be remedied, before the Guarantee is issued.
3. All waterproofing materials and system components must be supplied by Bauder Ltd, unless otherwise stated. Any sub-standard materials or un-authorised alternatives will be rejected. Any building work which is the responsibility of the roofing contractor and has a bearing on the life of the Bauder System must be carried out by properly trained and qualified tradesmen.
4. Any structural damage, peculiarities or details discovered that might affect the performance of the Bauder system, should be reported immediately to the client's representative and Bauder Limited in order that they may assist in overcoming the problem.
5. At the end of each working day, the new waterproofing should be terminated with a secure and waterproof temporary seal, which will be left in situ, utilising Bauder self-adhesive underlayer material (e.g. BauderTEC Sprint DUO) to create the seal. The Bauder self-adhesive underlayer should be extended onto the flat AVCL by a minimum of 200mm from the exposed edge of the insulation and heat activate to ensure a satisfactory seal. To ensure that no moisture contamination of the system can occur between each working period, it is essential that the night seal is properly and securely bonded.
6. Where building works are to be carried out by other trades, following completion of the waterproofing, the contractor must make adequate provision for supplying protection to prevent damage to the new membranes. The final inspection will not be carried out by the Bauder Site

Technician or the Bauder nominated Independent surveyor until all associated trades are complete and the roof areas are clear from all debris and protection layers.

7. It is imperative that the Bauder Approved Contractor conforms to the workmanship criteria as listed above. Any deviation will result in the contract being considered unguaranteeable.
8. All mechanical and electrical work to plant and equipment should be carried out by competent mechanical and electrical qualified tradesmen. All plant is to be reinstated and re-commissioned on completion of the roofing works in accordance with the client's detailed specification.
9. Where building works are to be carried out by other trades, following completion of the waterproofing, the contractor must make adequate provision for supplying protection to prevent damage to the new waterproofing.
10. If any items of plant/equipment are to be situated on the finished roof, a sacrificial layer of Bauder capping sheet is to be loose laid beneath. This is to extend a minimum 25mm past the point of contact on all sides. In the case of heavy items it may be necessary to introduce a load-spreading slab, please contact Bauder for further advice.
11. All lead work to be carried out by skilled tradesmen and in accordance with current codes of practice and the recommendations of the Lead Contractor Association.

561 Site Inspections

1. Bauder Site Technicians will carry out regular inspections of the project during the course of the works.
2. Bauder must be notified when the roof is ready for final inspection and all related works and snagging complete. See also clause 910.

562 Health & Safety Information - Roofing Work

1. Follow the advice shown in the "Responsible Specification Checklist" produced by the National Federation of Roofing Contractors.
2. Suitable precautions must be taken to prevent accidents occurring when roofing systems are being installed.
3. The contractor must ensure that adequate measures are taken to effectively prevent injury to members of the public, contractors and any other persons who may be affected by the works including the public.
4. Where microwave equipment is installed at roof level, care must be taken to prevent persons working on the roof from being exposed to large doses of microwave radiation.
5. Similarly, the contractor should liaise with the client to ensure that there are no extract outlets situated on the roof where noxious or harmful emissions could affect persons working. Suitable precautions will be necessary to prevent exposure where this situation arises.
6. The contractor is responsible for providing adequate firefighting equipment in the form of extinguishers during work on the roof. These should be kept in easily accessible locations and be suitably signed.
7. Whenever possible, access to the roof should be made via internal staircases rather than by temporary means. Where this is not available, it is the responsibility of the contractor to ensure a safe means of access, egress and a safe workplace.

As far as roofs are concerned, edge protection in the form of scaffolding or a fixed structure should be in place to a height of 1.1 metres in accordance with the Workplace (Health, Safety and Welfare) Regulations 1992.

Failing this, the hierarchy of controls should be applied from the Work at Height Regulations 2005. Means of access should be by fixed ladder, passenger hoist or scaffolding.

8. The contractor must ensure that suitable written method statements and risk assessments are available for the work being undertaken. In particular, it is essential that manual handling methods be fully assessed as roofing materials are heavy and can cause serious injury

9. The contractor must ensure that suitable information about the roof covering is provided to the Client at the end of the work to ensure that work in future can be carried out safely. This information will form part of the Safety File.
10. All persons working on the roof should be provided with, and wear, suitable personal protective equipment and wet weather gear. Training must be provided to all contract staff on the safe use of the equipment.
11. The installer must observe Product Safety Datasheets, relevant to the materials being used as well as completing and complying with COSHH risk assessments.
12. We draw your attention to your duties under the Construction (Design and Management) Regulations 2015. Regulation 4, Client's duties in relation to managing projects states that the client must make suitable arrangements for managing a project, including the allocation of sufficient time and other resources. Regulation 5, Appointment of the Principal Designer and the Principal Contractor states that where more than one contractor will be working on a project at any time, the client must appoint a Principal Designer and a Principal Contractor. Please note that although Bauder will assist with the roof waterproofing system design, we will not undertake the role of Principal Designer.
13. It is always the responsibility of the contractor to carry out a risk assessment on all aspects of the contract. The 'Safe2Torch' checklist is solely for guidance for the safe installation of torch-on reinforced bitumen membranes and use of gas torches in the workplace

Substrates/ air and vapour control layers/ warm deck roof insulation

610A Suitability of substrates

1. Substrates generally:
 - Secure and properly fastened to the sub-structure, clean, dry, smooth, free from frost, corrosion, contaminants, loose material, damage and protrusions.
 - Compatible with waterproofing / coating system.
2. Substrate design:
 - Plywood should conform to BS EN 636-3 condition of Use Service Class 3 (Exterior) and Use Class 3, minimum 18mm thickness as specified by client.
 - OSB/3 should conform to BS EN 300 & CPD/CE, minimum 18mm thickness as specified by client.
3. **Setting out:** The new deck, should be fixed directly to either the joists or firings using non corroding ring shank nails/recommended screw fasteners.
 - All board edges should be fully supported by either the joists, firings or noggins.
 - Board joints to be close butted and staggered in accordance with BS 8217. Expansion gaps between boards on long and short edges to be 3mm.
4. **Fasteners:** Type: As recommended by panel manufacturer.
5. **Fastener heads:** Flush with, or below the boards surface.
6. **Moisture content and stability of substrate:** The deck should be protected from wetting during transportation, storage and installation on site before the Waterproofing System is installed. Before installing the waterproofing system, moisture readings should be taken into the timber deck and not just the surface. The results should be recorded. Readings should be no more than 20%.
7. **Deck falls:** Refer to clauses 110/120/130 above (where appropriate) for roof specific requirements.
 - No deflections or back-falls shall be present if the deck is designed to achieve a zero falls finished surface.
 - Falls to comply with the drainage requirements of BS 6229:2018 and current codes of practice BS 8217:2005.
 - The design should take account of construction tolerances, permitted deviations and deflections under load, as per Item 4.4 of BS6229:2018.
 - An Engineer's deflection analysis and site level survey should be consulted before commencement of waterproofing. Measures to rectify back-falls or deflection shall be

- undertaken by the deck installer/supplier prior to commencement of the waterproofing system.
8. **Preliminary work:** Complete including:
 - Formation of independent upstands, kerbs, box gutters, sumps, grooves, chases and expansion joints.
 - Fixing of battens, fillets and anchoring plugs/strips as required.
 9. **Preparation:** The approved waterproofing contractor is to inspect the installed deck and report any issues, including insufficient falls, that may have a detrimental effect upon the waterproofing system to both the Clients representative and Bauder Limited.
 10. **Where specifying/installing Bauder Un-insulated (Cold) Roof System - Fire protection layer:** Supply and Install 6mm Magply Magnesium Oxy Sulphate Board directly over the deck. Boards to be close butted and staggered, and installed using suitable stainless-steel fixings at 600mm centres, or adhered with **Bauder PU Insulation Adhesive (Tin)** using 10-20mm wide beads at 250mm centres (150mm centres at roof perimeters), all in accordance with manufacturers fixing instructions.
This product is available directly from suitable Builders Merchants. Please contact Bauder for a list if necessary.
IMPORTANT NOTES:
 - Under no circumstances should other brands / types of Magnesium Oxide (MgO) board be used as these will not have been fire tested as part of the Bauder system.
 - **The inclusion of Magply Magnesium Oxy Sulphate board is to ensure the roof meets fire test BROOF(t4).****Alternative:** Fire Retardant to Euroclass B, FR Grade OSB/3 Deck, minimum 18mm or as specified by the client.
 11. **Priming:** Before priming and application of the membrane, the substrate shall be clean and dry, free from surface water, ice, snow or frost, dust, dirt, oil, grease, or any foreign matter detrimental to the adhesion of the waterproofing system.

640 Fixing timber trims

1. **Fasteners:** type/length appropriate and suitable to particular deck substrate.
2. **Fixing centres (maximum):** 500 mm

641 Installing Preformed metal hard edge insulation protection angles

1. **Location:** Use to provide hard edge protection at all internal gutter channels on warm roofs where the insulation from the flat area steps down to meet the insulation in the gutter sole.
2. **Preparation:** Surface to be inspected and cleaned if necessary, using white spirit to remove any contaminants, dirt or dust & primed with the specified Bauder primer. Prepared material to be thoroughly dry before use
3. **Installation:** The 50 x 50 mm galvanised mild steel angle to be adhered to the exposed leading edge of the insulation using the specified Bauder primer along the upper surface/edge. The purpose being to retain the metal in position to prevent any post-installation movement.

660B Applying Primer

1. **Primer:** **Bauder Activator-Primer (Canister), APR01-Black.**
2. **Purpose:** Substrate primer to seal and prepare dry surfaces of a variety of common substrate material prior to the application of Bauder self-adhesive bitumen membranes.
3. **Before application:** All surfaces must be dry, clean and free from dust, dirt, oil, grease and loose material.
4. **Application method:** Spray Applied to provide even and full coverage. Avoid pooling. Never attempt torching within 10 min of primer application, even if the surface appears dry.
5. **Application rate:** - 300mm wide spray
 - Coverage: Approx. 96 g/m²
 - Two coats may be required for very porous substrates.

6. **Application temperature:** +5 - +30°C
7. **Drying time:** Approx. 5 - 10 mins, dependent upon ambient temperature and material porosity.
8. **Coats:** Fully bond. Allow volatiles to dry off thoroughly between coats.
9. **Re-application:** Necessary after 4 hours exposure if waterproofing has not yet been applied, to maintain adhesion performance.
10. **Caution:** Use only outdoors in well ventilated areas or with respiratory apparatus and keep away from all sources of ignition. Take necessary precautions to avoid the solvent vapour from entering the buildings ventilation system.

670G Laying Air and Vapour Control Layer

1. **Attachment:** Generally, cold applied and fully bonded to substrate in accordance with Bauder requirements.

Metal Decks: With metal decks the sheets should run in the direction of the crowns/troughs, with laps formed on the crowns of the deck to ensure that they are fully supported in accordance with Bauder requirements. To ensure side laps of the air and vapour control layer (AVCL) are fully supported, they will need to have the width cut to suit the profile of the deck or larger laps should be formed. To support end laps, cut approximately 200mm strip off the end of the AVCL roll and apply it taut across the troughs where the roll ends will meet.

Timber Boarded / Cross Laminated Timber Decks: First layer random nailed, as clause 355/720. Second layer, **BauderTEC KSD FBS** cold applied and fully bonded to substrate in accordance with Bauder requirements.

2. **Side and end laps:** minimum 100 mm, laid with all laps heat sealed to provide a continuous bitumen bead extrusion. Installation methods as recommended by Bauder.
3. **Penetrations:** Fully seal using bonding methods recommended by Bauder.
4. **Edges of insulation at roof edges, abutments, upstands, kerbs, penetrations and the like:** Enclose, with air and vapour control layer:
The air and vapour control layer must be dressed up all upstands and to the full extent of the detail. This is to ensure that the detail is fully encapsulated to reduce the risk to exposed combustible materials. The contractor is also to form all details in such a way that the underlayer is fully bonded to the air and vapour control layer to the full extent of the detail ensuring in a minimum fully bonded 100mm lap with the air and vapour control layer – please see Bauder Bituminous detail drawings.
 - Care should be taken to ensure adhesion when the temperature is below + 5° C.

680B Laying warm roof insulation BauderRock

1. **Setting out**
 - 1.1. **Long edges:** Fully Supported.
 - 1.2. **End edges:** Fully support.
 - 1.3. **Joints:** Close butted together.
 - 1.4. **End joints:** Stagger.
2. **Bedding:** Bonded to the upper surface of the air and vapour control layer using either Bauder PU Insulation Adhesive - Tin or Bauder PU Insulation Adhesive – Twin Cartridge. The adhesive should be applied in strips following the direction of the board length giving 4 no. continuous and equally spaced adhesive beads within each board width.(increase to 6 no. at roof perimeter)*. Adhesive bead widths are stated on appropriate product label and datasheet
IMPORTANT NOTE - When using Bauder PU Insulation Adhesive - Tin, a 'water-overmist' must be applied to the wet adhesive beads prior to installation of the insulation board. The mist can be applied using a hand-held pressure sprayer with a pump-action to achieve a very fine mist of water over the surface of the insulation adhesive. This will start the curing process of the adhesive in this type of application.
Avoid excessive over-wetting. Where there is excessive over-wetting, please remove from the

insulation board with a dry cloth prior to application of the insulation board.
Please leave 'open' for 5 minutes before placing the insulation board into the wet adhesive.
Installation instructions are included on the appropriate Product Data Sheet.

3. **Multiple board layers:** Where the total thickness of insulation required is greater than can be achieved by a single standard board, then additional boards can be adhered to make up the total thickness required. These additional boards should be bonded using either Bauder PU Insulation Adhesive - Tin or Bauder PU Insulation Adhesive – Twin Cartridge, applied in strips following the direction of the board length giving 4 no. (increase to 6 no. at roof perimeter)* giving continuous and equally spaced adhesive beads within each board width. The second layer of boards should be laid off-set and staggered.

Adhesive bead widths are stated on appropriate product label and datasheet.
IMPORTANT NOTE - When using Bauder PU Insulation Adhesive - Tin, a 'water-overmist' must be applied to the wet adhesive beads prior to installation of the insulation board.

The mist can be applied using a hand-held pressure sprayer with a pump-action to achieve a very fine mist of water over the surface of the insulation adhesive. This will start the curing process of the adhesive in this type of application.

Avoid excessive over-wetting. Where there is excessive over-wetting, please remove from the insulation board with a dry cloth prior to application of the insulation board.

Please leave 'open' for 5 minutes before placing the insulation board into the wet adhesive.
Installation instructions are included on the appropriate Product Data Sheet.

4. **Protection to exposed edges of insulation:** Reduced thickness treated timber batten (or equivalent plywood construction), or where timber materials are not permitted on site for this purpose, a suitable non-combustible alternative should be used, a minimum width of 150mm and 10mm less in thickness than the insulation to accommodate the build-up of the waterproofing layers – all securely fixed to the deck. Outer edges chamfered at changes in level.
5. **Completion:** Boards must be in good condition, well-fitting and stable.
6. **IMPORTANT NOTES:** Bauder Activator-Primer (Canister), APR01-Black, must be applied to the uppermost surface of insulation prior to installation of the self-adhesive underlayer.
 - Where two layers of boards are to be installed, the unfaced board should be used as the first layer.
 - Foil to foil installation (e.g. FA-TE to FA-TE or FA to TEC KSD Foil) must not be carried out using the 6.5kg Bauder PU Insulation Adhesive – Tin.

680C Installing BauderRock insulated upstands

1. **Bedding:** Bonded to the upper surface of the air and vapour control layer using either Bauder PU Insulation Adhesive – Tin or Bauder PU Insulation Adhesive – Twin Cartridge. The adhesive should be applied in strips following the direction of the board length giving 6 no. continuous and equally spaced adhesive beads within each board width.

Adhesive bead widths are stated on appropriate product label and datasheet.

Upstand insulation boards should be installed before the insulation to the flat areas so that the vertical upstand insulation is retained both at the base and at the top.
Adhesive bead widths are stated on appropriate product label and datasheet.

BauderROCK Only: We recommend three no. fixings are used, one in each top corner and one in the centre of the board to retain the board in place whilst the adhesive is curing.

Use thermally broken tube washers with a minimum 45mm round head, and fasteners.

Tube size and fastener type for insulation and substrate as recommended by fixing supplier.

Screw fix through into the underlying substrate, ensuring that washer is pulled in tight to the insulation.

Where the surface is uneven or difficult to bond to, it is permissible to use suitable thermally broken fixings.

IMPORTANT NOTE - When using Bauder PU Insulation Adhesive - Tin, a 'water-overmist' must be applied to the wet adhesive beads prior to installation of the insulation board.

The mist can be applied using a hand-held pressure sprayer with a pump-action to achieve a very fine mist

of water over the surface of the insulation adhesive. This will start the curing process of the adhesive in this type of application.

Avoid excessive over-wetting. Where there is excessive over-wetting, please remove from the insulation board with a dry cloth prior to application of the insulation board.

Please leave 'open' for 5 minutes before placing the insulation board into the wet adhesive.

Installation instructions are included on the appropriate Product Data Sheet.

2. **Protection to exposed edges of insulation:** Treated timber battens (or where timber materials are not permitted on site for this purpose, a suitable non-combustible alternative should be used) or Bauder Insulated upstand brackets (as appropriate to given detail situation) must be used at all right angled edges e.g. top edges of parapet walls or abutment upstands.
3. **Encapsulation seal:** Provision must be allowed for forming a minimum 100mm lap seal between the air and vapour control layer and underlayer where the insulation finishes.
4. **Completion:** Boards must be in good condition, well-fitting and stable.
5. **IMPORTANT NOTES:**
 - Bauder Activator-Primer (Canister), APR01-Black, must be applied to the facing of the uppermost surface of insulation prior to installation of the self-adhesive underlayer.
 - BauderROCK only: Where two layers of boards are to be installed, the un-faced board should be used as the first layer.
 - Foil to foil installation (e.g. FA-TE to FA-TE or FA to TEC KSD Foil) must not be carried out using the 6.5kg Bauder PU Insulation Adhesive – Tin.

680D Laying warm roof insulation Type A

1. **Setting out**
 - 1.1. **Long edges:** Fully Supported.
 - 1.2. **End edges:** Fully support.
 - 1.3. **Joints:** Close butted together.
 - 1.4. **End joints:** Stagger.
2. **Bedding:** Bonded to the upper surface of the air and vapour control layer (AVCL) using suitable Bauder Polyurethane Insulation Adhesive. (Product selection assistance available from Bauder). The adhesive should be applied in strips following the direction of the board length. Giving continuous and equally spaced adhesive beads within each board width as indicated below:
 - 600mm width insulation boards - 2 no: (increase to 3 no. at roof perimeter)*
 - 800mm width insulation boards - 3 no: (increase to 4 no. at roof perimeter)*
 - 1000mm width insulation boards - 4 no: (increase to 6 no. at roof perimeter)*Adhesive bead widths or adhesive spray patterns and coverage rates are stated on appropriate product label and datasheet.
3. **Multiple board layers:** Where the total thickness of insulation required is greater than can be achieved by a single standard board, additional boards can be adhered to the previous layer (s) to make up the total thickness required.

Foil to Foil Insulation Boards Only: These additional boards should be bonded using either **Bauder Activator Primer (Canister) – APR01-Black** or **Bauder PU Insulation Adhesive – Twin Cartridge**.

Foil to Tissue Faced Boards: These additional boards should be bonded using suitable Bauder Insulation Adhesive.

Adhesives applied in strips following the direction of the board length giving continuous and equally spaced adhesive beads within each board width (as indicated above). The second layer of boards should be laid off-set and staggered.

4. **Protection to exposed edges of insulation:** Reduced thickness treated timber batten as clause 640 (or equivalent plywood construction), a minimum width of 150mm and 10mm less in thickness than the insulation to accommodate the build-up of the waterproofing layers – all securely fixed to the deck. Outer edges chamfered at changes in level.
5. **Completion:** Boards must be in good condition, well-fitting and stable.
6. **IMPORTANT NOTES:** **Bauder Activator-Primer (Canister), APR01-Black**, must be applied to the uppermost surface of insulation prior to installation of the self-adhesive underlayer.
 - Foil to Foil installation (e.g. FA-TE to FA-TE) must not be carried out using the 6.5kg **Bauder PU Insulation Adhesive – Tin**.
 - Foil to AVCL installation (e.g. FA G16 Tapered directly to KSD FBS) must not be carried out using **Bauder Activator Primer (Canister) – APR01-Black**.
7. **Adequate falls:** The inclusion of flat board insulation will not improve upon the existing fall present within the roof structure. The deck should be inspected by the installer to ascertain that the falls provided are adequate to prevent standing water. In the event of any queries Bauder Ltd should be contacted.

681B Installing Warm Roof Insulation (Insulated Upstands)

1. **Bedding:** Bonded to the upper surface of the air and vapour control layer (AVCL) using suitable Bauder Polyurethane Insulation Adhesive. The adhesive should be applied in strips following the direction of the board length giving 3 no. continuous and equally spaced adhesive beads within each board width. Upstand insulation boards should be installed before the insulation to the flat areas so that the vertical upstand insulation is retained both at the base and at the top. Adhesive bead widths are stated on appropriate product label and datasheet.
Note: Where the surface is uneven or difficult to bond to, it is permissible to use suitable thermally broken fixings.
2. **Multiple board layers:** Where the total thickness of insulation required is greater than can be achieved by a single standard board, then additional boards can be adhered to the previous layer(s) to make up the total thickness required. These additional boards should be bonded using suitable Bauder Insulation Adhesive.

Foil to Foil Insulation Boards Only: These additional boards should be bonded using either **Bauder Activator Primer (Canister) – APR01-Black** or **Bauder PU Insulation Adhesive – Twin Cartridge**.

Foil to Tissue Faced Boards: These additional boards should be bonded using suitable Bauder Insulation Adhesive.

Adhesive bead widths or adhesive spray patterns and coverage rates are stated on appropriate product label and datasheet.

The adhesive should be applied in strips following the direction of the board length giving 3 no. continuous and equally spaced adhesive beads within each board width. Upstand insulation boards should be installed before the insulation to the flat areas so that the vertical upstand insulation is retained both at the base and at the top.

The second layer of boards should be laid off-set and staggered.

Note: Where the surface is uneven or difficult to bond to, it is permissible to use suitable thermally broken fixings.

3. **Insulated upstand bracket:** Bauder insulated upstand support brackets must be used at all vertical abutment wall upstands (where the wall cavity is insulated) in conjunction with 30mm Bauder insulation. These are to be fixed at 400mm centres using suitable fixings through the air and vapour control layer, so that the top edge is a minimum of 300mm above the surface of the deck. A 3mm gap should be left between adjacent sections. The detail is to be carried out in accordance with the Bauder detail drawing, where provided.
4. **Protective hard edges:** treated timber battens or Bauder Insulated upstand brackets (as appropriate to given detail situation) must be used at all right-angled edges e.g. top edges of parapet walls or abutment upstands.
5. **Encapsulation seal:** Provision must be allowed for forming a minimum 100mm lap seal between the air and vapour control layer and underlayer where the insulation finishes.
6. **Important Notes:**
 - **Bauder Activator-Primer (Canister), APR01-Black, must be applied to the facing of the uppermost surface of insulation prior to installation of the self-adhesive underlayer.**
 - Foil to foil installation (e.g. FA-TE to FA-TE) must not be carried out using **the 6.5kg Bauder PU Insulation Adhesive – Tin.**
 - Foil to AVCL installation (e.g. FA G16 Tapered directly to KSD FBS) must **not** be carried out using **Bauder Activator Primer (Canister) – APR01-Black.**

Waterproof membranes/ accessories

710 Laying reinforced bitumen membranes generally

1. **Direction of laying:** Unrolled up the slope.
 - 1.1. Where practicable, install so that water drains over and not into laps.
2. **Side and end laps:** 100 mm, with the exception of mineral surfaced membranes, where side laps are 80 mm, but the head laps to remain 100 mm.
3. **Head and side laps:** Offset.
4. **Intermediate and top layer/ capsheet:** Fully bond.
5. **Successive layers:** Apply without delay. Do not trap moisture.
6. **Strips of bitumen membrane for 'linear' details:** Cut from length of roll e.g. gutter sole pieces
7. **Detail flashings:** to be cut from width of roll.
8. **Completed coverings:** Firmly attached, fully sealed, smooth, weatherproof and free-draining.

715A Laying reinforced bitumen membranes on roofs pitched more than 5°

1. **18mm WBP Plywood or OSB3 Capping Sheet Retention Plate:** When insulation is to be incorporated within the system, the retention plate should be installed to allow provision for mechanical fixing of the top leading edge of the capping sheet. The plate to run continuously across the roof slope coinciding with where the top leading edge of the capping sheet occurs. A continuous run of the retention plate will be required for each row where the cap sheet head laps are staggered. Head laps should not be fixed in a continuous line, unless the roof is less than the length of the roll from the ridge to the gutter eaves.
2. **Retention plate:** Retention plate to be installed in accordance with the relevant Bauder guidance sheet.
 - **Locations:** In accordance with the Bauder Installation and Quality Assurance Manual.
 - **Size:** 18mm x 300mm x 2400mm.

- **Insulation thickness:** Where the mechanical retention plate is to be incorporated, the insulation below needs to be 20mm thinner to ensure the plate finishes virtually flush with the main area insulation. Mechanically fix the plate through to the deck using the specified thermally broken fasteners, as clause 354.
 - **Priming: Bauder Activator-Primer (Canister), APR01-Black,** to seal and prepare the surface of the mechanical retention plate prior to the application of Bauder membranes, as clause 660B..
3. **Setting out:** Parallel to roof slope, with successive layers carried over ridges.
 - 3.1. **Lengths (maximum):** As per roll length
 - 3.2. **End laps:** Half stagger and align on alternate bitumen membranes.
 4. **Additional fixing of bitumen membranes:** Mechanically fix all head laps of the capping sheet using 5 no. equally spaced fixings across the 1m wide sheet, using fasteners, as clause 355A. If there is any doubt as to the exact requirements, please contact Bauder Limited before proceeding

747A Self-adhesive bonding of reinforced bitumen membranes

1. **Bond:** Full over whole surface, with no air pockets

Please note that **Bauder Activator-Primer (Canister), APR01-Black**, must be applied to the uppermost layer of Bauder Insulation prior to installation of the self-adhesive underlayer.

2. **Underlayer:** Cold applied and fully bonded by removing the peel off release film. The side laps are to be 100mm and must be **laid red over blue**, and heat sealed/torched (depending on 'Torch-Free' & 'Safe to Torch' zones) and rolling with the **Bauder Long Handled Lap Roller** to extrude a continuous bead of bitumen. Head laps to be 100mm and staggered, side laps to be 80mm and heat sealed/torched (depending on 'Torch-Free' & 'Safe to Torch' zones) to extrude a continuous bead of bitumen. The underlayer must be taken up all upstands, edge details, in accordance with current British Codes of Practice and Bauder recommendations. The underlayer must be fully heat sealed/torched (depending on 'Torch-Free' & 'Safe to Torch' zones) with the air and vapour control layer to the full extent of the detail ensuring a minimum 100mm fully bonded lap - please see Bauder Bituminous detail drawings.
3. **Alternative underlayer for detail work:** For detailing to un-insulated abutment upstands, where the waterproofing is to be applied to rough or uneven non-combustible surfaces i.e. brickwork or concrete, it is permissible for the installing contractor to use the Bauder underlayer appropriate to the specified system where this product is considered to be better for application to these surfaces. For all other situations, and particularly to vertical insulation, the Bauder Self-Adhesive Underlayer appropriate to the specified system must be used.
4. **Provision for prevention of wind uplift (where required):** Mechanically fix as per the corresponding project specific wind load calculation, using appropriate thermally broken fasteners (for uninsulated roofs as clause 355A and for warm roofs as clause 355B), fixed through to the deck.
5. **Fixing bitumen membranes to vertical upstands:** Please refer to clause 775.

750A Laying mineral faced reinforced bitumen torch-on capping sheet - 'Safe to Torch' zone

1. **Bond:** Full over whole surface, with no air pockets.
2. **Excess compound at laps of top layer/ capping sheet:** Leave as a continuous bitumen bead extrusion.
3. **Laying top layer:** Fully bonded to the underlayer by torching in the approved Bauder manner. Head laps to be 100 mm, side laps to be 80 mm. All laps to upstands, edge details, flashings, etc. to be 100 mm.
4. **Fixing of bitumen membranes to vertical upstands:** Please refer to clause 775.
5. **Final Inspection:** The finished roof must be thoroughly inspected by the Bauder Site Technician. This is to ensure that any remedial treatment that is necessary can be carried out prior to issuing the guarantee. Failure to ensure the instigation of this inspection will result in the issuing of the Bauder guarantee being put in jeopardy.

6. **IMPORTANT NOTE:** The mineral slate finish is a natural product, so the raw material may differ in colour and shade, over which Bauder has no control. There may also be colour variations between each roll of membrane.

773 Torch free & 'safe to torch' zones - alternative membranes and application

1. For detailing application in locations constructed from or within the 'Torch-Free' & 'Safe to Torch' zones to potentially combustible materials or otherwise where it is considered appropriate by the contractor necessary to minimise the potential risk.
2. **Primers: Bauder Activator-Primer (Canister), APR01-Black**, must be used when using Bauder self-adhesive membranes and a 'TORCH-FREE' application is required.
3. **Underlayers:** It is permissible to use a Bauder self-adhesive membrane so long as this product is a recognised component of the system specified.

Acceptable alternatives underlayers are listed below: -

- **BTRS: BauderFLEX G4E** to be replaced with **BauderTEC KSA DUO**
- **BauderFlex: Bauder EGV 35 TF** to be replaced with **BauderTEC Sprint DUO**
- **BTRS Plus: N/A**
- **System Airtech: N/A**

NB: Where surface is uneven or not suitable for a self-adhesive membrane and where the surface is of a non-combustible material and is not required to be a 'Torch-Free' or 'Safe to Torch' zone – it is permissible to use a Torch Applied underlayer, so long as the product is a recognised component of the system specified.

Acceptable alternative underlayers are listed below:

- **BTRS: BauderTEC KSA DUO** to be replaced with **BauderFLEX G4E**
- **BauderFlex: BauderTEC Sprint DUO** to be replaced with **Bauder EGV 35 TF**
- **BTRS Plus: N/A**
- **System Airtech: N/A**

4. **Capping sheets:** Where appropriate, the installing contractor can use **Bauder KSO-P SN / KSO SN / KSO ALP SN** self-adhesive capping sheet, applied using the hot air hand tools approved for use with bituminous systems. Please note that **Bauder Activator-Primer (Canister), APR01-Black**, must be applied to the underlayer prior to installation of the self-adhesive capping sheet. **BauderTEC KSO-P SN** is only available in one colour – Charcoal Grey
BauderTEC KSO SN is only available in one colour – Grey Slate
BauderTEC KSO ALP SN is only available in one colour – Brown

Self-adhered membranes - For upstands using only self-adhesive membranes, these will need to be mechanically fixed at the top leading edge of the membrane with Bauder aluminium peel bar regardless of the upstand height. This will be for System Airtech and for areas referred to as 'Torch-Free' zones. Please refer to Bauder Bituminous Standard Detail Drawings.

5. **Green roof notes:** Please note it is strictly only permissible to use self-adhesive capping sheet for flashings and detailing work when installing **BauderGREEN XF301 Sedum Blanket** or Hard landscaping finishes.
6. **Approved Hot Air Equipment:** The **BauderTEC KSO-P SN / KSO SN / KSO ALP SN** membrane must be applied using the approved hot air hand tools. The list of permissible hot air electrical equipment suppliers for installing Bauder waterproofing membranes are stated below. These are available either for purchase or hire from the below companies:

HOT AIR WELDING EQUIPMENT

LEISTER

Contact: Welwyn Tool Group, Tel 01707 331 111, <http://www.welwyntoolgroup.co.uk>

SIEVERT

Contact: Lister Gas Pro, Tel 0800 801 046, ch300@lister.co.uk

775 Skirtings and upstands

1. **Angle fillets:** Fix by bitumen bonding or nailing.

2. **Venting first layer of bitumen membrane:** Stop at angle fillet. Fully bond in bitumen for 300 mm strip around perimeters. Overlap onto upstand with strips of BS 8747, Class S1P1 bitumen membrane, fully bonded.
3. **Other layers of bitumen membrane:** Carry in staggered formation up upstand, with each layer fully bonded. Where practicable, carry top layer over top of upstand.
4. **Upstands**
 - 4.1. **At ends of rolls:** Form with bitumen membrane carried up without using separate strip.
 - 4.2. **Elsewhere:** Form with matching strips of bitumen membrane, maintaining laps.
 - 4.3. **Additional fixing of bitumen membranes:**

775 Skirtings and upstands

1. **Angle fillets:** **BauderPIR T KL 50 angle fillets (50 mm x 50 mm)** must be used at all right-angled upstands.
 - Warm Roof Construction: Provisionally bonded in suitable **Bauder Polyurethane Insulation Adhesive** or **Bauder Activator-Primer (Canister), APR01-Black**, and subsequently retained once the underlay detailing is applied.
 - Un-insulated/Inverted Roof Construction: Coat both the surface area to receive the angle fillet and the angle fillet with **Bauder Activator-Primer (Canister), APR01-Black**. Once both surfaces are tacky, place the angle fillet in the correct location and apply pressure to the angle fillet to ensure adhesion is achieved.
2. **Important note** - under no circumstances must fillets of an alternative material be incorporated (i.e. timber, cork, fibre, etc.) as this would invalidate the guarantee.
3. **Layers of bitumen membrane:** Carry in staggered formation up the upstand, with each layer fully bonded.
4. **Upstands: Elsewhere:** Form with matching strips of bitumen membrane, maintaining laps.
 - 4.1. **At ends of rolls:** Underlay layer only, form with bitumen membrane carried up without using separate strip
 - 4.2. **Elsewhere:** Form with matching strips of bitumen membrane, maintaining laps.
5. **Upstand details (150mm minimum height):** This must be taken from the finished roof surface. Please note that for landscaped roofs, this minimum height is measured from the finished landscape surface as opposed to the waterproofing surface. *Special attention should be paid to all structures, such as rooflights, counter-flashings, window and door cills, pipes etc. Bauder cannot take responsibility for water ingress over waterproofing details constructed below the recommended minimum height.
6. **Level thresholds:** Acceptable, providing conforms to BS6229:2018 and current NHBC Standards, chapter, 7.1.
7. **Flashings:** Separate flashings must always be formed. Capping sheet taken up the upstand in one piece will not be permitted.
8. **Green roof notes:** If the client should desire not to see a green mineral finish then it is permissible to install a piece of suitable colour Bauder bituminous capping sheet membrane.

The Bauder root resistant capping sheet must be taken a minimum 150mm from the finished landscaping surface*. The suitable coloured Bauder bituminous capping sheet must be lapped onto the Bauder root resistant capping sheet by a minimum 150mm and lapped onto the structure by a minimum of 100mm.

9. **Fixing of bitumen membranes to vertical upstands:** Where upstands terminate in excess of 250mm above the finished surface of the main field area capping sheet, screw fix through into the underlying substrate as per the requirements set out below and in accordance with the Bauder Reinforced Bitumen Membrane Installation Guide.

UNDERLAYER - INSULATED UPSTANDS ONLY:

Prior to the installation of the self-adhesive underlayer, ALL insulation board types used to create an insulated upstand require a coating of **Bauder Activator-Primer** applied to the surface of the

insulation board. The Activator-Primer should be applied at a rate of 96g/m². Insulated upstands exceeding 600mm in height will require additional fixings through the insulation board depending on the board type used, at a rate of:

- **BauderPIR FA-TE**, 5No. fixings, one in each corner and one in the centre
- **BauderROCK**, 1No. fixing, in the centre of the board.
- **BauderGLAS** 1No. fixing, in the centre of the board.

CAPPING SHEETS - ALL UPSTANDS:

Terminating on the vertical: Where the capping sheet is required to terminate on a vertical face, a **Bauder Aluminium peel bar** is to be used, mechanically fixed at 150mm centres (as per pre-drilled holes) with appropriate fixings suitable to the substrate. The termination should then be covered with the specified counter flashing material dressed over the top to ensure waterproofing integrity.

Should a traditional lead counter flashing be used, then a separating tape will need to be used between the aluminium bar and lead flashing.

Please note: A Bauder termination bar is not to be used in this scenario unless it has been specified by Bauder. Termination bars are not suitable for brickwork substrates. No additional membrane cover flashing is required.

Terminating on the horizontal: Where the capping sheet is taken onto the horizontal, e.g. on top of the parapet, the membrane must be fixed (using fixings appropriate and suitable to the substrate) with either:

- A **Bauder Aluminium peel bar**, mechanically fixed at 150mm centres (as per pre-drilled holes).
- 5No. screws and washers per capping sheet width, one fixing set in 50mm from the edge of either side of the sheet, with the three remaining fasteners equally spaced in between.
- A **Bauder GRP Trim**, mechanically fixed at 300mm centres (maximum) using screw fasteners of type appropriate to kerb or deck substrate. Nail fixing is not permitted.

Capping sheet cover flashing: - A separate flashing of Bauder capping sheet will be required to cover the fixings (please see Bauder detail drawings).

Please note: *Upstands using only self-adhesive membranes (E.g. System Airtech and areas referred to as 'Torch-Free' zones) will need to be mechanically fixed at the top leading edge of the membrane with a Bauder aluminium peel bar regardless of the upstand height, unless it is retained by a GRP Trim on the horizontal.*

Upstands that exceed approximately 1100mm: -

Bauder Ltd do not advocate the use of bitumen waterproofing on upstands exceeding approximately 1100mm due to consideration of the fire regulations. This should be carried out in an appropriate alternative as specified by the client. Please consult with Building Control for confirmation of upstand waterproofing requirements where the height exceeds 1100mm.

MECHANICAL FASTENER SUPPLIER

SFS Group Fastening Technology Ltd. 153 Kirkstall Road, Leeds, West Yorkshire. LS4 2AT

Tel: +44 (0)113 208 5500, Fax: +44 (0)113 208 5539, Email: uk.leeds@sfsintec.biz

Web: www.sfsintec.biz/uk

Mechanical Fasteners:

Type: IWF-5.2x35 screws together with associated IFC/IW-82x40 galvanised pressed steel washer plates.

777 Secondary weathering (pipe, ducting etc.)

1. Provision must be made to supply and install a secondary weathering flashing above all waterproof upstand detailing to pipe penetrations, balustrade posts, cable entry pipes, ventilation ducting, sun

pipes etc. This can take the form of a welded collar (where appropriate) or a bespoke galvanized cowl or hood sealed with a suitable sealant and fasteners. Solvent welded plastic collars fitted to plastic soil vent pipes.

784A Roof drainage outlets

1. **Product name:** **Bauder Bitumen Vertical Outlet**
2. **Material:** Cast polyurethane body with integral bituminous connection flange.
3. **Product size / reference:**
 - **Bauder Compact Vertical Outlet DN 100**, 6.1 litres/sec. flow rate, with vertical spigot designed to connect to standard 110mm pipework.
 - **Bauder Vertical Outlet DN 70**, 7.1 litres/sec. flow rate, with vertical spigot designed to connect to standard 75mm pipework.
 - **Bauder Vertical Outlet DN 150**, 7.2 litres/sec. flow rate, with vertical spigot designed to connect to standard 160mm pipework.
4. **Flow rate:** (Based upon vertical pipework and a 35 mm head of water pressure – according to BS EN 12056:3:2000).
5. **Pipe connection:** **Bauder High Capacity Vertical Outlet** are suitable for connection to:
 - uPVC “O” ring socketed soil grade pipe to BS 4514: 1983
 - Socketed and socket-less cast-iron pipework to BS 416:1973 and EN 887. Socketed pipework will require cold caulking or PVC to cast iron adaptors. Socket-less pipework can be connection using an appropriate SML mechanical coupling.
 - HDPE pipework with appropriate SML mechanical coupling
6. **Type of grate / fittings:** Supplied with a tough polyamide Dome Grate leaf guard.
Alternative: Bauder Locking Leaf Guard Long Leg: Additional component where higher durability or security is required.
7. **Bauder Extension/Compact Extension Unit :** When the outlet is used as part of a warm roof build-up and the insulation thickness exceeds 60 mm, an additional extension component must be used. The **Bauder Extension Unit** is suitable for use with Insulation:
 - 60mm - 220 mm.If the insulation is below 60mm a timber frame must be built to accommodate the outlet.
8. **Extension unit housing:** Polyurethane housing board 500 x 500 x 60mm with pre-molded opening to receive the **Bauder Extension Unit**. Thermal Conductivity 0.025 W/Mk. Compressive Strength 200 kPa and a Density of 50 kg/m³.
 - **Material:** Cast polyurethane body with integral bituminous connection flange.
 - **Fixing:** The extension unit must be mechanically fixed through the PUR rim to the structural deck.
 - **Type of grate/ fittings:** When required the Dome Grate leaf guard is transferred from the **Bauder Compact Vertical Outlet**.
9. **Bauder vertical outlet - Installation requirements::** These components that form part of the Bauder waterproofing system and for guarantee reasons, should only be installed by Bauder Approved installers. Connectivity to below deck drainage pipework to be the responsibility of the plumbing contractor.
10. **Fixing:** The outlet is to be secured through the rim to the structural deck by a minimum of four fasteners appropriate to obtain an adequate attachment to the deck substrate material. Some deck structures require preparatory works before the outlets can be installed: -
 - **Concrete decks** – the opening for the outlet to be either pre-cast or core-drilled so that the outlet can be installed at the same time as the air and vapour control layer. Provision for a 250 mm dia. opening is required.
 - **Profiled metal decks** – these also require a 250 mm dia. Opening cut into the decking, but in addition will require a 600 x 600 x 1.25 mm galvanised steel reinforcing plate secured to the deck before the outlet can be installed. This item has a pre-cut 250 mm dia. hole and is available from Bauder.

- For detailed information, refer to the Bauder Product Data Sheet and Bauder Detail Drawing.

784B Roof drainage outlets

1. **Product name:** **Bauder Bitumen Letterbox Parapet Outlet DN 205 including stainless-steel leaf guard**
2. **Material :** 600mm/1200mm Stainless steel rectangular spigot with integral bituminous connection flange.
3. **Product size / reference:** 205 mm width and 65mm height nominal opening, 210mm width and 70mm height horizontal spigot designed to connect to external hopper head.
4. **Angle of spigot:** 3°.
5. **Flow rate:** Based upon vertical pipework and a 35 mm head of water pressure – according to BS EN 12056:3:2000.
 - 2.02 litres/sec. (outlet discharging to external hopper and downpipe).
6. **Suitability:** These outlets are designed to be used as a through chute to external parapet walls kerbs in either warm, un-insulated or inverted roof applications, feeding to an external hopper-head and downpipe leading to a ground level open gully or soak-away. They are not suitable or intended for connection to internal pipework that is within a wall construction or boxed in and is inaccessible after construction.
7. **Detailing with vertical insulation:** **Vertical insulation and warm roofs:** a timber frame, 10 mm less in thickness than the insulation being installed, must be used to form an isolation frame around the outlet that enables the air and vapour control layer and underlayer to form a 100 mm overlap e.g. 20 mm thick timber used with 30 mm thick insulation. The timber frame should be mechanically secured to the deck/ upstands and in turn will provide a means of securing the outlet body to the roof.
8. **Type of grate / fittings:** supplied with a 345mm wide stainless steel leaf guard.
9. **Installation requirements:** These outlets are components that form part of the Bauder waterproofing system and for guarantee reasons, should only be installed by Bauder Approved installers.
10. **Fixing:** The deck or wall structure may require preparatory works before the outlets can be installed: -
Concrete decks – the opening for the outlet to be either pre-cast or core-drilled. The outlet body is designed to have a slight fall, so the penetration should be cut/ formed to reflect this. The outlet is to be secured to the structural deck/ upstand by a minimum of four fasteners through the outlet base plate to obtain an adequate attachment to the deck substrate material.
11. For detailed information, refer to the Bauder Technical Data Sheet & Bauder Detail Drawing.

785 Fixing perimeter trims

1. **Bituminous membranes:**
 - The first layer of membrane must be self-adhesive and dressed to the full extent of the detail using Torch-Free methods. This is to ensure that the detail is fully encapsulated to reduce the risk of fire to exposed combustible materials.
 - Dress the underlayer up and over the perimeter detail to provide a 25mm overhang.

Please refer to Bauder standard detail drawings.

2. **Trim**
 - 2.1. **Setting out:** 10mm gap between the back edge of the bottom of the drip to the fascia/wall and 3mm gap between abutting lengths of trim.
 - 2.2. **Fasteners:** Screw fasteners of type appropriate to kerb or deck substrate. Nail fixing is not permitted.

- 2.3. **Fixing:** 30mm from ends and at 300mm (maximum) centres, stagger fixed. Can be used to retain the capping sheet where the capping sheet is taken to the full extent of the detail – please see Bauder detail drawing.
- 150mm deep trims (type 6) – 3no. additional fixings per length of trim. The fixings are to be face fixed with screws and positioned 75mm down from the top edge, one fixing 100mm in from each end and one in the centre and capped with coloured matched plastic weathering caps. *A fixed timber packer will be required behind the face of the trim to help facilitate ease of fixing.*
 - *For roofs above 10 metres in height – the 100mm deep trim (type 4) will require face fixing, as per 150mm trim above. A fixed timber packer will be required behind the face of the trim to help facilitate ease of fixing.*
- 2.4. **Jointing sleeves / bridging pieces:** All lengths should be close butt jointed using an internal jointing sleeve. This must be provided to each joint.
- 2.5. **Corner pieces:** Purpose made.
3. **Completion**
- 3.1. **Contact surfaces:** Prime with **Bauder Primer**.
- 3.2. **Joints:** Cover with 200mm long pads of bitumen membrane, bonded to trim.
4. **Completion of bitumen membrane**
- 4.1. **Top layer / capping sheet:** Butt joint to rear edge of trim.
- 4.2. **Cover strip:** Fully bond to trim and top layer/ capping sheet of bitumen membrane. Carry over roof edge upstand and lap 100 mm onto roof. The capping sheet is to be dressed tightly into the top lip of the trim, ensuring a bead of bitumen extrudes at the edge. Please see Bauder detail drawing.
5. **Wall / kerb joints:** The new trim must cover any open joint that may exist at the top of the kerb or wall, by a minimum distance of 20mm.

785A Fixing perimeter trims

1. **Bauder GRP Water Check Trim:** to be used on parapets and perimeter kerbs over 200mm in width – must not be used on its own as a check kerb.
2. **Bituminous membranes:**
- The first layer of membrane must be self-adhesive and dressed to the full extent of the detail using Torch-Free methods. This is to ensure that the detail is fully encapsulated to reduce the risk of fire to exposed combustible materials.
 - Dress the underlayer up and over the perimeter detail to provide a 25mm overhang.

Please refer to Bauder standard detail drawings.

3. **Trim**
- 3.1. **Setting out:** The bottom inside lip of the drip to sit flush with the fascia/wall, leaving a 3mm gap between abutting lengths of trim.
- 3.2. **Fasteners:** Screw fasteners of type appropriate to kerb or deck substrate. Nail fixing is not permitted.
- 3.3. **Fixing:** 30mm from ends and at 300mm (maximum) centres, stagger fixed. Can be used to retain the capping sheet where the capping sheet is taken to the full extent of the detail – please see Bauder detail drawing.
- 3.4. **Jointing sleeves / bridging pieces:** All lengths should be close butt jointed using an internal jointing sleeve. This must be provided to each joint.
- 3.5. **Corner pieces:** Purpose made.
4. **Completion**
- 4.1. **Contact surfaces:** Prime with **Bauder Primer**.

- 4.2. **Joints:** Cover with 200mm long pads of bitumen membrane, bonded to trim.
5. **Completion of bitumen membrane**
 - 5.1. **Top layer / capping sheet:** Butt joint to rear edge of trim.
 - 5.2. **Cover strip:** Fully bond to trim and top layer/ capping sheet of bitumen membrane. Carry over roof edge upstand and lap 100 mm onto roof. The capping sheet is to be dressed tightly into the top lip of the trim, ensuring a bead of bitumen extrudes at the edge.
6. **Wall / kerb joints:** The new trim must cover any open joint that may exist at the top of the kerb or wall, by a minimum distance of 20mm.

786 Lightning conductor clips

1. The lightning conductor is to be fixed using the specified conductor clips, incorporating a colour matched Bauder capping sheet pad, fully bonded to the main capping sheet at 1m centres by the approved Bauder roofing contractor. Lightning conductor retention using bituminous membrane strips are not acceptable.
2. Where lightning conductors are required to be fixed to brickwork, upstands, parapet walls etc., proprietary clips mechanically fixed should be utilised. Care should be taken to ensure that fixings do not penetrate the new waterproofing system.

Surfacing

863 Designated maintenance walkways

1. **Location:** Refer to indicative locations on roof plan to suit mansafe system by specialist
2. **Application:** Remove the selvedge lap prior to installing. Membrane to be fully bonded to the previous capping sheet layer, making provision for leaving intermittent 100 mm gaps between sheets to form drainage channels, particularly where the walkway layer impedes water run-off.

Completion

910 Inspection

1. **Interim and final roof inspections:** in accordance with the manufacturer's requirements for guarantee.
2. **Notification: Final Inspection:** It is the responsibility of the approved contractor to advise Bauder Ltd when the roof is ready for Final Inspection. The 'Final Inspection' of the waterproofing must be carried out and approved by Bauder Ltd prior to any landscaping products/materials being installed, otherwise a guarantee cannot be issued. Safe access to carry out this inspection must be provided.
3. **Other requirements:** Please also refer to preliminaries / general conditions.
4. **If project needs to follow NHBC requirements:** The waterproofing must be visually inspected and electronically tested for waterproofing integrity, faults rectified, and retested prior to the installation of any landscaping products. The results of the test(s) should be made available to the NHBC.

940 Completion

1. **Roof areas:** Clean.
2. **Outlets:** Clear.
3. **Work necessary to provide a weathertight finish:** Complete.
4. **Storage of materials on finished surface:** Not permitted.
5. **Completed membrane:** Do not damage. Protect from chemicals, traffic and adjacent or high-level working.

950A Guarantee

- 1.

- A 20 year system product, workmanship and design (including consequential loss) guarantee is to be provided upon completion following a Final Inspection by Bauder. Details regarding the full terms and conditions are available separately from Bauder Ltd upon request. This system must be installed by a Bauder Approved Contractor, to be eligible for guarantee. The system comprises the waterproofing membranes, insulation, air and vapour control layer, and attachment of these products.

Ω End of Section

K10

Gypsum board dry linings/ partitions/ ceilings

Types of dry lining

001 General Requirements

1. Refer to Anderson Acoustics Stage 4 acoustic report for detail requirements for partitions in relation to acoustics

125A Gypsum board partition systems

1. Description: Partition type P1
2. Manufacturer: [British Gypsum](#)
 - 2.1. Contact details
 - 2.1.1. Address: East Leake
Loughborough
Leicestershire
United Kingdom
LE12 6JU
 - 2.1.2. Telephone: +44 (0)844 800 1991
 - 2.1.3. Web: <https://www.british-gypsum.com>
 - 2.1.4. Email: bg.technical@saint-gobain.com
 - 2.2. Product reference: [GypWall Single Frame Enhanced - X606009 \(EN\)](#)
3. Framing: Base channel: Gypframe 72 DC 60 Deep Flange Floor & Ceiling Channel. Head channel: Gypframe 72 DC 60 Deep Flange Floor & Ceiling Channel. Stud: Gypframe 70 S 60 'C' Stud. Abutments and openings: Gypframe 70 S 60 'C' Stud.
4. Linings: Board side 1, Layer 1: Rigidur H 12.5mm. Board side 2, Layer 1: Rigidur H 12.5mm.
5. Fasteners: Screws side 1, Layer 1: Rigidur Screws 30mm. Screws side 2, Layer 1: Rigidur Screws 30mm.
6. Joint treatment: Fixing strap: Gypframe GFS1 Fixing Strap. Fixing T: Gypframe GFT1 Fixing 'T'.
7. Finish: To achieve the specified performances, the system should be finished using either one of our Thistle or ThistlePro plasters, or Gyproc jointing products. See the product range guides on the British Gypsum website for more information.
8. System accessories: Sealant: Gyproc Sealant.
9. Execution: Please review details on the British Gypsum website at: <https://www.british-gypsum.com/Specification/White-Book-Specification-Selector/internal-partitions-walls/GypWall-Single-Frame-Enhanced/X606009-EN#details>
10. Fire resistance: BS EN 1364-1, 30 / 30 Integrity / Insulation
11. Sound insulation: BS EN ISO 10140-2, 43 Rw dB
12. Strength grade: BS 5234-2, Severe duty (SD)
13. Partition height: 3800mm
14. Partition thickness: 97mm

125B Gypsum board partition systems

1. Description: Partition type P2
2. Manufacturer: [British Gypsum](#)
 - 2.1. Contact details
 - 2.1.1. Address: East Leake
Loughborough

Leicestershire
United Kingdom
LE12 6JU

- 2.1.2. Telephone: [+44 \(0\)844 800 1991](tel:+4418448001991)
- 2.1.3. Web: <https://www.british-gypsum.com>
- 2.1.4. Email: bg.technical@saint-gobain.com
- 2.2. Product reference: [GypWall Single Frame Enhanced - X606A005 \(EN\)](#)
- 3. **Framing:** Base channel: Gypframe 72 DC 60 Deep Flange Floor & Ceiling Channel. Head channel: Gypframe 72 DC 60 Deep Flange Floor & Ceiling Channel. Stud: Gypframe 70 AS 50 AcouStud. Abutments and openings: Gypframe 70 S 60 'C' Stud.
- 4. **Linings:** Board side 1, Layer 1: Gyproc WallBoard 15mm. Board side 1, Layer 2: Rigidur H 12.5mm. Board side 2, Layer 1: Gyproc WallBoard 15mm. Board side 2, Layer 2: Rigidur H 12.5mm.
- 5. **Fasteners:** Screws side 1, Layer 1: British Gypsum Drywall Screws 25mm. Screws side 1, Layer 2: Rigidur Screws 40mm. Screws side 2, Layer 1: British Gypsum Drywall Screws 25mm. Screws side 2, Layer 2: Rigidur Screws 40mm.
- 6. **Joint treatment:** Fixing strap: Gypframe GFS1 Fixing Strap.
- 7. **Finish:** To achieve the specified performances, the system should be finished using either one of our Thistle or ThistlePro plasters, or Gyproc jointing products. See the product range guides on the British Gypsum website for more information.
- 8. **System accessories:** Sealant: Gyproc Sealant.
- 9. **Execution:** Please review details on the British Gypsum website at: <https://www.british-gypsum.com/Specification/White-Book-Specification-Selector/internal-partitions-walls/GypWall-Single-Frame-Enhanced/X606A005-EN#details>
- 10. **Fire resistance:** BS EN 1364-1, 60 / 60 Integrity / Insulation
- 11. **Sound insulation:** BS EN ISO 140-3, 54 Rw dB
- 12. **Strength grade:** BS 5234-2, Severe duty (SD)
- 13. **Partition height:** 4000mm
- 14. **Partition thickness:** 127mm

125C Gypsum board partition systems

- 1. **Description:** Partition type P3
- 2. **Manufacturer:** [British Gypsum](#)
 - 2.1. **Contact details**
 - 2.1.1. **Address:** East Leake
Loughborough
Leicestershire
United Kingdom
LE12 6JU
 - 2.1.2. **Telephone:** [+44 \(0\)844 800 1991](tel:+4418448001991)
 - 2.1.3. **Web:** <https://www.british-gypsum.com>
 - 2.1.4. **Email:** bg.technical@saint-gobain.com
 - 2.2. **Product reference:** [GypWall Single Frame Enhanced - X606A007 \(EN\)](#)
- 3. **Framing:** Base channel: Gypframe 72 DC 60 Deep Flange Floor & Ceiling Channel. Head channel: Gypframe 72 DC 60 Deep Flange Floor & Ceiling Channel. Stud: Gypframe 70 AS 50 AcouStud. Abutments and openings: Gypframe 70 S 60 'C' Stud.
- 4. **Insulation:** Insulation, Layer 1: 25mm Isover Acoustic Partition Roll (APR 1200).
- 5. **Linings:** Board side 1, Layer 1: Gyproc SoundBloc 15mm. Board side 1, Layer 2: Rigidur H 12.5mm. Board side 2, Layer 1: Gyproc SoundBloc 15mm. Board side 2, Layer 2: Rigidur H 12.5mm.

6. **Fasteners:** Screws side 1, Layer 1: British Gypsum Drywall Screws 25mm. Screws side 1, Layer 2: Rigidur Screws 40mm. Screws side 2, Layer 1: British Gypsum Drywall Screws 25mm. Screws side 2, Layer 2: Rigidur Screws 40mm.
7. **Joint treatment:** Fixing strap: Gypframe GFS1 Fixing Strap.
8. **Finish:** To achieve the specified performances, the system should be finished using either one of our Thistle or ThistlePro plasters, or Gyproc jointing products. See the product range guides on the British Gypsum website for more information.
9. **System accessories:** Sealant: Gyproc Sealant.
10. **Execution:** Please review details on the British Gypsum website at: <https://www.british-gypsum.com/Specification/White-Book-Specification-Selector/internal-partitions-walls/GypWall-Single-Frame-Enhanced/X606A007-EN#details>
11. **Fire resistance:** BS EN 1364-1, 60 / 60 Integrity / Insulation
12. **Sound insulation:** BS EN ISO 140-3, 58 Rw dB
13. **Strength grade:** BS 5234-2, Severe duty (SD)
14. **Partition height:** 4000mm
15. **Partition thickness:** 127mm

125D Gypsum board partition systems

1. **Description:** Partition type P4
2. **Manufacturer:** [British Gypsum](#)
 - 2.1. **Contact details**
 - 2.1.1. **Address:** East Leake
Loughborough
Leicestershire
United Kingdom
LE12 6JU
 - 2.1.2. **Telephone:** +44 (0)844 800 1991
 - 2.1.3. **Web:** <https://www.british-gypsum.com>
 - 2.1.4. **Email:** bg.technical@saint-gobain.com
 - 2.2. **Product reference:** [GypWall Twin Frame Independent - L216017F \(EN\)](#)
3. **Framing:** Base channel: Gypframe 62 FEC 50 Folded Edge Standard Floor & Ceiling Channel. Head channel: Gypframe 62 FEC 50 Folded Edge Standard Floor & Ceiling Channel. Stud: Gypframe 60 I 70 'I' Stud. Abutments and openings: Gypframe 60 S 50 'C' Stud.
4. **Insulation:** Insulation, Layer 1: 50mm Isover Acoustic Partition Roll (APR 1200).
5. **Linings:** Board side 1, Layer 1: Gyproc SoundBloc 15mm. Board side 1, Layer 2: Rigidur H 12.5mm. Board side 2, Layer 1: Gyproc SoundBloc 15mm. Board side 2, Layer 2: Rigidur H 12.5mm.
6. **Fasteners:** Screws side 1, Layer 1: British Gypsum Drywall Screws 25mm. Screws side 1, Layer 2: Rigidur Screws 40mm. Screws side 2, Layer 1: British Gypsum Drywall Screws 25mm. Screws side 2, Layer 2: Rigidur Screws 40mm.
7. **Joint treatment:** Fixing strap: Gypframe GFS1 Fixing Strap.
8. **Finish:** To achieve the specified performances, the system should be finished using Gyproc jointing products. See the product range guides on the British Gypsum website for more information.
9. **System accessories:** Sealant: Gyproc Sealant.
10. **Execution:** Please review details on the British Gypsum website at: <https://www.british-gypsum.com/Specification/White-Book-Specification-Selector/internal-partitions-walls/GypWall-Twin-Frame-Independent/L216017F-EN#details>
11. **Fire resistance:** BS EN 1364-1, 90 / 90 Integrity / Insulation
12. **Sound insulation:** BS EN ISO 10140-2, 66 Rw dB / 57 Rw + Ctr dB

13. Strength grade: BS 5234-2, Severe duty (SD)
14. Partition height: 3600mm
15. Partition thickness: 225mm

165 Gypsum board wall lining systems

1. Manufacturer: [British Gypsum](#)
 - 1.1. Contact details
 - 1.1.1. Address: East Leake
Loughborough
Leicestershire
United Kingdom
LE12 6JU
 - 1.1.2. Telephone: [+44 \(0\)844 800 1991](tel:+4418448001991)
 - 1.1.3. Web: <https://www.british-gypsum.com>
 - 1.1.4. Email: bg.technical@saint-gobain.com
 - 1.2. Product reference: GypLyner Single
2. Support: Bracket type: Gypframe GL2 Bracket. Fixing method: Gypframe GL11 GypLyner Anchor. Channel connector: Gypframe GL3 Channel Connector. Base channel: Gypframe GL8 Track. Head channel: Gypframe GL8 Track. Stud: Gypframe GL1 Lining Channel.
3. Insulation: N/A
4. Linings: Board side 1, Layer 1: Rigidur H 12.5mm + 1 layer 12mm plywood
5. Fasteners: Screws side 1, Layer 1: British Gypsum Drywall Screws 25mm. Bracket fixing: British Gypsum Wafer Head Drywall Screws 13mm.
6. Joint treatment: Fixing strap: Gypframe GFS1 Fixing Strap. Fixing T: Gypframe GFT1 Fixing 'T'.
7. Finish: To achieve the specified performances, the system should be finished using either one of our Thistle or ThistlePro plasters, or Gyproc jointing products. See the product range guides on the British Gypsum website for more information.
8. System accessories: Sealant: Gyproc Sealant. Thermal sealant: Gyproc Sealant.
9. Execution: Please review details on the British Gypsum website at: <https://www.british-gypsum.com/Specification/White-Book-Specification-Selector/wall-linings/GypLyner-Single/B226008-A-EN#details>
10. Strength grade: Not Stated

215 Suspended ceiling system

1. Description: Plasterboard ceilings
2. Standard: To BS EN 13964.
3. Evidence of compliance: Submit Declaration of Performance (DoP).
4. Lining board: 12.5 mm plasterboard
 - 4.1. Finishing: tape and joint
 - 4.1.1. Primer/ Sealer: Primer to painted areas
 - 4.2. Accessories:
5. Suspension system: As recommended by lining manufacturer to complete the ceiling system and achieve specified performance.
 - 5.1. Grid type: Contractor's choice
6. Insulation: 50mm Isover General Purpose Roll
7. Access units: Required
8. Integrated services fittings: Hangers and housings for linear luminaires
9. Other requirements: Fire-stopping around service penetrations as section P12

10. System performance

10.1. Standard: To BS EN 13964.

10.2. Structural: The ceiling system must safely support all anticipated loads including services fittings.

10.2.1. Additional loads/ pressures: Luminaires as drawings and schedules

10.2.2. Deflection of grid between points of support (maximum): $0.0025 \times \text{span}$.

10.3. Fire performance

10.3.1. Reaction to fire: Not required

10.4. Fire resistance: Not required

216 Perforated Plasterboard ceiling

1. Description: Plasterboard ceilings

2. Standard: To BS EN 13964.

3. Evidence of compliance: Submit Declaration of Performance (DoP).

4. Lining board: 12.5mm Rigitone 8-15-20 plasterboard.

4.1. Finishing: tape and joint

4.1.1. Primer/ Sealer: Primer to painted areas

4.2. Accessories: Metal beads/ stops recommended by lining board manufacturer

5. Suspension system: As recommended by lining manufacturer to complete the ceiling system and achieve specified performance.

5.1. Grid type: Contractor's choice

6. Insulation: 50mm Isover General Purpose Roll + black fleece lining

7. Access units: Required

8. Integrated services fittings: Hangers and housings for linear luminaires

9. Other requirements: Fire-stopping around service penetrations as section P12

10. System performance

10.1. Standard: To BS EN 13964.

10.2. Structural: The ceiling system must safely support all anticipated loads including services fittings.

10.2.1. Additional loads/ pressures: Luminaires as drawings and schedules

10.2.2. Deflection of grid between points of support (maximum): $0.0025 \times \text{span}$.

10.3. Fire performance

10.3.1. Reaction to fire: Not required

10.4. Fire resistance: Not required

General/ preparation

305 Compliance with performance requirements

1. Testing/ Assessment: Submit UKAS accredited laboratory reports for the following:

2. Materials, components and details: As used in testing/ assessment reports. If discrepancies arise, give notice.

335 Additional supports

1. Framing: Accurately position and securely fix to give full support to:

1.1. Partition heads running parallel with, but offset from main structural supports.

1.2. Fixtures, fittings and service outlets. Mark framing positions clearly and accurately on linings.

- 1.3. Board edges and lining perimeters, as recommended by board manufacturer to suit type and performance of lining.

375 New wet laid bases

1. Dpcs: Install under full width of partitions/ freestanding wall linings.
 - 1.1. Material: Bituminous sheet or plastics.

Components

400 Gypsum boards generally

1. Standard
 - 1.1. Gypsum plasterboard to BS EN 520.
 - 1.2. Gypsum fibre board to BS EN 15283-2.
 - 1.3. Evidence of compliance: Submit Declaration of Performance (DoP).

430 Access panels

1. Description: metal faced access panels
2. Type: Gyproc Profilex Sealed Panel.
 - 2.1. Sizes: 300x300mm or 550x550mm or 550 x 1200
 - 2.2. Fire performance
 - 2.2.1. Reaction to fire: As suspended ceiling/ partition fire rating
 - 2.2.2. Fire resistance: As suspended ceiling/ partition fire rating
3. Frame: Beaded frame
4. Panel: metal PPC
5. Lock: Tamper Proof lock.

Installation

435 Dry linings generally

1. General: Use fixing, jointing, sealing and finishing materials, components and installation methods recommended by board manufacturer.
2. Cutting gypsum boards: Neatly and accurately without damaging core or tearing paper facing.
 - 2.1. Cut edges: Minimize and position at internal angles wherever possible. Mask with bound edges of adjacent boards at external corners.
3. Fixings boards: Securely and firmly to suitably prepared and accurately levelled backgrounds.
4. Finishing: Neatly to give flush, smooth, flat surfaces free from bowing and abrupt changes of level.

445 Ceilings

1. Sequence: Fix boards to ceilings before installing dry lined walls and partitions.
2. Orientation of boards: Fix with bound edges at right angles to supports and with ends staggered in adjacent rows.
3. Two layer boarding: Stagger joints between layers.

455 Metal framing for partitions/ wall linings

1. Setting out: Accurately aligned and plumb.
 - 1.1. Frame/ Stud positions: Equal centres to suit specified linings, maintaining sequence across openings.
 - 1.2. Additional studs: To support vertical edges of boards.

2. Fixing centres at perimeters (maximum): 600 mm.
3. Openings: Form accurately.
 - 3.1. Doorsets: Use sleeved or boxed metal studs and/ or suitable timber framing to achieve strength grade requirements for framing assembly and adequately support weight of door.
 - 3.2. Services penetrations: Allow for associated fire-stopping.

465 Staggered stud partitions

1. Horizontal frame members (noggins, bearers, etc.) and boards: Fix between alternate studs and not touching adjacent offset studs.

475 Metal furrings for wall linings

1. Setting out: Accurately aligned and plumb.
 - 1.1. Vertical furring positions: Equal vertical centres to suit specified linings, maintaining sequence across openings. Position adjacent to angles and openings.
 - 1.2. Additional vertical furrings: To support vertical edges of boards and at junctions with partitions.
 - 1.3. Horizontal furring positions: To provide continuous support to edges of boards.
2. Adhesive bedding to furrings
 - 2.1. Dabs: Length 200 mm (minimum). Located at ends of furrings and thereafter at 450 mm (maximum) centres.
 - 2.2. Junctions with partitions: Continuous bed with no gaps across cavity.

485 Suspended ceiling grids

1. Setting out: Accurately aligned and level.
 - 1.1. Grid members and hangers: Centres to suit specified linings and imposed loads.
 - 1.2. Additional grid members: Provide bracing and stiffening at upstands, partition heads, access hatches, etc.
2. Fixing: Securely at perimeters, grid joints, top and bottom hanger fixings.

505 Installing mineral wool insulation

1. Fitting insulation: Closely butted joints and no gaps. Use fasteners to prevent slumping or displacement.
2. Services
 - 2.1. Electrical cables overlaid by insulation: Sized accordingly.
 - 2.2. Ceilings: Cut insulation around electrical fittings, etc.

510 Sealing gaps and air paths

1. Location of sealant: To perimeter abutments and around openings.
 - 1.1. Pressurized shafts and ducts: At board-to-board and board-to-metal frame junctions.
2. Application: To clean, dry and dust free surfaces as a continuous bead with no gaps.
 - 2.1. Gaps greater than 6 mm between floor and underside of gypsum board: After sealing, fill with jointing compound.

530 Cavity fire barriers within partitions/ wall linings

1. Metal framed systems
 - 1.1. Material:
 - 1.2. Installation: Form accurately and fix securely with no gaps to provide a complete barrier to smoke and flame.

2. Adhesive fixed wall lining systems
 - 2.1. Material: Adhesive compound.
 - 2.2. Installation: Form in a continuous line with no gaps to provide a complete barrier to smoke and flame.

545 Cavity fire barriers within suspended ceilings

1. Type:
2. Fire resistance:
3. Ceiling void subdivision: Fix barriers
4. Fixing at perimeters and joints: Secure, stable and continuous with no gaps, to provide a complete barrier to smoke and flame.
5. Service penetrations: Cut and pack to maintain barrier integrity. Sleeve flexible materials. Adequately support services passing through barrier.
6. Ceiling systems for fire protection: Do not impair fire-resisting performance of ceiling system.

555 Fire-stopping at perimeters of dry lining systems

1. Material: Tightly packed mineral wool or intumescent mastic/ sealant.
2. Application: To perimeter abutments to provide a complete barrier to smoke and flame.

560 Joints between boards

1. Tapered edged gypsum boards
 - 1.1. Bound edges: Lightly butted.
 - 1.2. Cut/ unbound edges: 3 mm gap.
2. Square edged plasterboards: 3 mm gap.
3. Square edged gypsum fibre boards: 5 mm gap.

565 Vertical joints

1. Joints: Centre on studs.
 - 1.1. Partitions: Stagger joints on opposite sides of studs.
 - 1.2. Two layer boarding: Stagger joints between layers.

570 Horizontal joints

1. Surfaces exposed to view: Horizontal joints not permitted. Seek instructions where height of partition/ lining exceeds maximum available length of board.
2. Two layer boarding: Stagger joints between layers by at least 600 mm.
3. Edges of boards: Support using additional framing.
 - 3.1. Two layer boarding: Support edges of outer layer.

590 Fixing gypsum board to metal framing/ Furrings

1. Partitions/ Wall linings: Fix securely and firmly at the following centres (maximum):
 - 1.1. Single layer boarding: To all framing at 300 mm centres. Reduce to 200 mm centres at external angles.
 - 1.2. Multi-layer boarding: Face layer at 300 mm centres, and previous layers around perimeters at 300 mm centres.
2. Ceilings: 230 mm. Reduce to 150 mm at board ends and at lining perimeters.
3. Position of screws from edges of boards (minimum): 10 mm.
 - 3.1. Screw heads: Set in a depression. Do not break paper or gypsum core.

592 Fixing insulation backed plasterboard to metal furrings

1. **Fixing to furrings:** In addition to screw fixings apply continuous beads of adhesive sealant to furrings.

595 Deflection heads

1. **Fixing boards:** Do not fix to head channels.

610 Fixing gypsum board to timber

1. **Fixing to timber:** Securely at the following centres (maximum):
 - 1.1. **Nails:** 150 mm.
 - 1.2. **Screws to partitions/ wall linings:** 300 mm. Reduce to 200 mm at external angles.
 - 1.3. **Screws to ceilings:** 230 mm.
2. **Position of nails/ screws from edges of boards (minimum)**
 - 2.1. **Bound edges:** 10 mm.
 - 2.2. **Cut/ unbound edges:** 13 mm.
3. **Position of nails/ screws from edges of timber supports (minimum):** 6 mm.

Finishing

650 Level of dry lining across joints

1. **Sudden irregularities:** Not permitted.
2. **Joint deviations:** Measure from faces of adjacent boards using methods and straightedges (450 mm long with feet/ pads) to BS 8212, clause 3.3.5.
 - 2.1. **Tapered edge joints**
 - 2.1.1. **Permissible deviation (maximum) across joints when measured with feet resting on boards:** 3 mm.
 - 2.2. **External angles**
 - 2.2.1. **Permissible deviation (maximum) for both faces:** 4 mm.
 - 2.3. **Internal angles**
 - 2.3.1. **Permissible deviation (maximum) for both faces:** 5 mm.

670 Seamless jointing to gypsum boards

1. **Cut edges of boards:** Lightly sand to remove paper burrs.
2. **Filling and taping:** Fill joints, gaps and internal angles with jointing compound and cover with continuous lengths of paper tape, fully bedded.
3. **Protection of edges/ corners:** Reinforce external angles, stop ends, etc. with specified edge/ angle bead.
4. **Finishing:** Apply jointing compound. Feather out each application beyond previous application to give a flush, smooth, seamless surface.
5. **Nail/ screw depressions:** Fill with jointing compound to give a flush surface.
6. **Minor imperfections:** Remove by light sanding.

692 Rigid beads/stops

1. **Internal:** To BS EN 13658-1.
2. **External:** To BS EN 13658-2.

695 Installing beads/ Stops

1. **Cutting:** Neatly using mitres at return angles.
2. **Fixing:** Securely using longest possible lengths, plumb, square and true to line and level, ensuring full contact of wings with substrate.
3. **Finishing:** After joint compounds/ plasters have been applied, remove surplus material while still wet from surfaces of beads exposed to view.

725 Repairs to existing gypsum board

1. Performance of repairs must match original specified performances.
2. **Filling small areas with broken cores:** Cut away paper facing, remove loose core material and fill with jointing compound.
 - 2.1. **Finish:** Flush, smooth surface suitable for redecoration.
3. **Large patch repairs:** Cut out damaged area and form neat hole with rectangular sides. Replace with matching gypsum board.
 - 3.1. **Fixing:** Use methods to suit type of dry lining, ensuring full support to all edges of existing and new gypsum board.
 - 3.2. **Finishing:** Fill joints, tape and apply jointing compound to give a flush, smooth surface suitable for redecoration.

Ω End of Section

K13

Rigid sheet fine linings and panelling

Types of lining and panelling

170 Acoustic panel lining

1. Description: Acoustic panels to Classrooms and the Heart Space as shown on 29 series elevations
2. Substrate: Plasterboard Partitions
3. Panels
 - 3.1. Manufacturer: Saint Gobain Ecophon,
Old Brick Kiln,
Ramsdell, Tadley
RG26 5PP. UK
Tel: +44 (0)1256 855208
www.ecophon.com/uk
 - 3.1.1. Product reference: Akusto Wall C Super G
 - 3.2. Thickness: 40mm
 - 3.3. Finish: Colour - As per finishes schedule
4. Installation
 - 4.1. Method of fixing panels: As per Ecophon installation method
5. Included features:
6. Trims: Ecophon Connect Thinline profile - Colour TBC

General requirements

210 Advance registration

1. Materials registered in advance by the employer: Obtain from supplier named in Preliminaries section A56.
 - 1.1. Ordering: Supersede employer's registration and take over responsibility by an order to the supplier covering price, supply and delivery to suit progress of the work.

260 Environmental conditions

1. General requirements prior to starting work specified in this section: Building weathertight; wet trades completed and affected areas dried out.
2. Temperature and humidity before, during and after fixing lining/ panelling: Maintained at levels approximating to those which will prevail after building is occupied.

Fabrication/ fixing/ finishing

350 Fixing linings and panelling

1. Setting out: Accurate, true to line and level, free from undulations and lipping, with lines and joints aligned, straight and parallel unless specified otherwise.
2. Movement allowance: Adequate for future moisture and temperature movement of boards.
3. Fixing of panels: Secure, to prevent pulling away, bowing, or other movement during use.
4. Methods of fixing and fasteners: As section Z20 unless specified otherwise.
5. Trims: Wherever possible, to be in unjointed lengths between angles or ends of runs.
 - 5.1. Running joints: Where unavoidable, submit proposals for location and method of jointing.
 - 5.2. Angle joints: Mitred, unless specified otherwise.

Ω End of Section

K32

Panel cubicles/ duct and wall linings/ screens

To be read with preliminaries/ general conditions.

110 Design

1. Complete the design of the fire stopping system using generic products such as those specified below. To allow for all necessary Fire stopping, as shown on drawings and as required to achieve the Fire Strategy.
2. Make reference to 18 series Fire Strategy drawings and all recommendations from fire strategy report
3. Submit drawings, technical information, calculations and manufacturers' literature.
4. Refer to product manufacturer literature for installation requirements and best practice.
5. Fire-stopping shall only be installed by a third-party accredited company, or a trained individual who has been assessed as competent through third-party accreditation.
6. All fire stopping, fire dampers and fire protection measures shall be documented using photographic records.
7. On completion, all passive protection measures installed shall be labelled, photographed, scheduled and recorded on plans by the third-party installer.
8. On Completion, the Contractor and the Employer shall have a comprehensive record and audit trail to demonstrate compliance which the end user can manage and update during the buildings' life (note: this information shall be housed within the H&S file and in the Regulation 38 Statement).

112 Panel cubicle systems Type A

1. Manufacturer: [Venesta Washroom Systems Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: 1st Floor, Units 19-23
St George's Square
St Georges Centre
Gravesend
Kent
United Kingdom
DA11 0TA
 - 1.1.2. Telephone: [+44 \(0\)1474 353333](tel:+441474353333)
 - 1.1.3. Web: <https://www.venesta.co.uk/>
 - 1.1.4. Email: marketing@venesta.co.uk
 - 1.2. Product reference: [V2 | Centurion Full Height \(SGL\)](#)
2. Panels: 13 mm solid grade laminate (SGL).
3. Panel supports: Pilasters fixed to floor using mounting blocks.
4. Pilasters: High strength extruded fluted satin anodized aluminium pilasters which extend down to floor level and are fixed to floor using mounting blocks.
5. Doors: 13 mm solid grade laminate (SGL).
6. Lippings: Black, polished and radiused.
7. Hardware: Satin anodized aluminium rise and fall butt hinge with outward opening hinge option available, satin anodized aluminium u-channels, satin anodized aluminium slide operated bolt with circular faceplate and emergency release facility. Faceplate and keep with concealed fixings.
8. System accessories:
9. Headrail: High strength extruded satin anodized aluminium headrail.

10. **Fixings:** Continuous satin anodized aluminium u-channel partition to wall fixing.
11. **Cubicle height (overall):** 2700mm from FFL to top of headrail.
12. **Cubicle depth:** 1700 mm.
13. **Clearance:** Floor clearance at pilasters and partitions, 0 mm; Floor clearance at doors, 5 mm; Ceiling, 20 mm (minimum).
14. **Colour/ Finish:**

150 Pre-plumbed installation frames

1. **Manufacturer:** [Venesta Washroom Systems Ltd](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** 1st Floor, Units 19-23
St George's Square
St Georges Centre
Gravesend
Kent
United Kingdom
DA11 0TA
 - 1.1.2. **Telephone:** [+44 \(0\)1474 353333](tel:+44201474353333)
 - 1.1.3. **Web:** <https://www.venesta.co.uk/>
 - 1.1.4. **Email:** marketing@venesta.co.uk
 - 1.2. **Product reference:** [Panelling | Frameduct - 100 mm Base Rail with Locks \(SGL\)](#)
2. **Materials:** Aluminium, nylon, steel, solid grade laminate.
3. **Panel:** High density solid grade laminate with decorative surface, both sides (SGL).
4. **Thickness:** 13 mm nominal.
5. **Edge treatment:** Black, radiused and polished.
6. **Flashgaps:** High density solid grade laminate (SGL) with decorative surface both sides with 6 mm extruded clip-on packer in clear UPVC.
7. **Framing:** Extruded aluminium base and headrail sections and rolled galvanized ultra steel wall channels.
8. **Base rails:** 100 mm high powder coated two part aluminium baserail to secure subframe units.
9. **Sub frame:** Rolled galvanized ultra steel subframe with riveted aluminium bracing bars, fitted to site frame with nylon click-fix locator brackets.
10. **Colour/ Finish:**

180 Duct panel support framing– site-fabricated softwood

1. **Description:**
2. **Framing:** Softwood, free from decay and active insect attack and with no knots wider than half the width of the section.
 - 2.1. **Finished size:**
 - 2.2. **Moisture content at time of fixing (maximum):** 18%.
 - 2.3. **Spacing (centres):**
 - 2.4. **Method of fixing:**
3. **Treatment:**

210 Samples

1. **General:** Before placing orders submit representative samples of the following:
2. **Delivered materials/ products:** To match samples.

220 Control samples

1. **General:** Complete samples as part of finished work and obtain approval of appearance before proceeding.
2. **Types:**
 - 2.1. **Locations:**

250 Installation

1. **Programming:** Do not install cubicles or duct/ wall panels before building is weathertight, wet trades have finished their work, wall and floor finishes are complete, and the building is well dried out.
2. **Accuracy:** Set out to ensure frames and/ or panels and doors are plumb, level and accurately aligned.
3. **Modifications:** Do not cut, plane or sand prefinished components except where shown on drawings.
4. **Fixing:** Secure components using methods and fasteners recommended by the cubicle/ panel manufacturer. Prevent pulling away, bowing or other distortions to frames, panels and doors.
5. **Moisture and thermal movement:** Make adequate allowance for future movement.

Ω End of Section

K40

Demountable suspended ceilings

Types of ceiling system

115A Unit/ modular suspended ceiling system

1. **Description:** Ceiling grid to corridors and other non noise sensitive
2. **Standard:** To BS EN 13964.
3. **Evidence of compliance:** All ceilings kits to be UKCA/ UKNI/ CE marked. Submit Declaration of Performance (DoP).
4. **Ceiling system manufacturer:** Knauf AMF Ceilings Ltd, South West Industrial estate, Peterlee, Co. Durham, U.K. Sacha Conte Tel: 07786 001278;
e-mail: sacha.conte@knauf.com; web: www.knaufceilingsolutions.com
 - 4.1. **Product reference:** Thermatex Acoustic SK, or equal approved.
5. **Ceiling**
 - 5.1. **Infill units:** AMF mineral fibre ceiling tile with acoustic fleece-coated facing.
 - 5.2. **Ceiling module:** 1800mm x 600mm in corridors
600 x 600mm all other spaces
 - 5.3. **Colour:** White
 - 5.4. **Soffit height above finished floor level:** Refer 17 series drawings
 - 5.5. **Weight:** 4.6kg/m²
 - 5.6. **Technical properties:** - Building material class: A2-s1,d0 as per EN13501-1
- Fire Resistance
F30 – F120 as per DIN 4102 part 2 (as per test certificate).
REI30 – REI120 as per EN 13501 part 2 (as per test certificate).
- Sound absorption: to EN ISO 354.
 $\alpha_w = 0.65$ (H) as per EN ISO 11654
NRC = 0.70 as per ASTM C 423
- Sound attenuation:
Dn,f,w = 38dB as EN ISO 10848 (19mm thickness, as per test certificate).
Dn,f,w = 40dB as EN ISO 10848 (semi-concealed system, as per test c.).
- Humidity resistance: up to 95% relative humidity.
- Light reflection: for white similar to RAL 9010, glare-free approx. 88%.
- Thermal conductivity: $\lambda = 0.052 - 0.057$ W/mK as per DIN 52612.
- Air permeability: PM1 (≤ 30 m³/hm²) as per DIN 18177.
- Indoor air: Indoor Air Comfort GOLD Status
- Clean room classification: class 3 as per ISO 14644-1.
6. **Grid**
 - 6.1. **Form:** 24mm wide Knauf AMF Grid System C. DONN or VENTATEC® Performance T24 GK exposed grid system, hot dipped galvanised steel 'T' section with pre painted white capping. Hot dipped galvanised steel 'T' sections with pre-painted white capping to white RAL 9010, or equal approved.
 - 6.2. **Exposure:** Exposed grid
 - 6.3. **Spacing:** Main runners (ref: T24/38-3600) are to be installed at 1200mm centres and suspended from soffit using hangers at 1200mm centres. The first hanger at no more than 500mm from the perimeter of the ceiling.
1200mm cross profiles (ref: PH1200) to be installed between main runners spaced at 600mm centres, creating a 1200mm x 600mm module.
7. **Access:** Infill units fully demountable
8. **Suspension system:** Ø2mm pre-straightened HDG steel wire or equal approved, to manufacturer's recommendations.

9. **Perimeter trim:** QIC YA15 plastered-in shadow trim
10. **Accessories:** TBD by others
11. **Integrated services fittings:** by others, if required
12. **Other requirements:** AMF-T Safety-Sleeve, hold-down clips if pressure washing is required. Or equal approved.

115B Unit/ modular suspended ceiling system

1. **Description:** Classrooms and ICT
2. **Standard:** To BS EN 13964.
3. **Evidence of compliance:** All ceilings kits to be UKCA/ UKNI/ CE marked. Submit Declaration of Performance (DoP).
4. **Ceiling system manufacturer:** Knauf AMF Ceilings Ltd, South West Industrial estate, Peterlee, Co. Durham, U.K. Sacha Conte Tel: 07786 001278;
e-mail: sacha.conte@knauf.com; web: www.knaufceilingsolutions.com
 - 4.1. **Product reference:** ADAGIO HD+ 35mm (Class A 43dB) or equal approved.
5. **Ceiling**
 - 5.1. **Infill units:** AMF mineral fibre ceiling tile with acoustic fleece-coated facing.
 - 5.2. **Ceiling module:** 600 x 600mm
 - 5.3. **Colour:** White
 - 5.4. **Soffit height above finished floor level:** Refer 17 series drawings
 - 5.5. **Weight:** 4.6kg/m²
 - 5.6. **Technical properties:** - Building material class: A2-s1,d0 as per EN13501-1
6. **Grid**
 - 6.1. **Form:** 24mm wide Knauf AMF Grid System C. DONN or VENTATEC® Performance T24 GK exposed grid system, hot dipped galvanised steel 'T' section with pre painted white capping. Hot dipped galvanised steel 'T' sections with pre-painted white capping to white RAL 9010, or equal approved.
 - 6.2. **Exposure:** Exposed grid
 - 6.3. **Spacing:** Main runners (ref: T24/38-3600) are to be installed at 1200mm centres and suspended from soffit using hangers at 1200mm centres. The first hanger at no more than 500mm from the perimeter of the ceiling.
1200mm cross profiles (ref: PH1200) to be installed between main runners spaced at 600mm centres, creating a 1200mm x 600mm module.
7. **Access:** Infill units fully demountable
8. **Suspension system:** Ø2mm pre-straightened HDG steel wire or equal approved, to manufacturer's recommendations.
9. **Perimeter trim:** QIC YA15 plastered-in shadow trim
10. **Accessories:** TBD by others
11. **Integrated services fittings:** by others, if required
12. **Other requirements:** AMF-T Safety-Sleeve, hold-down clips if pressure washing is required. Or equal approved.

135A Acoustic baffle suspended ceiling systems (A)

1. **Ceiling system manufacturer:** [Saint-Gobain Ecophon](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** Old Brick Kiln
Ramsdell
Tadley

Hampshire
RG26 5PP

- 1.1.2. Telephone: [+44 \(0\)1256 855250](tel:+441256855250)
- 1.1.3. Web: <https://www.ecophon.com/uk>
- 1.1.4. Email: technical@ecophon.co.uk
- 1.2. Product reference: [Solo™ Rectangle acoustic raft \(M637 wire - Solo™ Rectangle acoustic raft - 1200 x 600\)](#)
- 2. Ceiling support:
- 3. Support fasteners: Connect™ Absorber anchor.
- 4. Ceiling units: Acoustic high-density glass wool panels.
- 5. System accessories:
- 6. Size (l x w): as shown on RCP
- 7. Thickness: 40 mm.
- 8. Colour/ Finish: Black
- 9. Indoor climate: Recommended by the Swedish Asthma and Allergy Association. Certified as A+ by French VOC. Emission classification of building materials: M1.
- 10. Moisture resistance: Class C, relative humidity 95% and 30°C, according to EN 13964:2014.
- 11. Fire safety: To BS EN 13501-1, Class A2-s1, d0.
- 12. Sound absorption: To EN ISO 354:2003, Class A absorption.
- 13. Cleaning: The ceiling system can withstand daily dusting and vacuum cleaning, and weekly wet-wiping.
- 14. InstallationMethod: M637.

General/ performance

205 Compliance with performance requirements

- 1. Testing/ assessment: Submit UKAS accredited laboratory reports for the following:
- 2. Materials, components and details: Use those used in the test and identified in the assessment reports. If discrepancies arise, give notice.

Components

240 Samples

- 1. General: Submit representative samples of the following:

245 Standards

- 1. Steel panels: To BS EN 10346.
- 2. Aluminium sheet, strip and plate: To BS EN 485-1 and -2.
- 3. Aluminium bars, tubes and sections: To relevant parts of BS EN 515, BS EN 573, BS EN 755 and BS EN 12020.

250A Suspension system

- 1. Manufacturer: SAS International, 28 Suttons Business Park, London Road, Reading, RG6 1AZ.
Tel: +44 (0)118 929 0900
Fax: +44 (0)118 929 0900
- 2. Extent of system: The suspension system shall be as perimeter trims.
- 3. Top fixings: Fixings are to be selected and supplied by the appointed specialist ceiling contractor.
- 4. Hangers: SAS Emac hangers depth to suit

5. **Grid type:** Exposed beam grid - TBC to match existing school
 - 5.1. **Finish:** polyester powder coated
 - 5.2. **Colour:** RAL 9010 20% gloss - TBC with school

250B Suspension system

1. **Manufacturer:** Knauf AMF Ceilings Ltd, South West Industrial estate, Peterlee, Co. Durham, U.K.
Sacha Conte Tel: 07786 001278;
e-mail: sacha.conte@knauf.com; web: www.knaufceilingsolutions.com
2. **Form:** System B 2.2 CD60 or MF7/5 channel
3. **Exposure:** Concealed
4. **Spacings:** Carrier Frame Profile CD60 / MF5/7 to be installed at 900mm centres and suspended from soffit using hangers normally at 1200mm centres. First hanger no more than 125mm from the perimeter of the ceiling. Main-Runner CD60 / MF5 channel to be installed at 600mm centres.
5. **Hangers:** Generally from quick Hangers or Nonius Hangers. Alternative options available.
6. **Perimeter trims::** None required. Black timber shadow batten can be used
7. **Accessories/Fixings:** Heradesign Screws – 50/4.5 (coloured onsite with touch up paint to match panels)
8. **Insulation::** As necessary to achieve acoustic requirements
9. **Humidity resistance::** Up to 95%.
10. **Fire spread rating for materials exposed both to the room and the void, in accordance with BS EN 13964::** B-s1, d0.
11. **Other requirements:** Installation to be in accordance with BS EN 13964 and any other relevant current legislation.
12. **Other information:** Installation Guide (<http://www.knaufamf.com/wmv/?id=18752>)
Please check current lead times. Standard products have a lead time of 4-6 weeks.

287 Coated woven glass fibre cloth flexible cavity barriers Type A

1. **Manufacturer:** [Siderise Group](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** Forge Industrial Estate
Maesteg
Bridgend
Mid Glamorgan
United Kingdom
CF34 0AH
 - 1.1.2. **Telephone:** [+44 \(0\) 1656 730 833](tel:+44201656730833)
 - 1.1.3. **Web:** <https://www.siderise.com/>
 - 1.1.4. **Email:** projects@siderise.com
 - 1.2. **Product reference:** [Siderise CBX Flexible Acoustic Barriers for Suspended Ceilings](#)
2. **Thickness:** Nominal 40 mm.
3. **Fire performance:** B-s1, d0.
4. **Sound insulation rating:** 33 dB Rw.
5. **Material and finish:** Foil faced glass wool and polymeric barrier.
6. **Core:** Polymeric barrier central membrane.
7. **Weight:** Surface weight: 11 kg/m².

Execution

302 Control samples

1. **General:** Complete areas as part of the finished work in the following locations:
2. **Approval:** Obtain before completing areas of similar work.

305 Setting out

1. **General:** Completed ceiling should present, over the whole of its surface exposed to the room below, a continuous and even surface, jointed (where applicable) at regular intervals.
2. **Infill and access units, integrated services:** Fitted correctly and aligned.
3. **Edge/ perimeter infill units size (minimum):** Half standard width or length.
4. **Corner infill units size (minimum):** Half standard width and length.
5. **Grid:** Position to suit infill unit sizes. Allow for permitted deviations from nominal sizes of infill unit.
6. **Infill joints and exposed suspension members:** Straight, aligned and parallel to walls, unless specified otherwise.
7. **Suitability of construction:** Give notice where building elements and features to which the ceiling systems relate are not square, straight or level.

310 Bracing

1. **General:** Secure, with additional bracing and stiffening to give a stable ceiling system resistant to design loads and pressures.

315 Protection

1. **Loading:** Do not apply loads for which the suspension system is not designed.
2. **Ceiling materials:** When necessary, remove and replace correctly using special tools and clean gloves, etc. as appropriate.

320 Top fixing

1. **Building structure:** Verify suitability.
2. **Structural soffit:**
 - 2.1. **Suitability to receive specified fixings:** Evaluate and confirm.
3. **Fixing generally:** In accordance with BS EN 13964.
4. **Fixing to**
 - 4.1. **Concrete:** Drill and insert suitable expanding anchors.
 - 4.2. **Aerated concrete:** Fix through from the top of concrete units and provide a system of primary support channels.
 - 4.3. **Structural steel:** Drill, or use suitable proprietary clips/ adaptors.
 - 4.4. **Metal roof decking:** Fix to sides of liner tray corrugations.
 - 4.5. **Timber:** Fix to side of joists at least 50 mm from bottom edge. If ceiling system is intended for fire protection, fix into top third of joists.
 - 4.6. **Hollow structural members:** Submit fixing proposals.

325 Installing hangers

1. **Wire hangers:** Straighten and tension before use.
2. **Installation:** Install vertical or near vertical, without bends or kinks. Do not allow hangers to press against fittings, services, or insulation covering ducts/ pipes.

3. **Obstructions:** Where obstructions prevent vertical installation, either brace diagonal hangers against lateral movement, or hang ceiling system on an appropriate rigid sub-grid bridging across obstructions and supported to prevent lateral movement.
4. **Extra hangers:** Provide as necessary to carry additional loads.
5. **Fixing**
 - 5.1. **Wire hangers:** Tie securely at top with tight bends to loops to prevent vertical movement.
 - 5.2. **Angle/ strap hangers:** Do not use rivets for top fixing.
6. **Spacings:**

330 Installing timber edge battens

1. **Fixing:** Firmly to perimeter wall or other building structure.
 - 1.1. **Fasteners:**
 - 1.2. **Fixing centres (maximum):**

335 perimeter trims

1. **Jointing:** Neat and accurate, without lipping or twisting.
 - 1.1. **External and internal corners:** Mitre joints generally. Overlap joints at internal corners are
 - 1.2. **Intermediate butt joints:** Minimize. Use longest available lengths of trim. Align adjacent lengths.
2. **Fixing:** Fix firmly to perimeter wall, edge battens or other building structure.
 - 2.1. **Fasteners:**
 - 2.2. **Fixing centres:**

340 Exposed grids

1. **Grid fixings:**
2. **Main runners:** Install level. Do not kink or bend hangers.
 - 2.1. **Spliced joints:** Stagger.
 - 2.2. **Wire hangers passing through main runners:** Use sharp bends and tightly wrapped loops.
 - 2.3. **Angle/ strap hangers:** Do not use rivets for bottom fixing.
 - 2.4. **Angular displacement of long axis of one runner in relation to next runner in line with it:** Not visually apparent.
3. **Cross members supported by main runners or other cross members:** Install perpendicular to intersecting runners.
4. **Cross tees:** Flat and coplanar with flanges of main runners after panel insertion.
 - 4.1. **Cross tees over 600 mm long, cut and resting on perimeter trim:** Provide an additional hanger.
5. **Holding down clips:** Locate to manufacturer's recommendations.
 - 5.1. **Fire-protecting/ resisting ceiling systems:** Use clip type featured in the fire test/ assessment.

345 Concealed grids

1. **Grid fixings:**
2. **Primary support channels:** Install level. Do not kink or bend hangers.
 - 2.1. **Wire hangers wrapped around primary channels:** Twice wrapped. Loops tightly formed.
 - 2.2. **Angle/ Strap hangers:** Do not use rivets for bottom fixing.
3. **Splines:** Locate between infill units to assist levelling of adjacent units and to resist air movement at joints.

4. Spring-tee grids: Do not omit primary channel.

355 Installing infill units

1. General
 - 1.1. Perimeter infill units: Trimmed, as necessary, to fully fill space between last grid member and perimeter trim. Prevent subsequent movement.
 - 1.2. Deeply textured infill units: Minimize variations in apparent texture and colour. In particular, avoid patchiness.
2. Concealed grids: Install infill units uniformly, straight and aligned. Avoid dimension creep.
 - 2.1. Infill units around recessed luminaires and similar openings: Prevent movement and displacement.

365 Installing metal infill units

1. Sound absorbing pads: Fit to prevent upward air movement through infill units. Cut or fold pads in cut perimeter infill units to full unit size. Reseal cut pads.
2. Perimeter infill units: Firmly wedge cut units into perimeter trim, or clip down.

375 Board ceiling systems

1. Cut boards: Neat and accurate.
2. Fixing to grid
 - 2.1. Board edges: Fully support. Screw to grid members. Set heads of screws below surface of boards and fill flush with surface.
 - 2.2. Boards applied in two or more layers: Stagger joints.
3. Movement joints: Provide as appropriate for the area of ceiling system and/ or to coincide with movement joints in surrounding structure.

385 Upstands and bulkheads

1. Vertical ceiling systems: Support and brace to provide alignment and stability.
2. High upstands: Provide support at base of upstand.

390 Openings in ceiling materials

1. General: Neat and accurate. To suit sizes and edge details of fittings. Do not distort ceiling system.

395 Integrated services

1. General: Position services accurately, support adequately. Align and level in relation to the ceiling and suspension system. Do not diminish performance of ceiling system.
2. Small fittings: Support with rigid backing boards or other suitable means. Do not damage or distort the ceiling.
 - 2.1. Reaction to fire of additional supporting material: Not less than ceiling material.
3. Services outlets
 - 3.1. Supported by ceiling system: Provide additional hangers.
 - 3.2. Independently supported: Provide flanges to support ceiling system.

401 Ceiling-mounted luminaires

1. Support:
 - 1.1. Independently supported luminaires: Suspension adjusted to line and level of ceiling.
 - 1.2. Ceiling supported luminaires: Modifications and/ or extra support required:

2. **Surface mounted luminaires:** Units installed so that in event of a fire the designed grid expansion provision is not affected.
3. **Modular fluorescent recessed luminaires:** Compatible with ceiling module. Extension boxes must not foul ceiling system.
4. **Recessed rows of luminaires:** Provide flanges for support of grid and infill units, unless mounted above grid flanges. Retain in position with lateral restraint.
5. **Fire-protecting/ resisting ceiling systems:** Luminaires must not diminish protection integrity of ceiling system.
6. **Access:** Provide access for maintenance of luminaires.

406 Trunking

1. **Recessed trunking:** Provide flanges for support of grid and infill units, unless mounted above grid flanges. Retain in position with lateral restraint.

411 Mechanical services

1. **Fan coil units**
 - 1.1. **Inlet/ Outlet grilles:** Trim ceiling grid and infill units to suit.
 - 1.2. **Space beneath:** Sufficient for ceiling system components.
 - 1.3. **Suspension and connections:** Permit accurate setting out and levelling of fan coil units.
2. **Air grilles and diffusers**
 - 2.1. **Setting out:** Accurate and level.
 - 2.2. **Linear air diffusers:** Retain in place with lateral restraint. Provide flanges for support of grid and infill units.
 - 2.3. **Grille/ Diffuser ceiling joints:** Provide smudge rings and edge seals.
3. **Smoke detectors and PA speakers**
 - 3.1. **Ceiling infill units:** Scribe and trim to suit.
 - 3.2. **Independent suspension:**
 - 3.3. **Flexible connections:** Required.
4. **Sprinkler heads:** Carefully set out and level.

415 Installing insulation

1. **Fitting:** Fit accurately and firmly with butted joints and no gaps.
2. **Insulation within individual infill units:** Fit closely. Secure to prevent displacement when infill units are installed or subsequently lifted.
 - 2.1. **Dustproof sleeving:** Reseal, if cut.
3. **Width:** Lay insulation in the widest practical widths to suit grid member spacings.
4. **Services:** Do not cover electrical cables that have not been sized accordingly. Cut insulation carefully around electrical fittings, etc. Do not lay insulation over luminaires.
5. **Sloping and vertical areas of ceiling system:** Fasten insulation, to prevent displacement.

421 Ceiling systems intended for fire protection

1. **Junctions of ceiling systems with perimeter abutments and service penetrations:** Seal gaps with tightly packed mineral wool or intumescent sealant to prevent penetration of smoke and flame.
2. **Ceiling system/ wall junctions:** Maintain protective value of ceiling system.
 - 2.1. **Fixings and grounds:** Noncombustible.
 - 2.2. **Metal trim:** Provide for thermal expansion.
3. **Access and access panels:** Maintain continuity of fire protection.

425 Installing cavity fire barriers

1. Maximum ceiling void dimension in any direction:
2. Fixing:
 - 2.1. General: Fix barriers securely to channels or angles at abutments to building structure.
 - 2.2. At perimeters and joints: Provide permanent stability and continuity with no gaps to form a complete barrier to smoke and flame.
3. Joints: Form to preserve integrity in fire.
4. Service penetrations: Cut barriers neatly to accommodate services. Fit fire-resistant sleeves around flexible materials. Fill gaps around services to fire barrier manufacturer's recommendations to maintain barrier integrity. Adequately support services passing through the barrier.
5. Ceiling systems intended for fire protection: Do not impair fire-resisting performance of ceiling system.
6. Ceiling systems not intended for fire protection: Do not mechanically interlink barriers with ceiling system.

431 Installing sound barriers

1. Material:
2. Setting out: Align accurately with partition heads.
3. Fixing: Fix tightly at perimeters and joints using methods recommended by barrier manufacturer, including steel support sections as appropriate. Completed installation to be stable, secure and continuous, with no gaps.
4. Gaps at junctions with partition heads, ceiling system, structural soffit, walls, ducts, pipes, etc.: Seal with mineral wool or suitable sealant.

435 Installing air plenum barriers

1. Fixing: Fix tightly at perimeters and joints using methods recommended by the barrier manufacturer to ensure permanent stability.
2. Edges and joints: Seal effectively to prevent air leakage.

500 Electrical continuity and earth bonding

1. Inclusion in finished work:

Completion

505 Tools

1. Access tools: At Completion, supply one set of the following:

520 User instructions

1. Contents: Include the following:
 - 1.1. Correct methods for removing and replacing infill units and other components.
 - 1.2. Cleaning methods and materials.
 - 1.3. Recommendations for redecoration.
 - 1.4. Ceiling systems intended for fire protection: Limitations placed on subsequent alterations and maintenance procedures, to ensure that their fire performance is not impaired.
 - 1.5. Maximum number, position and value of point loads that can be applied to ceiling system after installation.

530 Spares

1. General: At practical completion, supply the following:

Ω End of Section

L10

Windows/ rooflights/ screens/ louvres

General

110 Evidence of performance

1. **Certification:** Provide independently certified evidence that all incorporated components comply with specified performance requirements.

115 Timber procurement

1. **Timber (including timber for wood-based products):** Obtained from well managed forests and/ or plantations in accordance with:
 - 1.1. The laws governing forest management in the producer country or countries.
 - 1.2. International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).
2. **Documentation:** Provide either in accordance with chain of custody certification scheme requirements:
 - 2.1. Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied.
 - 2.2. Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood-based products.
3. **Chain of custody certification scheme:**
 - 3.1. **Other evidence:**

120 Pre-construction survey

1. **Procedure:** 1. The Window Fabricator/Installer (Specialist Contractor) is expected to make a pre tender visit to site order to ascertain all relevant conditions, structural details and site layout. No additional claim will be entertained for items that would be apparent during the pre-tender site visit. The Specialist Contractor must allow in his tender for the replacement of all items specified and/or required.

The units supplied are to be manufactured to suit prepared openings.

Notwithstanding any information within this Specification, all framing and infills shall be capable of withstanding the design wind loadings calculated in accordance with BS EN 1991-1-4:2005 +A1:2010 and imposed loads as defined in BS6399-1-1:2002 or BS EN 1991-1-7 +A1:2014, and the Specialist Contractor shall carry out calculations to demonstrate this.

The Specialist Contractor shall provide drawings to the Main Contractor depicting all profiles, glazing, weather seals, gasket fixings and sealants to be used and the relationship of the above to the adjacent structural details for each window type.

Allow for anomalies and variations in the size of the openings, and for out-of-square openings. This is to include for the manufacture of “specials” as necessary.

Window configurations are to be depicted within the attached Window Schedule/Drawings as listed below.

The Specialist Contractor is to provide drawings showing the relationship of framing to structure, including all profiles, sealants, fixings, trims and weather seals.

2. **Designated items:**
3. **Primary support structure:** Carry out survey sufficient to verify that required accuracy and security of erection can be achieved.

4. **Timing:** Before fabrication.

140 Control samples

1. **Procedure**
 - 1.1. Finalize component details.
 - 1.2. Fabricate one of each of the following designated items as part of the quantity required for the project.
 - 1.3. Obtain approval of appearance and quality before proceeding with manufacturer of the remaining quantity.
2. **Designated items:**

Products

205 Window materials specification

1. **Minimum BRE 'Green Guide to Specification Online' rating:**

205 Window Materials Specification. Aluminium Windows Type A

1. **Minimum BRE 'Green Guide to Specification Online' rating:** All frames and other sections to be extruded aluminium alloy specification 6060 T6 compliant with EN 12020, EN 573-3, EN 515 and EN 775-1 to 9.

Technal window is made from Hydro CIRCAL®, recycled low carbon aluminium, a high quality aluminium made with a minimum of 75% recycled end-of-life aluminium (post-consumer scrap). Hydro CIRCAL® has a carbon footprint of 2.3kg of CO₂/kg of aluminium on average.

The thermal barrier section is achieved using two separate aluminium extrusions and two 40mm wide glass reinforced polyamide extrusions mechanically jointed to form a single compound profile. The sections forming the windows are to incorporate a thermal break, achieved using a high strength, glass reinforced polyamide barrier to PA66 GF25.

The thermal break is to be applicable to all profiles, including vents, couplers and cills.

Joints to be sealed with sealants as specified by the aluminium systems company fabrication manual.

All fabrication to be strictly in accordance with the system company's Fabrication and Specification Manuals and all current Technical Bulletins.

330 Aluminium windows

1. **Standard:** Non-fire and/ or smoke-rated windows to BS EN 14351-1 and BS 4873
2. **Manufacturer:** Hydro Building Systems Uk Ltd
 - 2.1. **Product reference:** Dualframe 75Si. Top Hung Casement
3. **Finish as delivered:** Polyester powder coating to BS EN 12206-1.
4. **Thermal performance (U-value maximum):** The window specified is to achieve a U value (U_w) of 1.3-1.4W/m²K when glazed with the materials outlined in Clause 336 below. U value to be calculated according to the smaller window configuration defined in BS EN14351-1; Conventions for U value calculations.
5. **Acoustic performance rating:** 27 dB $R_w + C_{tr}$ (details to be submitted for planning)
6. **Weather Performance:**
 - 6.1. **Air Permeability:** To BS 6375-1, Class 4.
 - 6.2. **Watertightness:** To BS 6375-1, Class 9A.

6.3. **Resistance to wind load:** Class A5 (Deflection 1/300 at 2000Pa pressure) in accordance with BS EN 12210

7. **Glazing details:** Correct glass thickness is always subject to calculation by the glazing sub-contractor.

SBD Glazing requirements: Glazing at ground floor. Minimum P3A rating. Laminate face to internal space.

All glazing in WCs and bathrooms shall have obscured glass to the inner pane.

All glass within 800mm from FFL shall be toughened or laminated. (Below 1500mm if within a door or 300mm of a door)

All glass and glazing shall conform to:

BS EN12600: 2002 Glass in building. Pendulum test. Impact test method and classification for flat glass

BS EN 1279: Glass in Buildings. Insulating Glass Units.

Part 1:2004 Generalities, dimensional tolerances and rules for the system description

Part 2:2002 Long term test method and requirements for moisture penetration

Part 3:2002 Long term test method and requirements for gas leakage rate and for gas concentration tolerances.

Part 4:2002 Methods of test for the physical attributes of edge seals

Part 5:2005+A2:2010 Evaluation of conformity

Part 6:2002 Factory production control and periodic tests.

Manifestation design and location to be confirmed by the Contract Administrator.

Recommendations of the Glass and Glazing Federation should be adhered to.

Note:

Heatsoak Testing Procedure

We recommend Heatsoak testing is carried out where required in accordance with European Standard BS EN 14179: Heat soaked thermally toughened soda lime silicate safety glass. The emphasis of this procedure is to ensure that the glass temperature is maintained at $290^{\circ}\text{C} \pm 10^{\circ}\text{C}$ during the holding phase of the process cycle.

The heat soak process cycle consists of a heating phase, a holding phase and a cooling phase. In summary the procedure consists of controlled heating of a complete volume of toughened glass from ambient room temperature to a temperature of $290^{\circ}\text{C} \pm 10^{\circ}\text{C}$ and then holding it at that temperature for a minimum period of 2 hours before then allowing it to cool naturally. The procedure is carried out at various locations in audited ovens to ensure that the glass temperature during the holding phase is maintained at $290^{\circ}\text{C} \pm 10^{\circ}\text{C}$.

Thermal Safety Check Requirement

As with all laminated or non heat treated glass types we strongly recommend that a thermal safety check is carried out by via the Installer. This should be carried out where blinds, external brise soleil, ceiling trench or similar may trap warm air pockets close to the non heat treated pane. It is to be noted that laminated or non heat treated glass are not compatible in spandrel areas.

Full Height Barrier glazing must comply with; BS6180: 2011 Barriers in and about buildings. Code of practice.

Please note that if glazed areas form a full height barrier then the glass may need to comply with BS6180:2011 which means that the inner pane may be required to meet relevant POINT, U.D.L and LINE LOADS. If this is the case then the inner pane of glass would have to be a class A non break – either minimum 8.8mm thick or 10mm toughened or greater depending upon acoustic , wind loading , size & dimension of DGU's.

8. **Beading:** Internal for sash

9. **Ironmongery/ accessories:** The Specialist Contractor is to obtain the written confirmation of the Contract Administrator as to the type and position of all ironmongery before commencing manufacture.
10. **Fixing:** To suit site conditions and building construction type. Quantity, type and size of fixings to be determined by window fabricator/ installer
11. **Other requirements:** Integrated louvre's as L10/650 connected to plenum box to M&E specification. Where there are no duct, A1 rated insulated backing to be provided to blanking plate

460 Framed rooflights

1. **Manufacturer:** [Glazing Vision Ltd](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** Glazing Vision Ltd
Sawmills Road
Diss
Norfolk
United Kingdom
IP22 4RG
 - 1.1.2. **Telephone:** +44 (0)1379 658300
 - 1.1.3. **Web:** <https://www.glazingvision.co.uk/>
 - 1.1.4. **Email:** salesuk@glazingvision.com
 - 1.2. **Product reference:** Flushglaze Fixed Flat Rooflight
2. **Form:** Flat.
3. **Frame**
 - 3.1. **Material:** Aluminium.
 - 3.2. **Finish**
 - 3.2.1. **Coating:** Polyester powder coating.
 - 3.2.2. **Colour:** RAL 7015, slate grey.
4. **Glazing or infill**
 - 4.1. **Type:** Glass units.
 - 4.2. **Construction:** 36.8mm double glazed unit (silicone sealed)
 - 4.3. **Colour:** Clear.
5. **Operation**
 - 5.1. **Type:** Fixed.
6. **Accessories:**
7. **Execution:** Installation of framed rooflights.
8. **Height:**
9. **Thickness:** 36.8mm
10. **Type:**
11. **Seals:** Silicone, poly-butyl and EPDM rubber seals.
12. **Security:** Secured by Design tested to LPS2081 Issue 1:2015.
13. **Outer:** 8mm HST Toughened soft coat low E (U Value 1.1W/m²K)
14. **Spacer:** 16mm Argon, Black warm edge spacer, silicone sealed 7mm min. bite
15. **Inner:** 12.8mm HST Toughened PVB Laminated Tested to class 1 (non-fragile) in accordance with CWCT TN66/67
16. **Thermal performance (U-value maximum):** 1.1 W/m²K
17. **G-value:** 0.4
18. **Light transmittance:** 0.68-0.7

19. Fire performance
 - 19.1. Fire resistance: 30 Mins Integrity from Inside E30
20. Must be designed to comply with the following:
 - a) Tested to class 1 (non-fragile) in accordance with CWCT TN66/67
 - b) DFE TN 65 Thermal fracture of glass.
 - c) DFE TN 66 Safety and fragility of overhead glazing: guidance on specification.
 - d) DFE TN 67 Safety and fragility of overhead glazing: testing and assessment.
 - e) ACR [M] 001:2014 Test For non-fragility of large element roofing assemblies. 5th edition.
 - f) The Minimum Life Expectancy of 25 years

510 Glazed wood screens. 30mins Fire Rated

1. Location: As per Glazed Screen Schedule
2. Timber: Generally to BS EN 942.
 - 2.1. Species: In Accordance with report requirements to suit required fire performance noted below
 - 2.2. Appearance Class: J2 for glazing beads, J30 or better for all other members
 - 2.3. Moisture content on delivery: 9-13%
3. Assembly adhesive: Manufacturer's standard
4. Joinery workmanship: As section Z10.
5. Finish as delivered: Prepared and primed as section M60
6. Fire performance: 30 minutes fire resistance and 30 minutes insulation
7. Glazing details: 15mm Pilkington Pyrostop 30-10. (to achieve 36 dB Rw+Ctr acoustic performance)
8. Special features/ other requirements: Manifestation to Building Regulations Part K
9. Fixing: Screw and pellet
10. Architraves: 18mm x 15mm SW pencil round. Primed for site painting

511 Glazed wood screens. 60mins Fire Rated

1. Location: As per Glazed Screen Schedule
2. Timber: Generally to BS EN 942.
Details to Pilkington test report Chilt/A07245
 - 2.1. Species: In Accordance with report requirements to suit required fire performance noted below
 - 2.2. Appearance Class: J2 for glazing beads, J30 or better for all other members
 - 2.3. Moisture content on delivery: 9-13%
3. Assembly adhesive: Manufacturer's standard
4. Joinery workmanship: As section Z10.
5. Finish as delivered: Prepared and primed as section M60
6. Fire performance: 60 minutes fire resistance, and 60 minutes insulation
7. Glazing details: Pyrostop 60-101, 23mm thick
8. Special features/ other requirements: Manifestation to Building Regulations Part K
9. Fixing: Screw and pellet
10. Architraves: 18mm x 15mm SW pencil round. Primed for site painting

512 Glazed wood screens. Non Fire Rated

1. Location: As per Glazed Screen Schedule
2. Timber: Generally to BS EN 942.

- 2.1. Species: Manufacturer's standard
- 2.2. Appearance Class: J2 for glazing beads, J30 or better for all other members
- 2.3. Moisture content on delivery: 9-13%
3. Assembly adhesive: Manufacturer's standard
4. Joinery workmanship: As section Z10.
5. Finish as delivered: Prepared and primed as section M60
6. Fire performance: n/a
7. Glazing details: Laminated clear glazing, thickness to meet acoustic requirements
8. Special features/ other requirements: Manifestation to Building Regulations Part K
9. Fixing: Screw and pellet
10. Architraves: 18mm x 15mm SW pencil round. Primed for site painting

650 Aluminium louvre panel units

1. Manufacturer: [Duco Ventilation & Sun Control NV](#)
 - 1.1. Contact details
 - 1.1.1. Address: Bedrijvenlaan 2
Veurne
Belgium
8630
 - 1.1.2. Telephone: [+32 58 33 00 66](tel:+3258330066)
 - 1.1.3. Web: <https://www.duco.eu/uk-ie/>
 - 1.1.4. Email: info@duco.eu
 - 1.2. Product reference: [DucoGrille Solid ISO F 30Z \(DucoGrille Solid ISO F 30Z P1\)](#)
2. Material: Aluminium.
3. Finish: Powder-coated.
4. Colour: RAL to match windows
5. Construction: Flanged frame.
6. Air intake and exhaust performance
 - 6.1. Application: supply/ intake
7. Weather performance
 - 7.1. Water penetration class (minimum): To BS EN 13030.
8. Louvre configuration
 - 8.1. Blade orientation: Horizontal.
 - 8.2. Blade pitch: 37.5 mm.
 - 8.3. Blade angle: Fixed.
9. Depth: 88 mm.
10. Frame width: 63 mm.
11. Thickness: 1.5 mm.
12. Free area: Visual free area: 60%; Physical free area: 34%.
13. InputFlowrate: 16.94.
14. ExhaustAirFlowrate: 18.26.
15. Perforation: P1.
16. Type: Z-shaped.
17. Mesh: Aluminium extrusions are perforated and act as insect mesh.
18. U-value: 0.9 W/m²K.

19. Plenum box: to suit M&E detail

Execution

710 Protection of components

1. **General:** Do not deliver to site components that cannot be installed immediately or placed in clean, dry floored and covered storage.
2. **Stored components:** Stack vertical or near vertical on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.

730 Priming/ sealing

1. **Wood surfaces inaccessible after installation:** Prime or seal as specified before fixing components.

750 Building in

1. **General:** Not permitted unless indicated on drawings.
 - 1.1. Brace and protect components to prevent distortion and damage during construction of adjacent structure.

765 Window installation generally

1. **Installation:** Into prepared openings. In accordance with BS 8213-4.
2. **Fixing of Frames:** As section Z20 using appropriate fixings.
Window framing to be securely fixed direct to the building structure, no further than 150mm from each corner and at centres not exceeding those laid down in procedures issued by the systems company.
3. **Distortion:** Install windows without twist or diagonal racking.

770 Damp-proof courses in prepared openings

1. **Location:** Ensure correct positioning in relation to window frames. Do not displace during fixing operations.

782 Fixing of aluminium frames

1. **Standard:** As section Z20.
2. **Fasteners:** To be determined by the fabricator/ installer to suit site conditions and building structure.
 - 2.1. **Spacing:** When not pre-drilled or specified otherwise, position fasteners not more than 250 mm from ends of each jamb, adjacent to each hanging point of opening lights, and at maximum 600 mm centres.

810 Sealant joints

1. **Sealant:** Low modulus neutral cure silicone sealant to BS EN ISO 11600: 2003 + A1: 2011.
Sealant backing to be provided and detailed in drawings.
2. **Application:** As section Z22 to prepared joints.
Seal all external joints between framing and structure.
Finish triangular fillets to a flat or slightly convex profile

820 Ironmongery

1. **Fixing:** In accordance with any third-party certification conditions applicable. Assemble and fix carefully and accurately using fasteners with matching finish supplied by ironmongery manufacturer. Do not damage ironmongery and adjacent surfaces.
2. **Checking/ adjusting/ lubricating:** Carry out at Completion and ensure correct functioning.

Ω End of Section

L20

Doors/ shutters/ hatches

General

0 Note

1.

000 Note

1. Aluminium profiles available from: -

Hydro Building Systems UK Ltd
Severn Drive
Tewkesbury
GL20 8SF

Tel: 01684 853500
www.technal.co.uk www.technal-library.co.uk

Harry Tamplin
Project Consultant

M: 07469 151268
T: 01684 853500
E: harry.tamplin@hydro.com

Profiles, dependent on product range are extruded from alloy 6005T6, 6063T6 or 6060T5 to BS EN12020, BS EN573-3, BS EN515 and BS EN755 - 1 to 9;

Fixing screws and components should be stainless steel, cadmium plated steel, or other corrosion resistant materials.

Glazing materials shall be Technal's standard gaskets and/or glazing tape.

Materials can be polyester powder coated to BS6496 available in a large range of colours or anodised in accordance with BS EN 12373 - 1 or BS3987 in natural silver, bronze Anolok or in a variety of colours.

The Technical Department within Technal can provide specialist advice and technical information on the whole range of Technal products and is fully equipped with up to date CAD facilities. The team of Project Consultants can visit architects and specifiers to assist with development of designs and details regarding the glazed areas of buildings.

Fabrication and installation of all Technal products should be performed by a suitable Technal fabricator/installers in accordance Technal fabrication and installation manuals and must also be to recognised trade standards using approved design drawings.

Double glazed sealed units glazed in accordance with BS6262 and Building Regulations. Single glazing can only be used where Building Regulations allow.

Details of all the current Technal systems are available from our websites on: -
www.technal.co.uk

110 Evidence of performance

1. The frames are to comply with all relevant British Standard Specifications, Codes of Practice, and Statutory Requirements (including all revisions and amendments), as well as the guides and recommendations laid down by the relevant trade organisation relating to their performance, constituent materials, methods of assembly and use. Any exceptions to the above are to be advised in writing by the specifier.

All frames and other sections to be extruded to BS EN 755-9: 2016, Specification 6060 T6 or 6063 T6.

All materials and ancillary products are to be used and fitted entirely in accordance with the instructions of the relevant manufacturer.

The door systems manufacturer must hold British Standard Kitemark licences for BS4873 "Specification for Aluminium Alloy Windows" and PAS 24-1:2016 "Enhanced Security performance requirements for doorsets and windows in the UK". Prior to commencement of work on site, the Specialist Contractor shall forward copies of the Kitemark Certificate or equivalent to the Contract Administrator.

The Door Fabricator/Installer (Specialist Contractor) is expected to make a pre tender visit to site/inspection of all relevant drawing and documents in order to ascertain all relevant conditions, structural details and site layout. No additional claim will be entertained for items that would be apparent during the pre-tender site visit and/or inspection of documents. The Specialist Contractor must allow in his tender for the replacement of all items specified and/or required.

112 Timber procurement

1. **Timber (including timber for wood-based products):** Obtain from well managed forests and/ or plantations in accordance with:
 - 1.1. The laws governing forest management in the producer country or countries.
 - 1.2. International agreements such as the 'Convention on International Trade in Endangered Species of wild fauna and flora (CITES)'.
2. **Documentation:** Provide either in accordance with chain of custody certification scheme requirements:
 - 2.1. Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied, or
 - 2.2. Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood-based products.
3. **Chain of custody certification scheme:**
 - 3.1. Other evidence:

115 Fire-resisting and smoke control pedestrian doors/ door assemblies/ doorsets

1. **UKCA/ UKNI/ CE marked fire-resisting and smoke control pedestrian doorsets:** To BS EN 16034 and in conjunction with BS EN 13241, BS EN 14351-1 and BS EN 14351-2.
2. **Door products:** As defined in BS EN 12519.
3. **Evidence of fire performance:** Provide certified evidence, in the form of a product conformity certificate, directly relevant fire test report or engineering assessment, that each door/ door assembly/ doorset supplied will comply with the specified requirements for fire-resisting and/ or smoke control if tested to BS 476-22, BS EN 1634-1, BS EN 1634-3 or is UKCA/ UKNI/ CE marked to BS EN 16034. Specified values should not be a combination of both standards. Such certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals.

4. Components, assemblies or sets will be marked to the relevant UKCA/ UKNI/ CE marking European product standard (hEN), national product standard and/ or third-party certification rating.

120 Non-fire-resisting pedestrian doors/ door assemblies/ doorsets

1. Provide certified evidence, in the form of a product conformity certificate or engineering assessment, that each pedestrian door/ doorset/ assembly supplied will comply with the specified requirements to BS EN 14351-1. Such certification must cover door and frame materials, glass and glazing materials and their installation, essential and ancillary ironmongery, hinges and seals.
2. Components and assemblies will be marked to the relevant UKCA/ UKNI/ CEI marking European product standard (hEN), national product standard and/ or third-party certification rating.

150A Site dimensions

1. The units supplied are to be manufactured to suit prepared openings.

Notwithstanding any information within this Specification, all framing and infills shall be capable of withstanding the design wind loadings calculated in accordance with BS6399 Pt2: 1997 or BS EN 1991-1-1 and imposed loads as defined in BS6399 Pt1: 1996 or BS EN 1991-1-4, and the Specialist Contractor shall carry out calculations to demonstrate this.

The Specialist Contractor is responsible for ensuring that all new doors are square and central in the opening, and that a perimeter gap shall be provided to allow adequate thermal expansion and contraction of the framing, consistent with the site location and limitations of the perimeter sealant used.

The Specialist Contractor shall provide drawings to the Main Contractor depicting all profiles, glazing, weather seals, gasket fixings and sealants to be used and the relationship of the above to the adjacent structural details for each door type.

Allow for anomalies and variations in the size of the openings, and for out-of-square openings. This is to include for the manufacture of “specials” as necessary.

Door configurations are to be depicted within the attached Door Schedule/Drawings as listed below.

The Specialist Contractor is to provide drawings showing the relationship of framing to structure, including all profiles, sealants, fixings, trims and weather seals.

170 Control samples

1. Procedure
 - 1.1. Finalize component details.
 - 1.2. Fabricate one of each of the following designated items as part of the quantity required for the project.
 - 1.3. Obtain approval of appearance and quality before proceeding with manufacture of the remaining quantity.
2. Designated items: Doorsets

Products

420 Wood doorsets

1. Description: Refer to internal door schedule
2. Manufacturer: Enfield Speciality Doors, 278 Alma Road, Enfield, Middx, EN3 7BB
Tel: 020 8805 6662 W: www.enfelddoors.co.uk
3. Door leaf:
4. Core: Solidcore

Rivington Street Studio Limited
10-10-2025

5. Thickness: 44mm and 54mm as required
6. Facings: Veneer : American White Oak
7. Lippings: 8mm minimum American White Oak Hardwood 650kg/m³
8. Finish: Pre Finished 2 coats Water Based S/G 50% Sheen Lacquer
9. Vision Panels:: where required (refer to door schedule)
 - o Beading: HW bolection, spray painted to match door laminate
 - o Size & Position: refer to door elevations
 - o Glass: 6.4mm laminated (NFR) 7mm Pyrodur (FD30) 15mm high impact laminated glass (Calm Room)
10. Frame and architraves:
11. Wood species: Softwood/Hardwood as required.
12. Finish as delivered: Factory Primed for site decoration
13. Extension linings: : Softwood as required
Finish: Factory Primed for site decoration
14. Architraves: Softwood **18 x50 mm** with one pencil round
 - o Finish: Factory Primed for site decoration
15. Ironmongery: Refer ironmongery schedule
Doors & linings to be supplied factory prepared for hinges
16. Fire performance
 - 16.1. Fire resistance: NFR & FD30(S) refer to door schedule
17. Acoustic performance: 30db & 35dB refer to door schedule
18. Mechanical Durability: DD171 Severe Duty Classification, BS EN 1192 Class 4
19. Other requirements: Use of MDF is not permitted
20. Security: Refer to door schedule for doors with enhanced security

480A Doorsets

1. Manufacturer: [Technal](#)
 - 1.1. Contact details
 - 1.1.1. Address: Hydro Building Systems UK Ltd
Severn Drive
Tewkesbury
Gloucestershire
GL20 8SF
 - 1.1.2. Telephone: [+44 \(0\)1684 853 500](tel:+441684853500)
 - 1.1.3. Web: www.technal.co.uk
 - 1.1.4. Email: info.uk@technal.com
 - 1.2. Product reference: Stormframe STII High Traffic Aluminium Commercial Door (STII Thermally broken shop-front system)
2. Size: Refer elevation drawings
3. Glazing: Double glazed units. P3A rating
4. Notwithstanding any information within this Specification, all framing and infills shall be capable of withstanding the design wind loadings calculated in accordance with BS6399 Pt2: 1997 or BS EN 1991-1-1 and imposed loads as defined in BS6399 Pt1: 1996 or BS EN 1991-1-4, and the Specialist Contractor shall carry out calculations to demonstrate this.

The thermal barrier section is achieved using two separate aluminium extrusions and two glass reinforced polyamide extrusions mechanically jointed to form a single compound profile. The sections forming the doors are to incorporate a thermal break, achieved using a high strength, glass reinforced polyamide barrier to PA6.6 GF25.

Ancillary profiles may incorporate a polyurethane resin thermal break.

The thermal break is to be applicable to all profiles, including vents, couplers and cills.

Sliding Doors

5. **Material:** All frames and other sections to be extruded from alloys 6060 T6 or T66 or 6063 T6 in accordance with BS EN 755-9: 2016 or BS EN 12020-2:2016.

Billet shall be supplied to the extruder under standard operating instructions, traceable and verified by a Notified Body

- A supporting EPD from an accredited third party EPD program operator with membership of Eco Platform for the aluminium billet, conforming to ISO 14025, ISO 21930 and EN 15804, must be made available on request. The Embodied Carbon Factor shall apply to Stages A1 to A3 using the avoided burden method and to include Scopes 1, 2 and 3 of the production and distribution process as defined by the GHG Protocol.

Aluminium profiles shall be extruded from REDUXA which is a certified, low carbon is new aluminium with a maximum carbon footprint of 4.0 kg CO₂-eq per kg aluminium.

BES6001

Aluminium systems supplier to have certification to BES6001 (Responsible Sourcing) for design, manufacture and supply, including thermal break and finishing processes

6. **Security:** PAS 24 rating minimum
7. **Overpanel:** Glazed over panel or Louvre as L10/650 as shown on drawing 4201
8. **Ironmongery:** Pull handles 600mm x 25mm Stainless steel offset D handles (except plantroom doors which are to have pad handles)

480B Aluminium door to plantroom

1. **Description::** As Clause 480A
2. **Except:** Powder coated insulated panel in lieu of glazing

481 Aluminium Doorsets Performance

1. Aluminium door system to have been tested to meet Exposure Category shown below as defined by BS 6375:1 2004/2009, and to have achieved the following:

Doors (Commercial)
Air Permeability Water Penetration Wind Resistance Category

STII Doors
Single Pivot Door with low threshold
Class 3 Class 3A Class B5 1200 Pa

* exposure category varies with width/ height of window and mullion/ transom profile used as these are the only unsupported members. An accurate figure can be obtained from the relevant Euro Stormframe 202e, BS EN 1991-1-4 or BS6399:Part2: 1997 calculations and the profile inertia figures given in the relevant product manual.

482 Aluminium doorsets - glazing

1. **Correct glass thickness is always subject to calculation by the glazing sub-contractor.**

SBD Glazing requirements: Glazing at ground floor. Minimum P3A rating

Correct glass thickness is always subject to calculation by the glazing sub-contractor.

All glass within 800mm from FFL shall be toughened or laminated. (Below 1500mm if within a door or 300mm of a door)

All glass and glazing shall conform to:

EN 12600:2002 Specification for Impact Performance;
BS6262: Parts 1-7:2005 Code of Practice for Glazing Buildings;
BS952-1:1995 Glass for Glazing; Classification
BS EN 1279 Glass in Buildings. Insulating Glass Units.
Part 1:2004 Generalities, dimensional tolerances and rules for the system description
Part 2:2002 Long term test method and requirements for moisture penetration
Part 3:2002 Long term test method and requirements for gas leakage rate and for gas concentration tolerances
Part 4:2002 Methods of test for the physical attributes of edge seals
Part 5:2005+A2:2010 Evaluation of conformity
Part 6:2002 Factory production control and periodic tests

Manifestation design and location to be confirmed by the Contract Administrator

Recommendations of the Glass and Glazing Federation should be adhered to.

Notes:

Heatsoak Testing Procedure

We recommend Heatsoak testing is carried out where required in accordance with European Standard BS EN 14179: Heat soaked thermally toughened soda lime silicate safety glass. The emphasis of this procedure is to ensure that the glass temperature is maintained at $290^{\circ}\text{C} \pm 10^{\circ}\text{C}$ during the holding phase of the process cycle.

The heat soak process cycle consists of a heating phase, a holding phase and a cooling phase. In summary the procedure consists of controlled heating of a complete volume of toughened glass from ambient room temperature to a temperature of $290^{\circ}\text{C} \pm 10^{\circ}\text{C}$ and then holding it at that temperature for a minimum period of 2 hours before then allowing it to cool naturally. The procedure is carried out at various locations in audited ovens to ensure that the glass temperature during the holding phase is maintained at $290^{\circ}\text{C} \pm 10^{\circ}\text{C}$.

Thermal Safety Check Requirement

As with all laminated or non heat treated glass types we strongly recommend that a thermal safety check is carried out by via the Installer. This should be carried out where blinds, external brise soleil, ceiling trench or similar may trap warm air pockets close to the non heat treated pane. It is to be noted that laminated or non heat treated glass are not compatible in spandrel areas.

Full Height Barrier glazing must comply with; BS6180: 2011 Barriers in and about buildings. Code of practice.

Please note that if glazed areas form a full height barrier then the glass may need to comply with BS6180:2011 which means that the inner pane may be required to meet relevant POINT, U.D.L and LINE LOADS. If this is the case then the inner pane of glass would have to be a class A non break – either minimum 8.8mm thick or 10mm toughened or greater depending upon acoustic , wind loading , size & dimension of DGU's.

Louvres to be manufactured from extruded aluminium Grade AA 6063 to (BS EN 755-9:) Blades to be secured to an outer frame, to be retained within the door system by the glazing beads and gaskets

Louvre blades to be individually screwed in place or retained within extruded aluminium holders. The use of plastic retention clips will not be permitted.

Louvres to be finished to match door framing
Free area minimum 50%.

100% High Density Polyethylene (HDPE) insect mesh to rear of louvers. Nominal 4mm x 4mm mesh size. Colour black.

Solid infill panels must comply with Building Regulations Approved Document B.

485 Aluminium doorsets - finish

1. Aluminium sections to be polyester powder-coated to BS EN 12206-1:2004

Colour: RAL Colour to be confirmed by the contract's administrator.

All finishing to be undertaken by TECHNAL prior to delivery to fabricator.

Execution

710 Protection of components

1. **General:** Do not deliver to site components which cannot be put immediately into suitable clean, dry, floored and covered storage.
2. **Stored components:** Stack near vertical on level bearers, separated with spacers to prevent damage by and to projecting ironmongery, beads, etc.
3. Any protective films must be removed within 3 months of delivery.

730 Priming/ sealing

1. **Wood surfaces inaccessible after installation:** Primed or sealed as specified before fixing components.

740 Corrosion protection

1. **Surfaces to be protected:** Aluminium alloy components in contact with preservative treated timber
2. **Protective coating:** Two coats of bitumen solution to BS 6949 or an approved mastic-impregnated tape.
 - 2.1. **Timing of application:** Before fixing components.

750A Fixing doorsets

1. As section Z20 using appropriate fixings.

Door framing to be securely fixed direct to the building structure, no further than 150mm from each corner and at centres not exceeding 450mm, as laid down in procedures issued by the systems company.

Frames are to be positioned to cover the cavity within the reveals, and level with the existing external window line wherever possible, ensuring that they are plumb, level and without bow.

The Specialist Contractor shall allow for all necessary making good of all work disturbed.

The Specialist Contractor is to allow for necessary measures to protect the occupants, fittings and finishes within the rooms for the duration of the works.

Allow for making good work to window/door openings, both internally and externally, including masonry, plaster, cladding and decorative finishes to reveals. No additional allowance will be made for costs associated with making good which would be visible on a site inspection.

Upon completion of the installation of each replacement door, all glazing, handles and all other surfaces are to be cleaned with a mild detergent. All components are to be checked for security of

fixings, adequacy of clearances, adjustment of hinges, locks etc. as may be necessary to leave the door units in good working order.

760 Building in

1. **General:** Not permitted unless indicated on drawings.

766 Door installation - wood doorsets

1. The opening may be lined with softwood/ Hardwood for FD30 which shall be continuous and the minimum width of 70mm. Each door frame jamb to be fixed through to the wall at not less than four points with steel or nylon fixings at maximum 500mm centres penetrating the wall to at least 50mm. Doorsets shall be installed as stated in BS8214: 2008 Table.2.

767 Intumescent

1. Any gaps between the back of the door frame and the structural surround should be fire stopped by means of Mineral or Glass wool packed to a depth of at least 10mm from the face of frame and closed by a bead of intumescent paste or intumescent mastic, BS 8214:2008 COP Fire door assemblies with non metallic leaves.

770 Damp-proof courses associated with built-in wood frames

1. **Method of fixing:** To backs of frames using galvanized clout nails.

780 Damp-proof courses in prepared openings

1. **Location:** Correctly positioned in relation to door frames. Do not displace during fixing operations.

790 Fixing of wood frames

1. **Spacing of fixings (frames not predrilled):** Maximum 150 mm from ends of each jamb and at 600 mm maximum centres.

800 Fixing of loose thresholds

1. **Spacing of fixings:** Maximum 150 mm from each end and at 600 mm maximum centres.

809 Fire-resisting and smoke control doors/ door assemblies/ doorsets/ roller shutters and curtains – accredited installer

1. **Installation:** By a firm currently registered under a third-party-accredited fire door installer scheme in accordance with instructions supplied with the product conformity certificate, test report or engineering assessment.

810 Fire-resisting and smoke control doors/ door assemblies/ doorsets/ roller shutters and curtains – contractor installed

1. **Gaps between frames and supporting construction:** Filled as necessary in accordance with requirements for certification and/ or door/ doorset manufacturer's instructions.

811 Fire-resisting and smoke control doorsets, industrial, commercial and garage doors

1. **Installation:** By manufacturer or their approved installers, in accordance with requirements of BS EN 16034 and in conjunction with BS EN 13241, including the Declaration of Performance (DoP) certification for the UKCA/ UKNI/ CE marked doorset.

820 Sealant joints

1. **Sealant**

- 1.1. **Manufacturer:** Contractor choice
 - 1.1.1. **Product reference:** Contractor choice
 - 1.1.2. Seal all external joints between framing and structure with low modulus neutral cure silicone sealant to BS EN ISO 11600: 2003 + A1:2011. Sealant backing to be provided and detailed in drawings.
- 1.2. **Colour:** TBC with Architect
- 1.3. **Application:** As section Z22 to prepared joints. Triangular fillets finished to a flat or slightly convex profile.

830 Fixing ironmongery generally

- 1. **Fasteners:** Supplied by ironmongery manufacturer.
 - 1.1. **Finish/ corrosion resistance:** To match ironmongery.
- 2. **Holes for components:** No larger than required for satisfactory fit/ operation.
- 3. **Adjacent surfaces:** Undamaged.
- 4. **Moving parts:** Adjusted, lubricated and functioning correctly at completion.

830A Ironmongery generally - aluminium doorsets

- 1. Assemble and fix carefully and accurately using fastenings with matching finish supplied by ironmongery manufacturer. Prevent damage to ironmongery and adjacent surfaces. At completion check, adjust and lubricate as necessary to ensure correct functioning.

840 Fixing ironmongery to fire-resisting door assemblies

- 1. **General:** All items fixed in accordance with door leaf manufacturer's recommendations ensuring that integrity of the assembly, as established by testing, is not compromised.
- 2. **Holes for through fixings and components:** Accurately cut.
 - 2.1. **Clearances:** Not more than 8 mm unless protected by intumescent paste or similar.
 - 2.2. **Lock/ latch cases for fire doors requiring > 60 minutes integrity performance:** Coated with intumescent paint or paste before installation.

850 Location of hinges

- 1. **Primary hinges:** Where not specified otherwise, positioned with centre lines 250 mm from top and bottom of door leaf.
- 2. **Third hinge:** Where specified, positioned
- 3. **Hinges for fire-resisting doors:** Positioned in accordance with door leaf manufacturer's recommendations.

860 Installation of emergency exit devices

- 1. **Standard:** Unless specified otherwise, install panic bolts/ latches in accordance with BS EN 1125: 2008.

Unless stated otherwise, install emergency exit devices in accordance with BS EN 179: 2008

Ω End of Section

L30

Stairs/ ladders/ walkways/ handrails/ balustrades

Preliminary information/ requirements

115 Timber procurement

1. Timber (including timber for wood-based products): Obtained from well managed forests and/ or plantations in accordance with:
 - 1.1. The laws governing forest management in the producer country or countries.
 - 1.2. International agreements such as the Convention on International Trade in Endangered Species of wild fauna and flora (CITES).
2. Documentation: Provide either in accordance with chain of custody certification scheme requirements:
 - 2.1. Documentary evidence (which has been or can be independently verified) regarding the provenance of all timber supplied or
 - 2.2. Evidence that suppliers have adopted and are implementing a formal environmental purchasing policy for timber and wood-based products.
3. Chain of Custody Certification scheme:
 - 3.1. Other evidence:

130 Site dimensions

1. Procedure: Before starting work on designated items take site dimensions, record on shop drawings and use to ensure accurate fabrication.
 - 1.1. Designated items:

Components

410 Permanently-fixed vertical ladder systems Type A

1. Manufacturer: [Eurosafes Solutions Ltd](#) or Similar
 - 1.1. Contact details
 - 1.1.1. Address: Unit 51b
Dorehouse Industrial Estate
Orgreave Drive
Sheffield
South Yorkshire
S13 9NR
 - 1.1.2. Telephone: [+44 \(0\)114 2507411](tel:+441142507411)
 - 1.1.3. Web: www.eurosafesolutions.co.uk
 - 1.1.4. Email: sales@eurosafesolutions.co.uk
 - 1.2. Product reference: [Ascent Access Ladders](#)
2. Rungs: Required.
3. Platforms
 - 3.1. Material: Galvanized mild steel powder coated to RAL colour to match windows TBC
4. Safety cage:
5. Assembly connectors: Assembly fixings supplied.
6. Superstructure connectors:
7. Execution: Installation of ladder and stair systems generally.
8. Head layout:

9. Step down:
10. Base fixing:
11. Safety:
12. Width: 400 mm.
13. Height:
14. Length:
15. Other requirements: Fixing into structure through cladding to be co-ordinated with Innovare

Installation

620 Priming/ Sealing/ Painting

1. Surfaces inaccessible after assembly/installation: Before fixing components, apply full protective/decorative treatment/coating system.

630 Corrosion protection of dissimilar materials

1. Components/ substrates/ fasteners of dissimilar materials: Isolate using washers/ sleeves or other suitable means to separate materials to avoid corrosion and/ or staining.

640 Installation generally

1. Fasteners and methods of fixing: To section Z20.
2. Structural members: Do not modify, cut, notch or make holes in structural members, except as indicated on drawings.
3. Temporary support: Do not use stairs, walkways or balustrades as temporary support or strutting for other work.
4. Applied finishes: Substrates to be even, dry, sound and free from contaminants. Make good substrate surfaces and prepare/ prime as finish manufacturer's recommendation before application.

Completion

920 Documentation

1. Contents
 - 1.1. Copies of structural design calculations/ test reports.
 - 1.2. General product information.
 - 1.3. Installation information.
 - 1.4. Inspection and maintenance reports.
2. Number of copies:
3. Submission:

Ω End of Section

M10

Cement based levelling/ wearing screeds

Types of screed

130 Proprietary quick drying levelling screeds

1. Manufacturer: [Flowcrete – a brand of CPG UK Ltd](#)
 2. Contact details
 - 2.1. Address: The Flooring Technology Centre
Booth Lane
Moston
Sandbach
Cheshire
CW11 3QF
 - 2.2. Telephone: [+44 \(0\)1942 251400](tel:+44(0)1942251400)
 - 2.3. Web: www.flowcrete.co.uk
 - 2.4. Email: flooringuk@cpg-europe.com
 3. Product reference: [Isocrete K-Screed - Standard](#)
 4. Certification: BBA Certificate: 91/2678.
 5. CompressiveStrength: >25 N/mm².
 6. Cure speed: Working time: 2 hours , Light traffic: 24 hours, Full traffic: 7 days, Curing under polythene: 7 days.
 7. Density: 1800–2000 kg/m³.
 8. Suitable environment: High traffic areas such as airports, shopping centres and hospitals where a high strength (Category A) screed is required.
 9. Thickness options: 75 mm.
Minimum 35mm cover over pipes
 10. Weight: 135 kg/m².
 11. Insulation : As Clause 291
 12. Mix
 - 12.1. Cement: Standards as clause 302.
 - 12.2. Aggregates: Standards as clause 305.
 - 12.3. Proportions: To manufacturer's recommendations.
 13. In situ crushing resistance (ISCR) category: A/B. (3/4 mm maximum indentation)
 - 13.1. Mass of test weight: 4kg
 14. Flatness/ Surface regularity class: Maximum permissible deviation: SR2 generally
 15. Finish: Trowelled as clause 540.
 - 15.1. To receive: Vinyl/ carpet/ resin floor finish
 16. Other requirements: Lay the screed in two layers, the bottom layer being a little wetter than usual to ensure full compaction around the heating pipes. Separation joints should be formed against walls and columns by using strips of compressible material. Where a thin floor finish is to be installed, overlay with a suitable thin section smoothing underlayment.
- Movement joints to be coordinated with underfloor heating and screed joints**

131 Proprietary quick drying levelling screed to Drama Block

1. Manufacturer: [Flowcrete – a brand of CPG UK Ltd](#)
2. Contact details

- 2.1. Address: The Flooring Technology Centre
Booth Lane
Moston
Sandbach
Cheshire
CW11 3QF
- 2.2. Telephone: +44 (0)1942 251400
- 2.3. Web: www.flowcrete.co.uk
- 2.4. Email: flooringuk@cpg-europe.com
3. Product reference: [Isocrete K-Screed - Standard](#)
4. Certification: BBA Certificate: 91/2678.
5. CompressiveStrength: >25 N/mm².
6. Cure speed: Working time: 2 hours , Light traffic: 24 hours, Full traffic: 7 days, Curing under polythene: 7 days.
7. Density: 1800–2000 kg/m³.
8. Suitable environment: High traffic areas such as airports, shopping centres and hospitals where a high strength (Category A) screed is required.
9. Thickness options: 75 mm.
10. Weight: 135 kg/m².
11. Insulation: As Clause 291, thickness to match existing
12. Mix
 - 12.1. Cement: Standards as clause 302.
 - 12.2. Aggregates: Standards as clause 305.
 - 12.3. Proportions: To manufacturer's recommendations.
13. In situ crushing resistance (ISCR) category: A/B. (3/4 mm maximum indentation)
 - 13.1. Mass of test weight: 4kg
14. Flatness/ Surface regularity class: Maximum permissible deviation: SR2 generally
15. Finish: Trowelled as clause 540.
 - 15.1. To receive: Vinyl/ carpet/ resin floor finish
16. Other requirements: Lay the screed in two layers, the bottom layer being a little wetter than usual to ensure full compaction around the heating pipes. Separation joints should be formed against walls and columns by using strips of compressible material. Where a thin floor finish is to be installed, overlay with a suitable thin section smoothing underlayment.

Generally/ preparation

205 Design life of screeds

1. Duration: The Life of the building
 - 1.1. Subject to reasonable wear and tear.
2. Location:
3. Condition of use: Subject to correct loading and traffic usage throughout duration.

210 Suitability of substrates

1. General
 - 1.1. Suitable for specified levels and flatness/ regularity of finished surfaces. Consider permissible minimum and maximum thicknesses of screeds.
 - 1.2. Sound and free from significant cracks and gaps.
2. Concrete strength: In accordance with BS 8204-1, Table 2.

3. **Cleanliness:** Remove plaster, debris and dirt.
4. **Moisture content:** To suit screed type. New concrete slabs to receive fully or partially bonded construction must be dried out by exposure to the air for minimum six weeks.

215 Surface hardness of substrates to receive polymer modified wearing screeds

1. **General:** Substrates must restrain stresses that occur during setting and hardening of wearing screeds.
2. **Test for surface hardness:** To BS EN 12504-2 using a rebound hammer with compliance values as follows:
 - 2.1. **Rebound hammer value (minimum)**
 - 2.1.1. **Screed thickness 15 mm or less:** 25.
 - 2.1.2. **Screed thickness greater than 15 mm:** 30.
3. **Report:** Submit details of areas where substrate surface hardness does not comply with these values.

220 Proprietary levelling/wearing screeds

1. **General:** Materials, mix proportions, mixing methods, minimum/ maximum thicknesses and workmanship must be in accordance with recommendations of screed manufacturer.
2. **Standard:**

230 Control samples

1. **General:** Complete areas of finished work and obtain approval of appearance before proceeding.
2. **Screed type:**
 - 2.1. **Location/ Size:**

250 Conduits under floating screeds

1. **Haunching:** Before laying insulation for floating screeds, haunch up in 1:4 cement:sand on both sides of conduits.

251 Conduits cast into or under screeds

1. **Reinforcement:** Overlay with reinforcement selected from:
 - 1.1. 500 mm wide strip of steel fabric to BS 4483, reference D49, or
 - 1.2. Welded mesh manufactured in rolls from mild steel wire minimum 1.5 mm diameter to BS 1052, mesh size 50 x 50 mm.
2. **Placing reinforcement:** Mid depth between top of conduit and the screed surface.
 - 2.1. **Width of reinforcement (minimum):** 300 mm.
3. **Screed cover over conduit (minimum):**

255 Pipe ducts/ trunking

1. **Preformed access ducts:** Before laying screed, fix securely to substrates and level accurately in relation to finished floor surface.

280 Unbonded construction

1. **Separation:** Lay screed over a suitable sheet dpm or a separating layer.
 - 1.1. **Type:**
2. **Installation of separating layer:** Lay on clean substrate. Turn up for full depth of screed at abutments with walls, columns, etc. Lap 100 mm at joints.

291 Floating construction ground floor

1. Insulation:
Type: 125mm Unilin XT/HYF to achieve minimum u value of **0.12 W/m²K**.
Installation: Lay with tight butt joints. Continue up at perimeter abutments for full depth of screed.

Edge strips at perimeter abutments:

Type: 8 mm PE foam strips with PE skirt to all internal surfaces.

Installation: To full depth of screed at perimeter abutments.

Separating layer:

Type: 500 gauge polythene sheet

Installation: Lay over insulation and turn up at perimeter abutments. Lap 100 mm at joints.

To be reviewed against underfloor heating system requirements

Batching/ mixing

302 Cements

1. Cement types: In accordance with BS 8204-1, clause 5.1.3.

305 Aggregates

1. Sand: To BS EN 13139.
 - 1.1. Grading limits: In accordance with BS 8204-1, Table B.1.
2. Coarse aggregates for fine concrete levelling screeds
 - 2.1. Standard: To BS EN 12620.
 - 2.2. Designation: 4/10.
3. Lightweight aggregates: In accordance with BS 8204-1, Annex A.

306 Proprietary polymer modified screeds

1. Cement types: In accordance with BS 8204-3.
2. Sand: To BS EN 13139:
 - 2.1. Grading limits:
3. Aggregates: In accordance with BS 8204-3.

307 Admixtures

1. Standard: In accordance with BS 8204-1, Table 1.
2. Calcium chloride: Do not use in admixtures.

310 Batching with dense aggregates

1. Mix proportions: Specified by weight.
2. Batching: Select from:
 - 2.1. Batch by weight.
 - 2.2. Batch by volume: Permitted on the basis of previously established weight:volume relationships of the particular materials. Use accurate gauge boxes. Allow for bulking of damp sand.

330 Mixing

1. Water content: Minimum necessary to achieve full compaction, low enough to prevent excessive water being brought to surface during compaction.
2. Mixing: Mix materials thoroughly to uniform consistency. Mixes other than no-fines must be mixed in a suitable forced action mechanical mixer. Do not use a free fall drum type mixer.

3. **Consistency:** Use while sufficiently plastic for full compaction.
4. **Ready-mixed retarded screed mortar:** Use within working time and site temperatures recommended by manufacturer. Do not retemper.

335 In situ crushing resistance (ISCR)

1. **Standards and category:** In accordance with BS 8204-1, table 4.
 - 1.1. **Testing of bonded and unbonded screeds:** To Annex D.
 - 1.2. **Testing of floating levelling screeds:** To Annex E.

340 Adverse weather

1. **Screeds surface temperature:** Maintain above 5°C for a minimum of four days after laying.
2. **Hot weather:** Prevent premature setting or drying out.

Laying

345 Level of screed surfaces

1. **Permissible deviation:** (allowing for thickness of coverings) ± 7 mm from datum.

350 Screeding to falls

1. **Minimum screed cover:** Maintain at the lowest point.
2. **Falls:** Gradual and consistent.
 - 2.1. **Gradient (minimum):**

355 Flatness/ Surface regularity of floor screeds

1. **Standard:** In accordance with BS 8204-1, Table 5.
2. **Test:** In accordance with BS 8204-1, Annex C.
3. **Sudden irregularities:** Not permitted.

375 Compaction of screeds

1. **General:** Compact thoroughly over entire area.
2. **Screeds over 50 mm thick:** Lay in two layers of approximately equal thickness. Roughen surface of compacted lower layer then immediately lay upper layer.

382 Stair screeds

1. **Construction:** Fully bonded to treads, risers and landings.
2. **Risers:** Form using fine finish formwork.
3. **Wearing screed surfaces:** Make good with compatible cement:sand mix. Wood float. When hardened remove laitance.

392 General reinforcement

1. **Steel fabric:** To BS 4483.
 - 1.1. **Type:**
2. **Installation:** In accordance with BS 8204-1.

395 Strip reinforcement

1. **Location:**
2. **Steel fabric:** To BS 4483.
 - 2.1. **Type**

2.2. Width of strips:

3. Installation: In accordance with BS 8204-1.

405 Joints in levelling screeds generally

1. Laying screeds: Lay continuously using 'wet screeds' between strips or bays. Minimize defined joints.
2. Daywork joints: Form with vertical edge.

406 Bay joints in levelling screeds

1. Screed type: As Clause 130A
2. Bay sizes
 - 2.1. Area (maximum): to manufacture's recommendations
 - 2.2. Length (maximum): to manufacture's recommendations
3. Location of bay joints: Coordinate with those required for substrate slab and floor covering.

415 Bay joints in bonded wearing screeds

1. Bay sizes
 - 1.1. Area (maximum):
 - 1.2. Length: Breadth ratio (maximum): 3:2.
2. Location of bay joints:

428 Heated screeds

1. Substrate slab: Irregularities not permitted.
2. Screed bays, movement joints and other joints: Coordinate with heating circuits.
3. Heating elements: Secure properly. Prevent displacement.
4. Screed laying: Compact thoroughly around heating elements. Do not damage them.

440 Crack inducing grooves in levelling screeds

1. Groove depth: At least half the depth of screed.
2. Cutting grooves: Straight, vertical and accurately positioned. Select from the following:
 - 2.1. Trowel cut as screed is laid.
 - 2.2. Saw cut sufficiently early after laying to prevent random cracking.

445 Crack inducing grooves in wearing screeds

1. Groove dimensions
 - 1.1. Depth: At least half the depth of wearing screed.
 - 1.2. Width:
2. Cutting grooves: Straight, vertical and accurately positioned. Saw cut sufficiently early after laying to prevent random cracking.

450 Sealant for sawn joints in wearing screeds

1. Type: to manufacture's recommendations
2. Preparation and application: As section Z22.

452 Sealing strips for sawn joints in wearing screeds

1. Manufacturer: to manufacture's recommendations
 - 1.1. Product reference:

- 1.2. Capping colour:

Finishing/curing

510 Finishing generally

1. **Timing:** Carry out all finishing operations at optimum times in relation to setting and hardening of screed material.
2. **Prohibited treatments to screed surfaces**
 - 2.1. Wetting to assist surface working.
 - 2.2. Sprinkling cement.

520 Wood floated finish

1. **Finish:** Slightly coarse, even texture with no ridges or steps.

530 Smooth floated finish

1. **Finish:** Even texture with no ridges or steps.

540 Trowelled finish to levelling screeds

1. **Floating:** To an even texture with no ridges or steps.
2. **Trowelling:** To a uniform, smooth but not polished surface, free from trowel marks and other blemishes, and suitable to receive specified flooring material.

550 Trowelled finish to wearing screeds

1. **Floating:** To an even texture with no ridges or steps.
2. **Trowelling:** Successively trowel at intervals, applying sufficient pressure to close surface and give a uniform smooth finish free from trowel marks and other blemishes.

560 Dewatered trowelled finish to wearing screeds

1. **Dewatering:** Immediately after compaction of wearing screeds, remove water using a vacuum process.
2. **Floating:** Without delay, power float to an even texture with no ridges or steps.
3. **Trowelling:** Successively trowel at intervals, applying sufficient pressure to close surface and give a uniform smooth finish free from trowel marks and other blemishes.

650 Curing

1. **General:** Prevent premature drying. Immediately after laying, protect surface from wind, draughts and strong sunlight. As soon as screed has set sufficiently, closely cover with polyethylene sheeting.
2. **Curing period (minimum):** Keep polyethylene sheeting in position for:
3. **Drying after curing:** Allow screeds to dry gradually. Do not subject screeds to artificial drying conditions that will cause cracking or other shrinkage related problems.

Ω End of Section

M50

Rubber/ plastics/ cork/ lino/ carpet tiling/ sheeting

Types of covering

110 Plastics sheet wall claddings Type A

1. Manufacturer: [Altro](#)
 - 1.1. Contact details
 - 1.1.1. Address: Works Road
Letchworth Garden City
Hertfordshire
SG6 1NW
 - 1.1.2. Telephone: [+44 \(0\)1462 480480](tel:+441462480480)
 - 1.1.3. Web: <https://www.altro.com/uk>
 - 1.1.4. Email: enquiries@altro.com
 - 1.2. Product reference: [Altro Whiterock Satins™](#)
2. Material: Virgin polyvinyl chloride (PVC-U).
3. Thickness: 2.5 mm.
4. Sheet size: 1220 x 3000 mm.
5. Finish: Satin.
6. Colour: White generally + feature colour to splashback
7. Fire performance: To EN 13501-1, class B-s3,d0; to BS 476-6, class 0; to BS 476-7, class 1.
8. Accessories:
9. Execution:

130 Pile carpet tiles

1. Carpet tiles
 - 1.1. Manufacturer: CFS
 - 1.1.1. Contact details
 - 1.1.1.1. Address: CFS Carpets & Furniture Store Ltd is Unit 5 Highgate Business Centre, Highgate Road, Birmingham, West Midlands, B12 8EA
 - 1.1.1.2. Telephone: [+44 \(0\)1625 428922](tel:+441625428922)
 - 1.1.1.3. Email: sales@cfs-ltd.com
 - 1.1.2. Product reference: New Typhoon carpet tile
 - 1.2. Style: durable, fine-rib, fibre-bonded commercial carpet tile
 - 1.3. Material: 100% Polypropylene fibre
 - 1.4. Classification
 - 1.4.1. Standard: To BS EN 1307.
 - 1.4.2. Level of use class: Classification: Heavy Commercial.
 - 1.5. Size: 500 x 500 mm.
 - 1.6. Colour and pattern: Smoke
 - 1.7. Total carpet thickness: nominal thickness 7.8–8.5 mm
 - 1.8. Primary backing: Non-woven.
 - 1.9. Secondary backing: Bitumen.
2. Other requirements: Laid and bonded using Acrylic Emulsion Styccobond F41 adhesive by F.Ball & Co Ltd.

158 Particle-based enhanced slip-resistant polyvinyl chloride (PVC) sheets

1. Manufacturer: [Forbo Flooring Systems](#)
 - 1.1. Contact details
 - 1.1.1. Address: PO Box 1
High Holborn Road
Ripley
Derbyshire
DE5 3NT
 - 1.1.2. Telephone: [+44 \(0\)800 093 5258](tel:+4418000935258)
 - 1.1.3. Web: <https://www.forbo-flooring.co.uk>
 - 1.1.4. Email: info.flooring.uk@forbo.com
 - 1.2. Product reference: [Surestep Original](#)
2. Standard: To BS EN ISO 10582 and BS EN 13845 & 14041.
3. Use class: Class 34 Commercial very heavy, Class 43 light industrial.
4. Slip potential
 - 4.1. Slip resistance value (SRV) (minimum)/ Pendulum test value (PTV) (minimum): DIN 51130, R10; BS EN 13893, DS ≥ 0.30 ; PTV Wet slider 96: ≥ 36 .
 - 4.2. Surface roughness (Rz) (minimum): ≥ 20 .
5. Recycled content: 20%.
6. Width: 2 m.
7. Thickness: Total: 2.0 mm; Wear layer: 0.7 mm.
8. Colour and pattern: 171032 smoke.
9. Dimension stability (to EN-ISO 23999): $<0.1\%$.
10. Residual indentation (to EN-ISO 24343-1): ≤ 0.05 mm.
11. Electrical resistance (to EN 1081 R1): $>1 \times 10^9 \Omega$.
12. Reaction to fire (to BS EN 13501): Bfl-s1.
13. Body voltage (to BS EN 1815): <2 kV.
14. Thermal conductivity (to BS EN 12524): 0.25 W/m·K.
15. Adhesive:
16. Fabricated underlay: Not required.
17. Castor chair continuous use (to EN 425): Pass.
18. Use in wet areas (to EN 13553): Yes.
19. Total weight (to EN ISO 23997): 2.75 kg/m².
20. Light fastness (to BS EN ISO 105 B02): ≥ 6 .
21. (to EN 13845 Annex C): Esf.
22. EN 13845 Annex D 50.000 revolutions $<10\%$ loss: Pass.
23. Resistance to chemicals (to EN ISO 26987): Very good.
24. Edging: Not required.
25. Skirtings: Coved/ timber as 15 series

General requirements

210 Workmanship generally

1. Base condition after preparation: All bases must be rigid, dry, sound, smooth and free from grease, dirt and other contaminants before coverings are applied

2. **Finished coverings:** Accurately fitted, tightly jointed, securely bonded, smooth and free from air bubbles, rippling, adhesive marks and stains
3. **Operatives:** Appropriately skilled and experienced for the type of work

220 Samples

1. **Covering samples:** Submit samples representative of each floor type prior to placing orders. Ensure that delivered materials match samples.

230 Control samples

1. **General:** Complete areas of finished work in approved locations as follows, and obtain approval of appearance before proceeding
 - 1.1. **Covering type:** Each covering type
 - 1.2. **Location:** To be agreed

250 Layout – sheet materials

1. **General:** Agree setting out of seams before ordering roll materials for sheeting types

255 Layout – tiles

1. **General:** Agree setting out of seams before ordering materials for covering types

270 Extra material

1. **Requirement:** Provide extra material for each type of floor covering installed
 - 1.1. **Quantity:** 5%
 - 1.2. **Timing:** Completion

330 Commencement

1. **Required condition of works prior to laying floor-covering materials**
 - 1.1. Building is weathertight and well dried out.
 - 1.2. Wet trades have finished work.
 - 1.3. Paintwork is finished and dry.
 - 1.4. Conflicting overhead work is complete.
 - 1.5. Floor service outlets, duct covers and other fixtures around which materials are to be cut are fixed.
 - 1.6. Provisions have been made to allow for ventilation during and after laying of materials.
2. **Notification:** Submit not less than 48 hours before commencing laying

340 Conditioning

1. **Prior to laying:** Condition materials in accordance with manufacturer's instructions.
2. **Conditioning time and temperature (minimum):** As recommended by manufacturer

350 Environmental conditions

1. **Temperature and humidity:** As per manufacturer's instructions
2. **Ventilation:** Maintain adequate provision before, during and after laying the floor covering

360 Floors with underfloor heating

1. **Commencement of laying:** Not before a period of 48 hours after heating has been turned off
2. **Post-laying start up of heating system:** Slowly return heating to its operative temperature not less than 48 hours after completing laying

- 2.1. Maximum temperature: As per manufacturer's instructions.

Preparing bases

410 New bases

1. **General:** Laying of coverings will be taken as joint acceptance by the Main Contractor and Subcontractor of the suitability of the bases and conditions within any given area.

420 Existing bases

1. **Notification:** Before commencing work, the contractor is to confirm that existing bases will, after preparation, be suitable to receive coverings.

430 New wet-laid bases

1. **Artificial base drying aids:** Not used for at least four days prior to moisture content testing
2. **Base moisture content test:** In accordance with [BS 5325](#), annex B or In accordance with [BS 8203](#), annex B
- 2.1. **Locations for readings:** in all corners, along edges, and at various points over the area being tested.
3. **Commencement of laying coverings:** Not until all readings show 75% relative humidity or less

440 Substrates to receive thin coverings

1. The specifications for trowelled finishes to receive thin floor coverings require:
- 1.1. A uniform, smooth surface free from trowel marks and other blemishes, and suitable to receive the specified floor finish material.
- 1.2. Adequate protection from construction traffic.
- 1.3. Allowance for making good by application of a smoothing compound by and to the satisfaction of the flooring subcontractor in the event of the surface being unsuitable due to inadequate finishing or protection.

460 Smoothing and levelling underlay compounds

1. **Manufacturer:** [Ardex UK Ltd](#)
- 1.1. **Contact details**
- 1.1.1. **Address:** Homefield Road
Haverhill
Suffolk
United Kingdom
CB9 8QP
- 1.1.2. **Telephone:** [+44 \(0\)1440 714939](tel:+441440714939)
- 1.1.3. **Web:** <https://ardex.co.uk>
- 1.1.4. **Email:** info@ardex.co.uk
- 1.2. **Product reference:** Arditec 'NA' smoothing compound.
- 1.3. Apply to base at a minimum thickness of 3mm in accordance with manufacturer's product data sheet incorporating aggregate for applications more than 12mm thick.
- 1.4. Allow to dry before laying floor tiling/sheeting

470 Bases from which existing floor coverings have been removed

1. **Substrate:** remove all traces of existing adhesive in accordance with new flooring recommendations

Laying coverings

610 Setting out tiles

1. **Method:** Set out from centre of area/ room, so that wherever possible tiles along opposite edges are of equal size and edge tiles are more than 50% of full tile width.
 - 1.1. **Requirement:** To be agreed with Architect

620 Colour consistency

1. **Finished work in any one area/ room:** Flooring free from banding or patchiness
 - 1.1. **Requirement:** Use material from the same batch by the same manufacturer

640 Adhesive fixing generally

1. **Primer:** Use a primer where recommended by adhesive manufacturer. Allow to dry thoroughly before applying adhesive.
 - 1.1. **Application:** As per manufacturer's instructions
2. **Adhesive**
 - 2.1. **Type:** As recommended by floor-covering manufacturer
 - 2.2. **Application:** Spread adhesive evenly and lay covering, pressing down firmly and rolling laterally and transversely (if recommended) to ensure full contact and a good bond overall. Reroll (if recommended) within 30 minutes.

Remove all surplus adhesive from exposed faces of coverings as the work proceeds.

Trowel ridges and high spots caused by particles on the substrate will not be accepted.

3. **Laying of floor covering:** In accordance with manufacturer's instructions
 - 3.1. **Fixtures:** Floor-covering material to be cut around all permanent fixtures

650 Seams

1. **Requirement:** In accordance with [BS 5325](#).
2. **Joints:** As per manufacturer's instructions

670 Laying borders and design strips

1. **Preparation:** Cut strips along the length of the sheet to prevent curl.
2. **Laying:**
 - 2.1. **Curl:** Not acceptable
 - 2.2. **Corners:** Mitre joints at corners

680 Seam welding – resilient floor coverings

1. **Floor covering:**
 - 1.1. **Method:**
2. **Commencement:** Do not commence welding of coverings until a minimum of 24 hours after laying or until adhesive has completely set.
3. **Welding:** Hot-weld using hot air welding gun (fitted with high speed welding nozzle) and Altro welding rod. Select colour of rod to match floor covering. Cut off surplus weld rod with spatula. Do not chemical weld.

Form a neat, smooth, strongly bonded joint, flush with finished surface.

- 3.1. **Trimming:** Cut groove, 3 mm wide by 2/3 the depth of material, evenly along each joint using:

- Altro hand grooving tool or
- Automatic grooving machine fitted with diamond blade.

740 Floor cover strips

1. Manufacturer: [Gradus](#)
 - 1.1. Contact details
 - 1.1.1. Address: Chapel Mill
Park Green
Macclesfield
Cheshire
SK11 7LZ
 - 1.1.2. Telephone: [+44 \(0\)1625 428922](tel:+44(0)1625428922)
 - 1.1.3. Web: <https://www.gradus.com>
 - 1.1.4. Email: imail@gradus.com
 - 1.2. Product reference: TH383 Threshold trims between carpet/ vinyl
2. Properties
 - 2.1. Material: Aluminium
 - 2.2. Height difference: To suit changes in level
3. Fixing: As per manufacturer's instructions. Secure with edge of covering gripped. Use matching fasteners where exposed to view.
 - 3.1. Requirement: After laying the floor covering

745 Laying of floor coverings on stairs

1. Neatly cut and tightly fit risers first, then treads
2. Fit tread and riser in one piece on stairs with coved tread/riser.
3. Cove former: Altro Type 20R. Securely bond to base and background with a suitable adhesive.

780 Trafficking after laying

1. Covering types: All
2. Traffic-free period: Until adhesive is set

Completion

861 Slip resistance testing

1. Testing authority: An approved laboratory
2. Floor covering(s) to be tested: All resilient floorings
3. Test: To [BS EN 16165](#)
 - 3.1. Floor covering condition: Wet

880 Leftover material

1. Spare covering material: Extra material may be needed for repairs and should therefore be handed over to the employer on completion of the project is wanted.

Waste should be disposed off safely. Refer to instructions on the material safety data sheet.

Ω End of Section

M60

Painting/clear finishing

Coating systems

110 Emulsion paint - new plasterboard

1. Manufacturer: [Dulux Trade, brand of AkzoNobel](#)
 - 1.1. Contact details
 - 1.1.1. Address: AkzoNobel Decorative Paints
Wexham Road
Slough
Berkshire
SL2 5DS
 - 1.1.2. Telephone: [+44 \(0\)333 222 7070](tel:+44(0)3332227070)
 - 1.1.3. Web: www.duluxtradepaintexpert.co.uk
 - 1.1.4. Email: project.support@akzonobel.com
 - 1.2. Product reference: Diamond Matt
2. Composition: Acrylic copolymer.
3. Sheen: Matt.
4. Colour: Colours to confirmed with school
5. Coverage: 5 m²/L (approx.).
6. Drying time: Recoatable after eight hours.
7. Volume solids: Medium (8-24.99%).
8. VOC content: Low.
9. Application method: Brush or roller.
10. Form: Liquid.
11. Capacity:
12. Surfaces: New internal plasterboard walls and ceilings
 - 12.1. Preparation: Ensure that all surfaces are dry, clean and free from debris. Follow manufacturer's recommendations.
13. Prime: One coat of Dulux Trade Alkali Resisting Primer
14. Initial coats: coat of thinned coat of Dulux Trade Diamond Matt
15. Finishing coats: 2 coats of Dulux trade Diamond Matt.
16. Spot prime any bare metal , metal fixing nail heads etc... with One coat of Dulux Trade Metalshield Quick Drying Metal Primer

111 Water-based finishing coats

1. Manufacturer: [Dulux Trade, brand of AkzoNobel](#)
 - 1.1. Contact details
 - 1.1.1. Address: AkzoNobel Decorative Paints
Wexham Road
Slough
Berkshire
United Kingdom
SL2 5DS
 - 1.1.2. Telephone: [+44 \(0\)333 222 7070](tel:+44(0)3332227070)
 - 1.1.3. Web: <https://www.duluxtradepaintexpert.co.uk/en>

- 1.1.4. Email: project.support@akzonobel.com
- 1.2. Product reference: [Dulux Trade Diamond Eggshell](#).
- 2. Composition: Acrylic copolymer.
- 3. Sheen: Eggshell.
- 4. Colour:
- 5. System code:
- 6. Capacity:
- 7. Form: Liquid.

112 Water-based finishing coats

- 1. Manufacturer: [Dulux Trade, brand of AkzoNobel](#)
 - 1.1. Contact details
 - 1.1.1. Address: AkzoNobel Decorative Paints
Wexham Road
Slough
Berkshire
SL2 5DS
 - 1.1.2. Telephone: [+44 \(0\)333 222 7070](tel:+44(0)3332227070)
 - 1.1.3. Web: www.duluxtradepaintexpert.co.uk
 - 1.1.4. Email: project.support@akzonobel.com
 - 1.2. Product reference: Diamond Satinwood
- 2. Composition: Acrylic copolymer.
- 3. Sheen: Satinwood
- 4. Colour:
- 5. Execution: Applying coating.
- 6. System code:
- 7. Form: Liquid.
- 8. Capacity: 5 L.
- 9. Sheen: Mid-sheen.
- 10. Prime: to manufacturers recommendations
- 11. Finishing coats: 2 coats of Dulux Trade Sterishield Diamond Satinwood of selected shade.

180 Concrete floor paints Type A

- 1. Manufacturer: [Dulux Trade, brand of AkzoNobel](#)
 - 1.1. Contact details
 - 1.1.1. Address: AkzoNobel Decorative Paints
Wexham Road
Slough
Berkshire
United Kingdom
SL2 5DS
 - 1.1.2. Telephone: [+44 \(0\)333 222 7070](tel:+44(0)3332227070)
 - 1.1.3. Web: <https://www.duluxtradepaintexpert.co.uk/en>
 - 1.1.4. Email: project.support@akzonobel.com
 - 1.2. Product reference: [High Performance Floor Paint](#)
- 2. Colour:
- 3. System code:

4. Form: Liquid.
5. Capacity: 5 L.
6. Composition: Epoxy resin.
7. Sheen: Mid-sheen.
8. Viscosity: 3.95 cm²/s.
9. Coverage: Up to 6 m²/L.

Generally

210 Coating materials

1. Manufacturers: Obtain materials from any of the following:
2. AKZO NOBEL DECORATIVE COATINGS LIMITED.
3. Selected manufacturers: Submit names before commencement of coating work.
4. Paint specification: Refer to finishes schedule 6511 for product

215 Handling and storage

1. Coating materials: Deliver in sealed containers, labelled clearly with brand name, type of material and manufacturer's batch number.
2. Materials from more than one batch: Store separately. Allocate to distinct parts or areas of the work.

220 Compatibility

1. Coating materials selected by contractor
 - 1.1. Recommended by their manufacturers for the particular surface and conditions of exposure.
 - 1.2. Compatible with each other.
 - 1.3. Compatible with and not inhibiting performance of preservative/fire-retardant pretreatments.

280 Protection

1. 'Wet paint' signs and barriers: Provide where necessary to protect other operatives and general public, and to prevent damage to freshly applied coatings.

300 Control samples

1. Sample areas of finished work: Carry out, including preparation, as follows:
2. Types of coating Location
 - 2.1. All
3. Approval of appearance: Obtain before commencement of general coating work.

320 Inspection by coating manufacturers

1. General: Permit manufacturers to inspect work in progress and take samples of their materials from site if requested.

321 Inspection of work stages

1. Programme for inspections: Submit as follows:
2. Types of coating Inspection at completion of
3. M60/
4. Inspection: Give prior notice when each stage is ready for inspection.

Preparation

400 Preparation generally

1. **Standard:** In accordance with BS 6150.
2. **Refer to any pre-existing CDM Health and Safety File.**
3. **Refer to CDM Construction Phase Plan where applicable.**
4. **Suspected existing hazardous materials:** Prepare risk assessments and method statements covering operations, disposal of waste, containment and reoccupation, and obtain approval before commencing work.
5. **Preparation materials:** Types recommended by their manufacturers and the coating manufacturer for the situation and surfaces being prepared.
6. **Substrates:** Sufficiently dry in depth to suit coating.
7. **Efflorescence salts:** Remove.
8. **Dirt, grease and oil:** Remove. Give notice if contamination of surfaces/ substrates has occurred.
9. **Surface irregularities:** Remove.
10. **Joints, cracks, holes and other depressions:** Fill flush with surface, to provide smooth finish.
11. **Dust, particles and residues from preparation:** Remove and dispose of safely.
12. **Water based stoppers and fillers**
 - 12.1. Apply before priming unless recommended otherwise by manufacturer.
 - 12.2. If applied after priming: Patch prime.
13. **Oil based stoppers and fillers:** Apply after priming.
14. **Doors, opening windows and other moving parts**
 - 14.1. Ease, if necessary, before coating.
 - 14.2. Prime resulting bare areas.

420 Fixtures and fittings

1. **Removal:** Before commencing work remove: surface mounted fittings.
2. **Replacement:** Refurbish as necessary, refit when coating is dry.

425 Ironmongery

1. **Removal:** Before commencing work: Remove ironmongery from surfaces to be coated.
2. **Hinges:**
3. **Replacement:** Refurbishment as necessary; refit when coating is dry.

430 Existing ironmongery

1. **Refurbishment:** Remove old coating marks. Clean and polish.

440 Previously coated surfaces generally

1. **Preparation:** In accordance with BS 6150, clause 11.5.
2. **Contaminated or hazardous surfaces:** Give notice of:
 - 2.1. Coatings suspected of containing lead.
 - 2.2. Substrates suspected of containing asbestos or other hazardous materials.
 - 2.3. Significant rot, corrosion or other degradation of substrates.
3. **Suspected existing hazardous materials:** Prepare risk assessments and method statements covering operations, disposal of waste, containment and reoccupation, and obtain approval before commencing work.

4. **Removing coatings:** Do not damage substrate and adjacent surfaces or adversely affect subsequent coatings.
5. **Loose, flaking or otherwise defective areas:** Carefully remove to a firm edge.
6. **Alkali affected coatings:** Completely remove.
7. **Retained coatings**
 - 7.1. Thoroughly clean to remove dirt, grease and contaminants.
 - 7.2. **Gloss-coated surfaces:** Provide key.
8. **Partly removed coatings**
 - 8.1. **Additional preparatory coats:** Apply to restore original coating thicknesses.
 - 8.2. **Junctions:** Provide flush surface.
9. **Completely stripped surfaces:** Prepare as for uncoated surfaces.

451 Previously coated surfaces – blast cleaning

1. **Operatives**
 - 1.1. Trained/ experienced in blast cleaning.
 - 1.2. Submit evidence of training/ experience on request.
2. **Dust and nuisance:** Minimize.

456 Previously coated surfaces – burning off

1. **Risk assessment and method statement:** Prepare, and obtain approval before commencing work.
2. **Adjacent areas:** Protect from excessive heat and falling scrapings.
3. **Exposed resinous areas and knots:** Apply two coats of knotting.
4. **Removed coatings:** Dispose of safely.

461 Previously coated wood

1. **Degraded or weathered surface wood:** Take back to provide suitable substrate.
2. **Degraded substrate wood:** Repair with sound material of same species.
3. **Exposed resinous areas and knots:** Apply two coats of knotting.

471 Preprimed wood

1. **Areas of defective primer:** Take back to bare wood and reprime.

481 Uncoated wood

1. **General:** Provide smooth, even finish with arrises and moulding edges lightly rounded or eased.
2. **Heads of fasteners:** Countersink sufficient to hold stoppers/fillers.
3. **Resinous areas and knots:** Apply two coats of knotting.

490 Previously coated steel

1. **Defective paintwork:** Remove to leave a firm edge and clean bright metal.
2. **Sound paintwork:** Provide key for subsequent coats.
3. **Corrosion and loose scale:** Take back to bare metal.
4. **Residual rust:** Treat with a proprietary removal solution.
5. **Bare metal:** Apply primer as soon as possible.
6. **Remaining areas:** Degrease.

500 Preprimed steel

1. Areas of defective primer, corrosion and loose scale: Take back to bare metal. Reprime as soon as possible.

511 Galvanized, sherardized and electroplated steel

1. White rust: Remove.
2. Pretreatment: Apply one of the following:
 - 2.1. Mordant solution to blacken whole surface.
 - 2.2. Etching primer recommended by coating system manufacturer.

521 Uncoated steel – manual cleaning

1. Oil and grease: Remove.
2. Corrosion, loose scale, welding slag and spatter: Remove.
3. Residual rust: Treat with a proprietary removal solution.
4. Primer: Apply as soon as possible.

531 Uncoated steel – blast cleaning

1. Oil and grease: Remove.
2. Blast cleaning
 - 2.1. Atmospheric conditions: Dry.
 - 2.2. Abrasive: Suitable type and size, free from fines, moisture and oil.
 - 2.3. Surface finish: To BS EN ISO 8501-1, preparation grade
3. Primer: Apply as soon as possible and within four hours of blast cleaning.

541 Uncoated aluminium/ copper/ lead

1. Surface corrosion: Remove and lightly key surface.
2. Pretreatment: Etching primer if recommended by coating system manufacturer.

580 Uncoated plaster

1. Nibs, trowel marks and plaster splashes: Scrape off.
2. Overtrowelled 'polished' areas: Key lightly.

590 Uncoated plasterboard

1. Depressions around fixings: Fill with stoppers/ fillers

601 Uncoated plasterboard – to receive textured coating

1. Joints: Fill, tape and feather out with materials recommended by textured coating manufacturer.

611 Wall coverings

1. Retained wall coverings: Check that they are in good condition and well adhered to substrate.
2. Previously covered walls: Wash down to remove paper residues, adhesive and size.

622 Organic growths

1. Dead and loose growths and infected coatings: Scrape off and remove from site.
2. Treatment biocide: Apply appropriate solution to growth areas and surrounding surfaces.
3. Residual effect biocide: Apply appropriate solution to inhibit re-establishment of growths.

631 Previously painted window frames

1. Paint encroaching beyond glass sight line: Remove.
2. Loose and defective putty: Remove.
3. Putty cavities and junctions between previously painted surfaces and glass: Clean thoroughly.
4. Finishing
 - 4.1. Patch prime, reputty as necessary, and allow to set.
 - 4.2. Seal and coat as soon as fully set.

640 External pointing to existing frames

1. Defective sealant pointing: Remove.
2. Joint depth: Approximately half joint width; adjust with backing strip if necessary.
3. Sealant
 - 3.1. Manufacturer:
 - 3.1.1. Product reference:
 - 3.2. Preparation and application: As section Z22.

645 Sealing of internal movement joints

1. General: To junctions of walls and ceilings with architraves, skirtings and other trims.
2. Sealant: Waterborne acrylic.
 - 2.1. Manufacturer:
 - 2.1.1. Product reference:
 - 2.2. Preparation and application: As section Z22.

651 Existing gutters

1. Dirt and debris: Remove from inside of gutters.
2. Defective joints: Clean and seal with suitable jointing material.

Application

711 Coating generally

1. Application standard: In accordance with BS 6150, clause 9.
2. Conditions: Maintain suitable temperature, humidity and air quality during application and drying.
3. Surfaces: Clean and dry at time of application.
4. Thinning and intermixing of coatings: Not permitted unless recommended by manufacturer.
5. Overpainting: Do not paint over intumescent strips or silicone mastics.
6. Priming coats
 - 6.1. Thickness: To suit surface porosity.
 - 6.2. Application: As soon as possible on same day as preparation is completed.
7. Finish
 - 7.1. Even, smooth and of uniform colour.
 - 7.2. Free from brush marks, sags, runs and other defects.
 - 7.3. Cut in neatly.
8. Doors, opening windows and other moving parts: Ease before coating and between coats.

720 Priming joinery

1. **Preservative treated timber:** Retreat cut surfaces with two flood coats of a suitable preservative before priming.
2. **End grain:** Coat liberally allow to soak in, and recoat.

730 Workshop coating of concealed joinery surfaces

1. **General:** Apply coatings to all surfaces of components.

731 Site-coating of concealed joinery surfaces

1. **General:** After priming, apply additional coatings to surfaces that will be concealed when fixed in place.
 - 1.1. **Components:**
 - 1.2. **Additional coatings:**

740 Concealed metal surfaces

1. **General:** Apply additional coatings to surfaces that will be concealed when component is fixed in place.
 - 1.1. **Components:**
 - 1.2. **Additional coatings:**

800 Glazing

1. **Etched, sand blasted and ground glass:** Treat or mask edges before coating to protect from contamination by oily constituents of coating materials.

Ω End of Section

N10

General fixtures/ furnishings/ equipment

Products

110 ICT perimeter benching

1. Description: ICT Benching
2. Manufacturer: Contractor's Choice of Joinery Subcontractor
3. Timber: To BS EN 942.
 - 3.1. Species: American White Oak Contractor's choice
 - 3.2. Appearance class: J2
 - 3.3. Moisture content on delivery: 9 to 13%
4. Other materials:
5. Finishes: Matt Formical laminate - colour as finishes schedule
6. Fixings: Timber rail fixed to partition on back edge and adjustable white metal leg supports at maximum centres
7. Joinery workmanship: As section Z10.
8. Metalwork materials and workmanship: As section Z11.
9. Worktop: 25mm laminate faced desk
10. Other requirements: Black Grommits spaced at 650mm CC

195 Pinboards

1. Manufacturer: [Forbo Flooring Systems](#)
 - 1.1. Contact details
 - 1.1.1. Address: PO Box 1
High Holborn Road
Ripley
Derbyshire
DE5 3NT
 - 1.1.2. Telephone: +44 (0)800 093 5258
 - 1.1.3. Web: <https://www.forbo-flooring.co.uk>
 - 1.1.4. Email: info.flooring.uk@forbo.com
 - 1.2. Product reference: [Bulletin Board Pinboard Linoleum](#)
2. Material: Pinboard linoleum.
3. Dimensions
 - 3.1. Thickness: 6.0 mm $\pm$ 0.25 mm.
 - 3.2. Size: 1.22 m.
4. Decorative facing: Surfacing linoleum.
5. Total weight: To EN-ISO 23997, 4.7 kg/m² $\pm$ 10%.
6. Flexibility (to EN-ISO 24344): To EN-ISO 24344, Ø 50 mm, according to method A.
7. Roll length: To EN-ISO 24341, $\leq$ 28 m.
8. Sound absorption coefficient: To EN-ISO 354, 0.1 α_w .
9. Thermal Conductivity: To EN 12524, 0.10 W/m·K.
10. Self-healing properties: LH000420, yes.
11. Force on pins: LH000421, 25 N, 10 N and 15 N.
12. Chemical resistance: To EN-ISO 26987.

13. Backing: Natural jute.
14. Dimensional stability: Dimensionally stable in all directions.
15. HasAntiStaticSurface: true
16. Sheen: Matt.
17. Recycled content (minimum): 43%.
18. Colour:

196 Lockable Pinboards

1. Manufacturer: Tamperproof
 - 1.1. Product reference: Fire Retardant Lockable Noticeboard
2. Material: felt/ aluminium/Fire Rated polycarbonate
3. Dimensions
 - 3.1. Size: 1200x1200/ 1800x 1200
4. Colour: TBC with school

235 Curtains

1. Description: Drama Studio Theatre Stage Curtain Track
2. Manufacturer: Direct Fabrics or equal and approved
<https://www.direct-fabrics.co.uk/>
 - 2.1. Product reference: Stage Curtain Track and fabric
3. Standard: To [BS 5867-1](#) and [BS 5867-2](#).
4. Fire performance: Flame retardant Curtain Fabrics - BS5867
5. Dimensions
 - 5.1. Width: Refer to furniture layout plan number r5050
 - 5.2. Drop: 2500mm
6. Heading style: Theatre track
7. Material
 - 7.1. Fabric: Black Wool Serge Acoustic Flame Retardant Curtain Fabric 150cm Wide 560gms
 - 7.2. Colour/ Pattern: Black
8. Integral accessories: Stage Curtain Track | Heavy Duty | Walk Along / Cord Operated | 60kg with

240A Blinds Type 01

1. Description: Manually operated made to measure glare control blind, see architects drawings and schedule for locations.

To ground floor level windows group room, classrooms and offices
2. Standard: To [BS EN 13120](#).
3. Manufacturer: Waverley Contract & Supply Ltd
Rowan House,
Guildford Road Trading Estate,
Farnham, GU9 9PZ
Tel. 01252 737973
www.waverley.co.uk
Contact: Dan Voller
Email. dan@waverley.co.uk
 - 3.1. Product reference: ShadeTech® RBL-G
4. Type: Vertical roller
5. Dimensions: Refer to schedule and drawings for locations

6. **Hardware::** 50mm x 65mm powder coated steel brackets
Self-lubricating low friction bearings.
45mm diameter reinforced aluminium roller barrel
Hembar: Fabric enclosed

7. **Material:** Fabric

7.1. **Finish/ Colour:** ShadeTech® BetaScreen® 90 3%

To meet DfE shading requirement 5-7% Light Transmission:

Colour Light Grey

8. **Fixing::** Blinds to be installed top/face fix with universal fixing brackets.
9. **Operation:** Manual, gear operation by detachable crank rods.
10. **Accessories/ Other requirements:** Mounting Rail, ensuring strong fixing ground into plasterboard.

5 years manufacturing defects & workmanship warranty on hardware only.

240B High-level electrically operated roller blinds

1. **Description:** Electrically operated made to measure blackout blind, see architects drawings for locations

To high level in Classrooms and Heart Space

2. **Standard:** To [BS EN 13120](#).

3. **Manufacturer:** Waverley Contract & Supply Ltd
Rowan House,
Guildford Road Trading Estate,
Farnham, GU9 9PZ
Tel. 01252 737973
www.waverley.co.uk
Contact: Dan Voller
Email. dan@waverley.co.uk

3.1. **Product reference:** ShadeTech® RBL-E

4. **Type:** Vertical roller
5. **Dimensions:** Refer to schedule and drawings for locations
6. **Hardware::** 50mm x 65mm powder coated steel brackets
Self-lubricating low friction bearings.
45mm diameter reinforced aluminium roller barrel
Hembar: Fabric enclosed
7. **Material:** Fabric
7.1. **Finish/ Colour:** ShadeTech® DeltaBlock® 90. Colour Light Grey
8. **Fixing::** Blinds to be installed top/face fix with universal fixing brackets.
9. **Operation:** Electrically operated via 230v radio motors. Wireless wall switch programmed for group up/down control.
10. **Accessories/ Other requirements:** Mounting Rail, ensuring strong fixing ground into plasterboard.

5 years manufacturing defects & workmanship warranty on hardware only.

2 years manufacturing defects & workmanship warranty on blind motors only.

240C High-level electrically operated roller blinds. Black out

1. **Description:** Electrically operated made to measure blackout blind, see architects drawings for locations

To high level drama studio

2. **Standard:** To BS EN 13120.
3. **Manufacturer:** Waverley Contract & Supply Ltd
Rowan House,
Guildford Road Trading Estate,
Farnham, GU9 9PZ
Tel. 01252 737973
www.waverley.co.uk
Contact: Dan Voller
Email. dan@waverley.co.uk
 - 3.1. **Product reference:** ShadeTech® RBL-E
4. **Type:** Vertical roller
5. **Dimensions:** Refer to schedule and drawings for locations
6. **Hardware::** 50mm x 65mm powder coated steel brackets
Self-lubricating low friction bearings.
45mm diameter reinforced aluminium roller barrel
Hembar: Fabric enclosed
7. **Material:** Fabric
 - 7.1. **Finish/ Colour:** ShadeTech® DeltaBlock® 90. Colour Black
8. **Fixing::** Blinds to be installed top/face fix with universal fixing brackets.
9. **Operation:** Electrically operated via 230v radio motors. Wireless wall switch programmed for group up/down control.
10. **Accessories/ Other requirements:** Mounting Rail, ensuring strong fixing ground into plasterboard.

5 years manufacturing defects & workmanship warranty on hardware only.

2 years manufacturing defects & workmanship warranty on blind motors only.

Execution - Not Used

Completion - Not Used

Ω End of Section

N11

Domestic kitchen fittings, furnishings and equipment

Products

310 Fitted kitchen units

1. **Description:** To staff office
2. **Standard:** To BS EN 14749.
3. **Manufacturer:** Howdens Joinery Co. International House 1st Floor, 66 Chiltern Street, London W1U 4JT (or similar CA approved.)
 - 3.1. **Product reference:** Greenwich Gloss
4. **Structural performance:** To BS 6222-2, test level H.
5. **Dimensions:** To BS EN 1116.
6. **Surface finishes:** To BS 6222-3.
7. **Doors and drawer fronts**
 - 7.1. **Material:** Matt finish
 - 7.2. **Finish and colour:** Colour TBC
 - 7.3. **Edges:** Fully encapsulated
 - 7.4. **Other requirements:** Hinges: Blum 170deg. zinc plated steel cushion close
Drawer Box: Blumotion epoxy coated metal sides & rear, 15mm MFC base, soft closing action
8. **Side panels, plinths and shelves**
 - 8.1. **Material:** Plastics laminate
 - 8.2. **Finish and colour:** Gloss PVC finish, colour as finishes schedule
 - 8.3. **Edges:** Fully encapsulated
9. **Accessories:** Legs, plinths as recommended by manufacturer.
Worktops as clause N11/340
Sample to be provided for client approval

340 Worktops

1. **Description:** Proposals to be submitted for approval and samples of finish to be provided prior to manufacture.
2. **Standard:** To BS 6222-3
3. **Material:** Solid core, moisture resistant chipboard with rolled edge plastic laminate finished with balancing veneer on underside.
4. **Dimensions:** 40mm
5. **Exposed edges:** Exposed ends generally shall be fitted with metal cover strips and bedded in mastic. - colour TBC
6. **Support:** On kitchen base units. Worktops adjoining units shall be fixed by steel angle

Black metal support legs as necessary to areas of workbenching elsewhere
7. **Other requirements:** Mitred corner jointing shall be provided at worktop corners bedded in mastic - colour TBC
All cut edges where aperture has been cut for installation of sink/ drainer unit, shall be sealed

350 Sinks, taps, traps and wastes

1. **Description:** Stainless steel sink with drainer
2. **Sinks**

- 2.1. Standard: To BS EN 13310.
- 2.2. Manufacturer: Franke Sissons (Or similar approved)
 - 2.2.1. Product reference: ASCONA ASX611
- 2.3. Overall size: 880x510
- 2.4. Material: Stainless Steel
- 3. Taps:
 - 3.1. Manufacturer: Armitage Shanks
 - 3.1.1. Product reference: Sandringham SL Sink Mixer (B4449AA)
- 4. Wastes:
 - 4.1. Standard: To BS EN 274-1, -2 and -3.
 - 4.2. Manufacturer: Contractor's choice
 - 4.2.1. Product reference: Manufacturer's recommendations
- 5. Traps:
 - 5.1. Standard: To BS EN 274-1, -2 and -3.
 - 5.2. Depth of seal (minimum): 75 mm.
- 6. Accessories: Chainstay/Overflow holes

Execution

610 Moisture content of wood and wood-based boards

- 1. Control and monitoring
 - 1.1. Method statement: Submit.

620 Installation generally

- 1. Fixings and adhesives: As section Z20.
- 2. Services:

630 Installing units and worktops

- 1. General: Well-fitting, stable and secure.

640 Installing appliances

- 1. Connections: Provide to electric, gas, and hot and cold water services.

650 Installing sinks, taps and wastes

- 1. Water supply: To BS EN 806-2 and -4.
- 2. Taps
 - 2.1. Fixing: Secure, watertight seal with the appliance.
 - 2.2. Positioning: Hot tap to left of cold tap as viewed by the user of the appliance.
- 3. Wastes
 - 3.1. Bedding: Waterproof jointing compound.
 - 3.2. Fixing: With resilient washer between appliance and backnut.

660 Sealant bedding and pointing

- 1. Application: As section Z22.
- 2. Bedding:
- 3. Pointing:

670 Installing trims and mouldings

1. Lengths: Un-jointed between angles or ends of runs.
2. Angle joints: Mitred.

Completion

910 General

1. Doors and drawers: Accurately aligned, not binding. Adjusted to ensure smooth operation.
2. Ironmongery: Checked, adjusted and lubricated to ensure correct functioning.

920 Appliance commissioning

1. Appliance operation, functions and controls: Verify.
2. Documentation: Submit guarantees, instruction manuals, etc.

Ω End of Section

N13

Sanitary appliances and fittings

Products

300 WC pans Type C

1. Pan
 - 1.1. Manufacturer: [Armitage Shanks](#)
 - 1.1.1. Contact details
 - 1.1.1.1. Address: Armitage
Old Road
Rugeley
Staffordshire
WS15 4BT
 - 1.1.1.2. Telephone: [+44 \(0\)870 122 8822](#)
 - 1.1.1.3. Web: [www.idealspec.co.uk](#)
 - 1.1.1.4. Email: [info@thebluebook.co.uk](#)
 - 1.1.2. Product reference: [Contour 21+ Back-to-Wall Rimless Toilet](#)
 - 1.2. Standard: To BS EN 997.
 - 1.3. Form: Back-to-wall, dual flush.
 - 1.4. Connecting dimensions: To BS EN 33.
 - 1.5. Pan height: 410 mm.
 - 1.6. Materials
 - 1.6.1. Body: Vitreous china.
 - 1.6.2. Finish and colour: White.
 - 1.7. Water supply, outlet and seat holes
 - 1.7.1. Outlet: Horizontal back outlet.
 - 1.8. Seat: Charcoal seat
2. Cistern
 - 2.1. Cistern: Conceala 2 Cistern Universal Height 6/4 Litre
3. Accessories: Contemporary Flush Plate Dual Flush Armitage Shanks. Stainless Steel finish

311 Accessible WC equipment packages

1. Description: Accessible WC Left Hand
2. Manufacturer: [Armitage Shanks](#)
 - 2.1. Contact details
 - 2.1.1. Address: Armitage
Old Road
Rugeley
Staffordshire
WS15 4BT
 - 2.1.2. Telephone: [+44 \(0\)870 122 8822](#)
 - 2.1.3. Web: [www.idealspec.co.uk](#)
 - 2.1.4. Email: [info@thebluebook.co.uk](#)
 - 2.2. Product reference: [Contour 21+ - Doc M Back To Wall Left Hand Pack](#)
3. Standards: In accordance with Approved Document M.
4. Form: Complete accessible WC Doc M package and fittings.

5. Arrangement: Back-to-wall pack.
6. Transfer handing: Left-hand.
7. Material and colour
 - 7.1. WC pans: Vitreous china to BS EN 997, white.
 - 7.2. WC cisterns: Grey plastics to BS EN 14055, concealed.
 - 7.3. WC seats: Plastics, no cover.
 - 7.4. Washbasins
 - 7.4.1. Material: Vitreous china to BS EN 14688, white.
 - 7.5. Handrails:
8. Water supply fittings: Lever-operated thermostatic mixer tap.
9. Integral accessories:

338 Cleaner's Sinks

1. Description: To Cleaner's stores
2. Standard: To BS 1206 – Fireclay sinks.
3. Manufacturer: Armitage Shanks
 - 3.1. Product reference: Alder Heavy Duty Cleaners' Sink high back and fitted stainless steel bucket grating S5900(01)
4. Size: 380 x 510 mm
5. Material: Fine Fireclay
6. Tap/ chainstay/ overflow holes: wall mounted taps, no chainstay, no overflow holes
7. Water supply fittings: Alterna Quadrant 1/2in Turn Bib Taps wall mounted with ceramic discs and lever handles.
 - 7.1. Water supply temperature (maximum): Include thermostatic mixing valve Ref: A5900AA.
 - 7.2. Flow rate (maximum): 9 litres per minute at 0.1 bar .
 - 7.3. Manufacturer:
 - 7.3.1. Product reference:
8. Wastes:
 - 8.1. Standards: To BS EN 274-1, -2 and -3.
 - 8.2. Manufacturer: as above
 - 8.2.1. Product reference:
 - 8.3. Size:
 - 8.4. Material: Brass
 - 8.5. Tail: to manufacturer's recommendations
9. Traps: Exposed
 - 9.1. Standards: To BS EN 274-1, -2 and -3.
 - 9.2. Manufacturer: as above
 - 9.2.1. Product reference: to manufacturer's recommendations
 - 9.3. Size:
 - 9.4. Material: Plastics, white
 - 9.5. Depth of seal (minimum): 75 mm.
10. Accessories: pair of stainless steel legs and aluminium alloy bearers ref S9233
11. Fixings: Domex screws (pair)

346 Wash troughs

1. Manufacturer: [Venesta Washroom Systems Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: 1st Floor, Units 19-23
St George's Square
St Georges Centre
Gravesend
Kent
United Kingdom
DA11 0TA
 - 1.1.2. Telephone: [+44 \(0\)1474 353333](tel:+44(0)1474353333)
 - 1.1.3. Web: <https://www.venesta.co.uk/>
 - 1.1.4. Email: marketing@venesta.co.uk
 - 1.2. Product reference: [Vanities | Solid Surface Wash Trough, Pre-Plumbed \(Solid surface wash trough with brackets and deck mounted taps\)](#)
2. Profile : DfE3 Thermo Profile: Thermoformed trough with deck mounted taps
3. Form: Floor-mounted.
4. Materials
 - 4.1. Body: Solid surface.
 - 4.2. Finish and colour: Alabaster - VEN403.
5. Size: 2500mm length
6. Water supply, overflow and waste holes
 - 6.1. Waste: TSWC109 (WT2) 1 ½ " Unslotted strainer waste, 80 mm tail.
7. Integral accessories:
8. Preplumbing: Pre-fixed and sealed to vanity unit by Venesta, all factory assembled.
9. Framing: Cantilever brackets manufactured from 3 mm thick brushed stainless steel, laser cut and folded with integral stainless steel fixing block (supplied loose).
10. Nominal Thickness: 13 mm.
11. Vanity top and panels:
12. Sensor:
13. Manual: VSN123
14. Trap: 7360800SC – 1 ½ " self-colour (white), plastic re-sealing bottle trap 75 mm seal.

379 Water Points

1. Description: Water Bottle Filling Stations. Waste and water supply plumbed in as part of contract
2. Manufacturer: TBC to be provided by school

438 Mirrors

1. Description: To WCs
2. Manufacturer: Pilkington
 - 2.1. Product reference: Optimirror™ OW Protect
3. Type: 6 mm clear float glass
4. Size: As shown on 26 series drawings
5. Edge treatment: Ground edge
6. Protective backing: safety backing

441 Paper roll dispensers

1. Manufacturer:
 - 1.1. Product reference: To Match Existing Schools dispensers
2. Material:
 - 2.1. Width:
 - 2.2. Finish/ colour:

458 Soap dispensers

1. Description: Plastic wall fixed units
2. Manufacturer:
 - 2.1. Product reference: To Match Schools Existing
3. Type:
4. Size:
5. Material:
6. Finish/ colour:

472 Hand dryers

1. Description: Wall fixed units
2. Standard: To BS EN 60335-2-23.
3. Type: High-velocity air
4. Manufacturer: TBC
 - 4.1. Product reference: To match existing school dryers

Execution

610 Installation generally

1. Standards: In accordance with BS 6465-1, -2 and -3.
2. Assembly and fixing: Surfaces designed to falls to drain as intended.
3. Fasteners: Non-ferrous or stainless steel.
4. Fixing: Fix appliances securely to structure. Do not support on pipework.
5. Jointing and bedding compounds: Recommended by manufacturers of appliances, accessories and pipes being jointed or bedded.
6. Appliances: Do not use. Do not stand on appliances.
7. Supply and discharge pipework: Fix before appliances.
8. On completion: Components and accessories working correctly with no leaks.
9. Labels and stickers: Remove.

613 Compatibility of components

1. General: Each sanitary assembly must consist of functionally compatible components, preferably obtained from a single manufacturer.
 - 1.1. Exceptions:

620 Noggings and bearers

1. Noggings, bearers, etc. to support sanitary appliances and fittings: Position accurately. Fix securely.

630 Tiled backgrounds other than splashbacks

1. **Timing:** Complete before fixing appliances.
2. **Fixing appliances:** Do not overstress tiles.

650 Installing WC pans

1. **Floor-mounted pans:** Screw fix and fit cover caps over screw heads. Do not use mortar or other beddings.
2. **Seat and cover:** Stable when raised.

670 Installing cisterns

1. **Cistern operating components:** Obtain from cistern manufacturer.
2. **Inlet and flushing valves:** Match to pressure of water supply.
3. **Internal overflows:** Into pan, to give visible warning of discharge.
4. **External overflows:** Fix pipes to falls and locate to give visible warning of discharge. Agree location where not shown on drawings.

710 Installing taps

1. **Fixing:** Secure against twisting.
2. **Seal with appliance:** Watertight.
3. **Positioning:** Hot tap to left of cold tap as viewed by user of appliance.

720 Installing wastes and overflows

1. **Bedding:** Waterproof jointing compound.
2. **Fixing:** With resilient washer between appliance and backnut.

725 Installing hand dryers

1. **Fused connection units**
 - 1.1. **Type:**
 - 1.2. **Engraving:** With 'HAND DRYER'.
 - 1.3. **Location:**
2. **Final connection:** Concealed.
 - 2.1. **Containment:**

755 Sealant bedding and pointing

1. **Bedding:**
2. **Pointing:**

Ω End of Section

N91

External signage and interpretation

Signage outline

120 Bespoke signage system

1. **Description:** Year 7 Bridge sign fixed to brick parapet
2. **Manufacturer:** The Graphics Art Group
graphicartsgroup.co.uk
sales@graphicartsgroup.co.uk
024 7667 3415

To be read equal and approved
3. **Form:** 'The Bridge' External Sign - 2D
2mm aluminium flat cut logo, size 3800mm width x height to suit, cut to shape and powdercoated to 3x standard RAL colours, brass locators, complete with template.
4. **Products/ integral features:** Fixed to brickwork secet fixed
5. **Materials:** Aluminium / Steel
6. **Finish:** PPC
 - 6.1. Full fabrication drawings to be provided for architect comment

System performance

205 Design of signage systems

1. **Description:**
2. **Design:** Complete detailed design and submit before commencing work.
3. **Proposals:** Submit drawings, schedules, technical information, calculations and manufacturer's literature before commencing work.

210 External signage generally

1. **Signage systems generally:** Complete to BS 559, including components, inserts, accessories and fixings necessary to complete the system.
2. **External signage:** To BS 559, clause 6.1.
3. **Content:** Signs including facing information, components, inserts, accessories and fixings necessary to complete the system.
4. **Geometric shapes, colours and layout:**
5. **Wind loads:** To BS EN 1991-1-4.

220 Safety signage requirements

1. **Description:**
2. **General:** To relevant parts of BS ISO 3864-1.
3. **Safety meaning:** To BS ISO 3864-1.
4. **Geometric shapes, colours and layout:** To BS ISO 3864-1 and BS ISO 3864-4.
5. **Water safety:** To BS EN ISO 7010.
6. **Escape route:** To BS ISO 16069.

230 Accessibility

1. **Design standard for disabled users:** In accordance with BS 8300-1.

2. Geometric shapes, colours and layout:
 - 2.1. Font: Helvetica Medium.

Products

305 Signage products generally

1. Materials: To BS 559.
2. Colorimetric and photometric properties: To BS ISO 3864-4.
3. Fabricated letters: To BS 559, clause 6.6.
4. Fixings: To BS 559, clause 6.11 and section Z12.

Materials

410 Aluminium

1. Description:
2. Standards: To relevant parts of BS EN 515 and BS EN 12020-2.
 - 2.1. Mechanical properties of extruded rod/ bar, tube and profiles: To BS EN 755-2.
 - 2.2. Tolerances on dimensions and form of seamless tubes: To BS EN 755-7.
 - 2.3. Profiles, tolerances on dimensions and form: To BS EN 755-9.
 - 2.4. Structural members: In accordance with BS EN 1999-1-1.
 - 2.5. Aluminium sheet: To BS EN 573-3.
3. Alloy, temper and thickness: Suitable for the application and specified finish.
4. Component thickness:
5. Perimeters:
6. Finish:
7. Additional requirements:

Fabrication

505 Signage samples

1. Sign type: Door numbers signs
 - 1.1. Action: Submit labelled samples.
 - 1.1.1. Timing:
 - 1.2. Conformity: Retain samples on site for the duration of the contract, or until instructed to remove them.
 - 1.3. Delivered product: To conform to labelled samples.

Execution/ erection/ installation

605 Method of fixing signs

1. Description:
2. Method of fixing:

Completion - Not Used

Ω End of Section

P10 Sundry insulation/ proofing work

Types of insulation

190 Insulation fitted between studs

1. Manufacturer: [ROCKWOOL Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: ROCKWOOL Ltd
Wern Tarw
Pencoed
Bridgend
United Kingdom
CF35 6NY
 - 1.1.2. Telephone: [+44 \(0\)1656 862621](tel:+441656862621)
 - 1.1.3. Web: <https://www.rockwool.com/uk/>
 - 1.1.4. Email: customersupportcentre@rockwool.com
 - 1.2. Product reference: Rockwool Steelframe Slab 180mm
2. General requirements: Insulation products generally.
3. Standard: To BS EN 13162, ISO 14001 and ISO 9001.
4. Thickness: 180mm
5. Edges: Square.
6. Thermal conductivity (maximum): 0.034 W/m·K.
7. Fire performance: Euroclass A1 non-combustible.
8. Vapour resistivity: 5.9 MN·s/g.
9. Size: 600 x 1200 mm.

217A Mineral wool slab insulation within brick cavity

1. Manufacturer: [ROCKWOOL Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: ROCKWOOL Ltd
Wern Tarw
Pencoed
Bridgend
United Kingdom
CF35 6NY
 - 1.1.2. Telephone: [+44 \(0\)1656 862621](tel:+441656862621)
 - 1.1.3. Web: <https://www.rockwool.com/uk/>
 - 1.1.4. Email: customersupportcentre@rockwool.com
 - 1.2. Product reference: [RainScreen Duo Slab®](#) (Rainscreen Duo-Slab, unfaced, >100 mm thick)
2. General requirements: Insulation products generally.
3. Standard: To BS EN 13162; ISO 14001.
4. Thickness: 200mm
5. Facing: Plain (unfaced).
6. Edges: Square.
7. Density: Manufacturer's standard.
8. Thermal conductivity (maximum): 0.035 W/m·K.

9. Compressive strength (minimum): Manufacturer's standard.
10. Fire performance: To BS EN 13501-1, A1.
11. Water vapour resistance: 5.9 MNs/gm.
12. Sound reduction: Rw 62 dB.
13. Installation requirements
 - 13.1. To manufacturer's recommendations

217B Mineral wool slab insulation within rainscreen

1. Manufacturer: [ROCKWOOL Ltd](#)
 - 1.1. Contact details
 - 1.1.1. Address: ROCKWOOL Ltd
Wern Tarw
Pencoed
Bridgend
United Kingdom
CF35 6NY
 - 1.1.2. Telephone: [+44 \(0\)1656 862621](tel:+44(0)1656862621)
 - 1.1.3. Web: <https://www.rockwool.com/uk/>
 - 1.1.4. Email: customersupportcentre@rockwool.com
 - 1.2. Product reference: [RainScreen Duo Slab®](#) (Rainscreen Duo-Slab, black tissue-faced to one side)
2. General requirements: Insulation products generally.
3. Standard: To BS EN 13162; ISO 14001.
4. Thickness: 200 mm.
5. Facing: Black tissue (one side).
6. Edges: Square.
7. Density: Manufacturer's standard.
8. Thermal conductivity (maximum): 0.034 W/m·K.
9. Compressive strength (minimum): Manufacturer's standard.
10. Fire performance: To BS EN 13501-1, A1.
11. Water vapour resistance: 5.9 MNs/gm.
12. Sound reduction: Rw 62 dB.
13. Installation requirements
 - 13.1. To manufacturer's recommendations

217C Below DPC Insulation Type A

1. Manufacturer: [Unilin Insulation UK Limited](#)
 - 1.1. Contact details
 - 1.1.1. Address: Park Road
Holmewood Enterprise Park
Chesterfield
Derbyshire
United Kingdom
S42 5UY
 - 1.1.2. Telephone: [+44 \(0\)371 2221033](tel:+44(0)3712221033)
 - 1.1.3. Web: www.unilininsulation.co.uk
 - 1.1.4. Email: info.ui@unilin.com

- 1.2. Product reference: xps
2. Material: Rigid Polystyrene
3. Recycled content: na
4. Thickness: As detailed
5. Installation requirements
 - 5.1. Joints: Butted, no gaps. Fit insulation tightly between/ around cladding supports.
 - 5.2. Fasteners:

218 Structural Below ground insulation board

1. Manufacturer: Ecological Building Systems
 - 1.1. Contact details
 - 1.1.1. Address: Ecological Building Systems UK Ltd
Cardewlees
Carlisle, Cumbria
CA5 6LF
United Kingdom
 - 1.1.2. Telephone: +44 (0)1228 711511
 - 1.1.3. Web: <https://www.ecologicalbuildingsystems.com/>
 - 1.1.4. Email: info@ecologicalbuildingsystems.com
 - 1.2. Product reference: Bosig phonotherm 200 (RG550)
2. Material: polyurethane
3. Recycled content: na
4. Thickness: 50mm
5. Installation requirements
 - 5.1. Joints: Butted, no gaps.
 - 5.2. Fasteners:

310 Vapour barrier

1. Description: A virgin polyethylene vapour control membrane designed to restrict the passage of warm, moist air from within a building from permeating into the building structure or the roof.

Visqueen Vapour Barrier provides a means of protecting the warm side of the thermal insulation incorporated in a building by creating a barrier to the movement of warm, moist air. It is a loose laid membrane designed for use in roofs, walls and floors subjected to humidity levels less than 60% at 20 degrees Celsius (BS 5250: 2002 class 2 and 3 conditions) e.g. offices and domestic dwellings with low occupancy.

Utilised below screed
2. Manufacturer: Visqueen Building Products, Heanor Gate Industrial Estate, Heanor, Derbyshire, UK, DE75 7RG.
Tel: 0845 302 4758
Email: enquires@visqueenbuilding.co.uk Web: www.visqueenbuilding.co.uk
 - 2.1. Product reference: Visqueen Vapour Barrier
3. Material: Virgin polyethylene
4. Installation requirements
 - 4.1. Setting out: Joints minimized.
 - 4.2. Method of fixing:
 - 4.3. Joints: At supports only, lapped 150 mm minimum.
 - 4.4. Openings: Membrane fixed to reveals.

- 4.5. **Joints and edges:** Sealed with double-sided tape with vapour resistivity not less than the air and vapour control layer.
- 5. **Penetrations:** Sealed.
- 6. **Other requirements:** Visqueen Double Sided Bonding Tape
Visqueen Vapour Tape
Visqueen Vapour Edge Tape

320 Breather membrane

- 1. **Manufacturer:** Submit proposals
 - 1.1. **Product reference:** Submit proposals
- 2. **Standard:** BS EN 13859-2
 - 2.1. **Characteristics:** Reaction to fire: To BS EN 13501-1, Class B-s3, d0 or better
- 3. **Installation requirements**
 - 3.1. **Setting out:** Joints minimized. Membrane to form a continuous barrier to prevent water, snow and wind blown dust reaching the substrate.
 - 3.2. **Method of fixing:**
 - 3.3. **Joints:** Lapped 100 mm minimum horizontally and 150 mm minimum vertically.
 - 3.4. **Openings:** Membrane fixed to reveals.
 - 3.5. **Bottom edges:** Membrane lapped over flashings, sills, etc. to allow free drainage to the exterior.
- 4. **Penetrations:** Sealed.

420 Open State Sleeved mineral wool cavity barriers Type A

- 1. **Manufacturer:** [ROCKWOOL Ltd](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** ROCKWOOL Ltd
Wern Tarw
Pencoed
Bridgend
United Kingdom
CF35 6NY
 - 1.1.2. **Telephone:** [+44 \(0\)1656 862621](tel:+441656862621)
 - 1.1.3. **Web:** <https://www.rockwool.com/uk/>
 - 1.1.4. **Email:** customersupportcentre@rockwool.com
 - 1.2. **Product reference:** [SP FireStop OSCB 25 & 44 \(ROCKWOOL SP Firestop OSCB 25\)](#)
- 2. **General requirements:** Insulation products generally.
- 3. **Cavity barriers**
 - 3.1. **Attachment:**
- 4. **Size:** 1000 mm.
- 5. **Fire performance:**
- 6. **Overall cavity width:** Refer to details
- 7. **Thickness:** 90 mm.

432 Mineral wool fire-stopping Type A

- 1. **Manufacturer:** [ROCKWOOL Ltd](#)
 - 1.1. **Contact details**
 - 1.1.1. **Address:** ROCKWOOL Ltd
Wern Tarw

Pencoed
Bridgend
United Kingdom
CF35 6NY

1.1.2. Telephone: [+44 \(0\)1656 862621](tel:+441656862621)

1.1.3. Web: <https://www.rockwool.com/uk/>

1.1.4. Email: customersupportcentre@rockwool.com

1.2. Product reference: [FirePro® SP FireStop EN \(SP FireStop EN 120\)](#)

2. Material

2.1. Standard: To BS EN 1366-4:2021.

2.2. Type: Rigid.

2.3. Density:

2.4. Facing: Metal foil laminate.

3. Fire performance: Fire resistance: EI 120 to BS EN 13501-2.

4. Recycled content: 20—20%.

5. Reaction to fire: Euroclass A1 non-combustible.

6. Thickness: 90 mm.

7. Width: 1200 mm.

8. Length: 1000 mm.

9. Colour: Black.

10. Accessories:

Ω End of Section

P12 Fire-stopping systems

General

110 Fire-stopping systems

1. **Description:** To individual and multiple service penetrations
2. **Penetration seal/ gap filler:** Submit proposals

210 Design

1. Complete the design of the fire stopping system using generic products such as those specified below. To allow for all necessary Fire stopping, as shown on drawings and as required to achieve the Fire Strategy.
2. Make reference to 18 series Fire Strategy drawings and all recommendations from fire strategy report
3. Submit drawings, technical information, calculations and manufacturers' literature.
4. Refer to product manufacturer literature for installation requirements and best practice.
5. Fire-stopping shall only be installed by a third-party accredited company, or a trained individual who has been assessed as competent through third-party accreditation.
6. All fire stopping, fire dampers and fire protection measures shall be documented using photographic records.
7. On completion, all passive protection measures installed shall be labelled, photographed, scheduled and recorded on plans by the third-party installer.
8. On Completion, the Contractor and the Employer shall have a comprehensive record and audit trail to demonstrate compliance which the end user can manage and update during the buildings' life (note: this information shall be housed within the H&S file and in the Regulation 38 Statement).

System performance

240 Fire performance

1. **Description:** To Building Regulation Requirements Approved Document Part B Vol 2 and to Building Bulletin 100 Requirements.
2. **Resistance to fire:** To BS EN 13501-2, EI 60 or better

260 Design life

1. **Effective design life:** Not less than 30 years, Manufacturer to provide technical information and relevant certificates.

Products

305 Product certification

1. **Certification:** For products specified generically, submit evidence of compliance with the specification.

Execution

610 Third-party-certified installer

1. **Certification:** For the technical competency of the installer of the evidence of compliance with a third-party installation certification scheme.

2. Acceptable evidence:

620 Workmanship generally

1. **Gaps:** Seal between building elements and services, to provide effective resistance to fire and the passage of smoke. Allow for capping sealants where required. Finish flush with surrounds.
2. **Adjacent surfaces:** Prevent overrun of filler, sealant or mortar on to finished surfaces.

650 Installing flexible intumescent gap sealer

1. **Fitting of strips:** Compress strips and fit into gap so that, as they decompress, the strips wedge themselves in the void.
2. **Joints**
 - 2.1. **Ends of strips:** Fit intumescent 'end piece' at both ends of run of fire stop laminate.
 - 2.2. **Joints between strips:** Fit two intumescent 'end pieces' at each butt joint.

660 Applying intumescent foam

1. **New joints:** Remove builders' debris, mortar droppings, grease, and other contaminants.
2. **Old joints:** Clean and remove existing sealant from each joint.
3. **Priming:** Lightly moisten substrate with water.
4. **Application:** Fill joint to approximately half its depth, and allow foam to expand to face of joint.

670 Applying intumescent mortar

1. **Sequence:** Install mortar after services are permanently installed.
2. **Loose dust and combustible materials:** Remove from the opening.
3. **Shuttering:** Install suitable shuttering panels to the faces of the opening.
4. **Temperature:** Do not apply mortar when it could be damaged by frost.
5. **Mortar cure:** Do not disturb mortar before final set has taken place.
6. **Shuttering:** Remove after mortar has cured.

680 Installing intumescent pillows

1. **Orientation of bags:** Following manufacturers instructions to comply with testing

690 Applying intumescent putty

1. **Sequence:** Install putty after services are permanently installed.
2. **Loose dust and combustible materials:** Do not disturb putty before final set has taken place.

710 Installing mineral wool batts

1. **Installing batts:** Fit tight into void between the penetrating services and the surrounding construction to form a solid barrier.
 - 1.1. **Brackets:** As per manufacturers recommendations
2. **Face of batts:** Flush with the surface of wall, floor or soffit.
3. **Joints between batts:** As per manufacturers recommendations
4. **Gaps between services and barrier:** Seal with fire-resisting sealant.

730 Fixing pipe collars

1. **Collar fixing:** As per manufacturers recommendations

740 Inserting sealant backing material

1. **Preparation:** Removed debris from service penetration.
2. **Installation:** As per manufacturers recommendations

745 Applying sealants generally

1. **Application:** As section Z22.

Completion

910 Cleaning

1. **Masking tapes:** Remove.
2. **Cleaning:** Clean off splashes and droppings. Wipe down finishes.

Ω End of Section

P20

Unframed isolated trims/ skirtings/ sundry items

To be read with preliminaries/ general conditions

120 Medium-density fibreboard - sill boards

1. Description: Sill boards
2. Standard: To BS EN 622-5.
 - 2.1. Type: MDF.H
 - 2.2. Formaldehyde class: To BS EN 622-1, class E1.
3. Reaction to fire rating: To BS 476-7, Class 1.
4. Thickness: 18mm
5. Edges: Pencil Round
6. Finish: Prepared and primed for site painting
7. Support/ Fixing: Countersunk screw fixed, filled and sanded.

170 Corner guards Type A

1. Description: To all external plasterboard corners to student areas
2. Manufacturer: [Gradus](#)
 - 2.1. Contact details
 - 2.1.1. Address: Chapel Mill
Park Green
Macclesfield
Cheshire
SK11 7LZ
 - 2.1.2. Telephone: [+44 \(0\)1625 428922](tel:+441625428922)
 - 2.1.3. Web: <https://www.gradus.com>
 - 2.1.4. Email: imail@gradus.com
 - 2.2. Product reference: [Adjustable Angle PVC-U Corner Guards \(CGFLX60N\)](#)
3. Material: PVC-U with PVC co-extruded apex and wing tips.
4. Finish
 - 4.1. Colour: (001) Chalk (LRV: 87.51)
5. Standard: BS 8300 and Building Regulations (E&W) Approved Document M guidelines.
6. Length:
7. Wings: 60 mm.
8. Fixing: Adhesive-fixed.
9. Configuration: Supplied without tape.

200 Medium-density fibreboard - skirtings

1. Description: Skirtings to areas of vinyl, carpet and barrier matting
2. Standard: To BS EN 622-5.
 - 2.1. Type: MDF.H
 - 2.2. Formaldehyde class: To BS EN 622-1, class E1.
3. Reaction to fire rating: To BS 476-7, Class 1.
4. Thickness: 15mm
5. Height: 100mm

6. Edges: Pencil Round
7. Finish: Prepared and primed for site painting
8. Support/ Fixing: Countersunk screw fixed, filled and sanded.
9. To be pieced-in where required to match existing areas

Execution

510 Installation generally

1. Joinery workmanship: As section Z10.
2. Metal workmanship: As section Z11.
3. Methods of fixing and fasteners: As section Z20 where not specified.
4. Straight runs: To be in one piece, or in long lengths with as few joints as possible.
5. Running joints: Location and method of forming to be agreed where not detailed.
6. Joints at angles:
7. Position and level: To be agreed where not detailed.

Ω End of Section

P21

Door/ window ironmongery

Pre-tender

10 Quantities and locations

1. Quantities and locations of ironmongery are to be scheduled by the supplier .
2. Fixing: As sections L10 and L20.

General

121 Ironmongery from single proprietary range

1. **Manufacturer:** 3V Architectural Hardware
Unit F, Silwood Park
Buckhurst Road
Ascot
Berkshire
SL5 7PW
England, UK

or approved similar
1.1. **Product reference:**
2. **Principal material/ finish:** Stainless steel
3. **Items unavailable within selected range:** Submit proposals.
4. **Schedule:** Refer to 3V specification and schedule

122 Ironmongery from listed proprietary ranges

1. **Source:** One only of the following manufacturers/ suppliers and ranges:
2. **Notification:** Submit details of selected range, manufacturer and/ or supplier.
3. **Principal material/ finish:**
4. **Items unavailable within selected range:** Submit proposals.

140 Samples

1. **General:** Before placing orders with suppliers submit labelled samples of the following: all .
1.1. **Conformity:** Retain samples on site for the duration of the Contract. Ensure conformity of ironmongery as delivered with labelled samples.

141 Sample boards

1. **General:** Before placing orders with suppliers submit a sample board, containing labelled samples of ironmongery and showing methods of fixing.
2. **Range:** Include
2.1. **Conformity:** Retain board on site in an approved location for the duration of the Contract. Ensure conformity of ironmongery as delivered with labelled samples.

170 Ironmongery for fire doors

1. **Relevant products:** Ironmongery fixed to, or morticed into, the component parts of a fire resisting door assembly.
2. **Compliance:** Ironmongery included in successful tests to BS 476-22 or BS EN 1634-1 on door assemblies similar to those proposed.

2.1. Certification:

3. Melting point of components (except decorative non-functional parts): 800°C minimum.

180 Strength class or category of duty for door ironmongery

1. Requirement:

2. General: Durability of ironmongery components to be compatible with stated category of duty of each door leaf.

2.1. Exclusions: Ironmongery with specific duty or 'category of use' defined elsewhere.

2.2. Documentation: Before placing orders with suppliers submit documentation showing product compliance with stated category of duty.

Door hanging devices - Not Used

Window hanging devices - Not Used

Door operating devices - Not Used

Door securing devices - Not Used

Window securing devices - Not Used

Door furniture - Not Used

Window furniture - Not Used

Ω End of Section

Q40 Fencing

Fencing systems

125 Metal mesh panel fencing systems Type A

1. Manufacturer: [Jacksons Fencing](#)
 - 1.1. Contact details
 - 1.1.1. Address: 209 Stowting Common
Ashford
Kent
TN25 6BN
 - 1.1.2. Telephone: [0800 408 4757](tel:08004084757)
 - 1.1.3. Web: www.jacksons-security.co.uk
 - 1.1.4. Email: sales@jacksons-fencing.co.uk
 - 1.2. Product reference: [Securi-Mesh® Fencing](#)
2. Category: 3.
3. Posts: Manufacturer's standard.
4. Post foundations: Over length set in concrete.
5. Panels: 2440 mm.
6. Fence topping: None.
7. System accessories: Matching gates.
8. Finish/ Colour: RAL to match existing building provisionally RAL 7012 TBC with site colour matching
9. Other requirements: Galvanised steel sheets fixed to side of enclosure facing residential property to provide acoustic baffle. PPC RAL colour to match fence

Gates, posts and stiles - Not Used

Accessories - Not Used

Execution

720 Setting posts in concrete

1. Standard: To [BS 8500-2](#).
2. Mix: Designated concrete not less than GEN1 or standard prescribed concrete not less than ST2
3. Holes: Excavate neatly and with vertical sides

Completion

930 Gates

1. Hinges, latches and closers: Adjust to provide smooth operation. Lubricate where necessary.
 - 1.1. Timing: Before handover.

Ω End of Section

R10

Rainwater drainage systems

General

110 Gravity rainwater drainage system

1. Rainwater outlets: Proprietary
2. Gutters: n/a
3. Pipework: Aluminium
4. Below ground drainage: Refer to S. Eng Specification and details
5. Disposal: Refer to S. Eng Specification and details
6. Controls: Refer to S. Eng Specification and details
7. Accessories:

System performance

210 Design

1. Design: Complete the design of the rainwater drainage system.
2. Standard
 - 2.1. To BS EN 12056-3, clauses 3–7, Annex A and National Annexes.
 - 2.2. To BS EN 12056-5, clauses 3, 4, 6 and 11.
3. Proposals: Submit drawings, technical information, calculations and manufacturers' literature.

221 Collection and distribution of rainwater

1. General: Complete, and without leakage or noise nuisance.

230 Design parameters - general

1. Roof and gutter construction and finish: Bitument Membrane Roof as J41
2. Design rate of rainfall: As BS EN 12056-3, National Annex NB.2.
 - 2.1. Category: 2
3. Design life of building: 60
4. Available capacity of existing below ground drainage (maximum):

Products

311 Aluminium eaves gutters

1. Manufacturer: [Alumasc Water Management Solutions](#)
 - 1.1. Contact details
 - 1.1.1. Address: Station Road
Burton Latimer
Kettering
Northamptonshire
NN15 5JP
 - 1.1.2. Telephone: [+44 \(0\)1536 383810](tel:+441536383810)
 - 1.1.3. Web: <https://www.alumascwms.co.uk/>
 - 1.1.4. Email: marketing@alumascwms.co.uk
 - 1.2. Product reference: [Alumasc Rainwater GX Joggle Gutter \(GXJ3\)](#)

2. Form: Extruded.
3. Third party product certification: BBA Agrément, to BS8530: 2010.
4. Profile: Joggle.
5. Size (nominal width): 150 mm.
6. Thickness or gauge: 2 mm.
7. Finish: Polyester powder coated to BS EN 12206-1: 2004.
8. Colour: RAL TBC
9. Film thickness (minimum): 120-160 Microns.
10. Integral accessories: Top straps. Outlets. Stop Ends.
11. Nominal height: 100 mm.
12. Fire rating (to BS EN 13501): A2 to BS EN 13501: 2018.
13. Type: Plain end with internal union.
14. Materials: Aluminium, LM6 Marine Grade.

360 Sealant for gutters

1. Type:

365 Proprietary rainwater outlets

1. Manufacturer: Refer to J41 bitumen roof specification
 - 1.1. Product reference:
2. Roof construction:
 - 2.1. Roof insulation thickness:
3. Type of grate/ Fittings:
4. Outlet: Type and direction to suit pipework with suitable adaptors and connections.
5. Accessories:

370 Aluminium rainwater pipes and fittings Type A

1. Standard:
2. Manufacturer: [Alumasc Water Management Solutions](#)
 - 2.1. Contact details
 - 2.1.1. Address: Station Road
Burton Latimer
Kettering
Northamptonshire
NN15 5JP
 - 2.1.2. Telephone: [+44 \(0\)1536 383810](tel:+441536383810)
 - 2.1.3. Web: www.alumascwms.co.uk
 - 2.1.4. Email: info@alumascwms.co.uk
 - 2.2. Product reference: Guardian Security non climb downpipe
3. Type/ Thickness : Extruded, 1.2 mm
4. Section: Round
5. Nominal size: 102mm dia
6. Finish: Polyester powder coating
7. Colour: RAL colour to match windows
8. Brackets: proprietry brackets as part of system - non climb
 - 8.1. Fixings:

8.1.1. Size:

9. Accessories: Rainwater hopper to match

880 Leftover material

1. Spare covering material: Clean, adhesive-free off-cuts to be removed from site and taken to an Altro Recofloor collection points for recycling.

Contact Altro for further details of this scheme and the Recowall recycling scheme. Altro Limited, tel: +44(0)1462 707600, fax: +44 (0)1462 707515, email: enquiries@altro.com, website www.altro.com

Custom made products - Not Used

Execution

600 Preparation

1. Work to be completed before commencing work specified in this section
- 1.1. Below ground drainage. Alternatively, make temporary arrangements for dispersal of rainwater without damage or disfigurement of the building fabric and surroundings.
 - 1.2. Painting of surfaces which will be concealed or inaccessible.

605 Installation generally

- 1. Electrolytic corrosion: Avoid contact between dissimilar metals where corrosion may occur.
- 2. Plastics and galvanized steel pipes: Do not bend.
- 3. Allowance for thermal and building movement: Provide and maintain clearance as fixing and jointing proceeds.
- 4. Protection
 - 4.1. Fit purpose made temporary caps to prevent ingress of debris.
 - 4.2. Fit access covers, cleaning eyes and blanking plates as the work proceeds.

610 Fixing and jointing gutters

- 1. Joints:
- 2. Brackets: Securely fixed.
 - 2.1. Fixings:
 - 2.1.1. Fixing centres:
 - 2.2. Additional brackets: Where necessary to maintain support and stability, provide at joints in gutters and near angles and outlets.
- 3. Roofing underlay: Dressed into gutter.

630 Installing rainwater outlets

- 1. Fixing: Secure. Fix before connecting pipework.
 - 1.1. Method:
- 2. Junctions between outlets and pipework: Accommodate movement in structure and pipework.

635 Fixing pipework

- 1. Pipework: Fix securely, plumb and/ or true to line.
- 2. Branches and low gradient sections: Fix with uniform and adequate falls to drain efficiently.
- 3. Externally socketed pipes and fittings: Fix with sockets facing upstream.
- 4. Additional supports: Provide as necessary to support junctions and changes in direction.

5. Vertical pipes

- 5.1. Provide a loadbearing support at least at every storey level.
 - 5.2. Tighten fixings as work proceeds so that every storey is self supporting.
 - 5.3. Wedge joints in unsealed metal pipes to prevent rattling.
6. Wall and floor penetrations: Isolate pipework from structure.
- 6.1. Pipe sleeves: As section P31.
 - 6.2. Masking plates: Fix at penetrations if visible in the finished work.
7. Expansion joint pipe sockets: Fix rigidly to buildings. Elsewhere, provide brackets and fixings that allow pipes to slide.

650 Jointing pipework and gutters

1. General: Joint with materials and fittings that will make effective and durable connections.
2. Jointing differing pipework and gutter systems: Use adaptors intended for the purpose.
3. Cut ends of pipes and gutters: Clean and square. Remove burrs and swarf. Chamfer pipe ends before inserting into ring seal sockets.
4. Jointing or mating surfaces: Clean and, where necessary, lubricate immediately before assembly.
5. Junctions: Form with fittings intended for the purpose.
6. Jointing material: Strike off flush. Do not allow it to project into bore of pipes and fittings.
7. Surplus flux, solvent jointing materials and cement: Remove.

675 Cutting coated pipework and gutters

1. Cutting: Recoat bare metal.

680 Fixing insulation to internal pipelines and gutters

1. Fixing: Secure and neat. Provide continuity at supports and leave no gaps. Fix split pipe insulation with the split on 'blind' side of pipeline.
 - 1.1. Method:
2. Timing: Do not fit insulation until completion of pipe airtightness or leakage testing.

685 Identification of internal rainwater pipework

1. Standard: In accordance with Water Regulations Advisory Scheme (WRAS) Information and guidance note 9-02-05 and BS 8515.

690 Electrical continuity - pipework

1. Joints in metal pipes with flexible couplings: Clips (or suitable standard pipe couplings) supplied for earth bonding by pipework manufacturer to ensure electrical continuity.

695 Electrical continuity - gutters

1. Joints in metal gutters: Purpose made links supplied by the gutter manufacturer to ensure electrical continuity.

700 Access for testing and maintenance

1. General: Install pipework and gutters with adequate clearance to permit testing, cleaning and maintenance, including painting where necessary.
2. Access fittings and rodding eyes: Position so that they are not obstructed.

Completion

900 Testing generally

1. Dates for testing: Give notice.
 - 1.1. Period of notice (minimum):
2. Preparation
 - 2.1. Pipework: Complete, securely fixed, free from defects, obstruction and debris before testing.
3. Testing
 - 3.1. Supply clean water, assistance and apparatus.
 - 3.2. Do not use smoke to trace leaks.
4. Records: Submit a record of tests.

905 Internal pipework test - England, Wales, Ireland and Northern Ireland

1. Preparation: Temporarily seal open ends of pipework with plugs.
2. Test apparatus: Connect a 'U' tube water gauge and air pump to pipework via a plug.
3. Testing: Pump air into pipework until gauge registers 38 mm.
4. Required performance
 - 4.1. Allow a period for temperature stabilization, after which the pressure of 38 mm is to be maintained without loss for at least 3 minutes.

915 Maintenance instructions

1. General: At completion, submit printed instructions recommending procedures for maintenance of the rainwater installation, including full details of recommended inspection, cleaning and repair procedures.

920 Immediately before handover

1. Construction rubbish, debris, swarf, temporary caps and fine dust which may enter the rainwater system: Remove. Do not sweep or flush into the rainwater system.
2. Access covers, rodding eyes, outlet gratings and the like: Secure complete with fixings.

Ω End of Section

Z10

Purpose made joinery

To be read with preliminaries/ general conditions.

110 Fabrication

1. **Standard:** To BS 1186-2.
2. **Sections:** Accurate in profile and length, and free from twist and bowing. Formed out of solid unless shown otherwise.
 - 2.1. **Machined surfaces:** Smooth and free from tearing, wooliness, chip bruising and other machining defects.
3. **Joints:** Tight and close fitting.
4. **Assembled components:** Rigid. Free from distortion.
5. **Screws:** Provide pilot holes.
 - 5.1. **Screws of 8 gauge (4 mm diameter) or more and screws into hardwood:** Provide clearance holes.
 - 5.2. **Countersink screws:** Heads sunk at least 2 mm below surfaces visible in completed work.
6. **Adhesives:** Compatible with wood preservatives applied and end uses of timber.

120 Cross section dimensions of timber

1. **General:** Dimensions on drawings are finished sizes.
2. **Maximum permitted deviations from finished sizes**
 - 2.1. **Softwood sections:** To BS EN 1313-1:-
 - 2.1.1. Clause 6 for sawn sections.
 - 2.2. **Hardwood sections:** To BS EN 1313-2:-
 - 2.2.1. Clause 6 for sawn sections.
 - 2.2.2. Clause NA.3 for further processed sections.

130 Preservative treated wood

1. **Cutting and machining:** Completed as far as possible before treatment.
2. **Extensively processed timber:** Retreat timber sawn lengthways, thickened, planed, ploughed, etc.
3. **Surfaces exposed by minor cutting and/ or drilling:** Treat as recommended by main treatment solution manufacturer.

140 Moisture content

1. **Wood and wood based products:** Maintained within range specified for the component during manufacture and storage.

210 Laminated plastics veneered boards/ panels

1. **Fabrication:** To British Laminated Plastics Fabricators Association Ltd (BLF) fabricating standards.
2. **Balancing veneer:** From decorative veneer manufacturer and of similar composition. Applied to reverse side of core material.
3. **Finished components:** Free from defects, including bow, twist, scratches, chipping, cracks, pimpling, indentations, glue marks, staining and variations in colour and pattern.
4. **Joints visible in completed work:** Tight butted, true and flush.

220 Wood veneered boards/ panels

1. Core material and veneers: Conditioned before bonding.
2. Setting out: Veneer features and grain pattern aligned regularly and symmetrically unless instructed otherwise.
3. Balancing veneer: Applied to reverse side of core material.
 - 3.1. Moisture and temperature movement characteristics: As facing veneer.
4. Veneer edges: Tight butted and flush, with no gaps.
5. Tolerance of veneer thickness (maximum): ± 0.5 mm.
6. Finished components: Free from defects, including bow, twist, scratches, chipping, splits, blebs, indentations, glue marks and staining.
7. Surface finish: Fine, smooth, free from sanding marks.

250 Finishing

1. Surfaces: Smooth, even and suitable to receive finishes.
 - 1.1. Arrises: Eased unless shown otherwise on drawings.
2. End grain in external components: Sealed with primer or sealer as section M60 and allowed to dry before assembly.

Ω End of Section

Z11

Purpose made metalwork

To be read with preliminaries/ general conditions.

310 Materials generally

1. Grades of metals, section dimensions and properties: To appropriate British Standards. When not specified, select grades and sections appropriate for the purpose.
2. Prefinished metal: May be used if methods of fabrication do not damage or alter appearance of finish, and finish is adequately protected.
3. Fasteners: To appropriate British Standards and, unless specified otherwise, of same metal as component being fastened, with matching coating or finish.

320 Steel long and flat products

1. Hot rolled structural steels (excluding structural hollow sections and tubes): To BS EN 10025-1.
2. Fine grain steels, including special steels: To BS EN 10025-3 and -4.
3. Steels with improved atmospheric corrosion resistance: To BS EN 10025-5.

330 Steel plate, sheet and strip

1. Plates and wide flats, high yield strength steel: To BS EN 10025-6.

340 Hot rolled steel plate, sheet and strip

1. Flat products, high yield strength for cold forming: To BS EN 10149-1, -2 and -3.
2. Carbon steel sheet and strip for cold forming: To BS EN 10111.
3. Narrow strip, formable steel and steel for general engineering purposes: To BS 1449-1.8 and BS 1449-1.14.

350 Cold rolled steel plate, sheet and strip

1. Steel sections: To BS EN 10162.
2. Flat products, high yield strength micro-alloyed steels for cold forming: To BS EN 10268.
3. Carbon steel flat products for cold forming: To BS EN 10130 and BS EN 10131.
4. Uncoated carbon steel narrow strip for cold forming: To BS EN 10139 and BS EN 10140.
5. Narrow strip steel for general engineering purposes: To BS EN 10132-1, -2, and -3.
6. Carbon steel flat products for vitreous enamelling: To BS EN 10209.

360 Coated steel flat products

1. Hot dip zinc coated carbon steel sheet and strip for cold forming: To BS EN 10346 and BS EN 10143.
2. Hot dip zinc coated structural steel sheet and strip: To BS EN 10143 and BS EN 10346.
3. Hot dip zinc-aluminium (za) coated sheet and strip: To BS EN 10346.
4. Hot dip aluminium-zinc (az) coated sheet and strip: To BS EN 10346.
5. Organic coated flat products: To BS EN 10169.

370 Steel structural hollow sections (SHS)

1. Non alloy and fine grain steels, hot finished: To BS EN 10210-1 and -2.
2. Non-alloy and fine grain steels, cold formed welded: To BS EN 10219-2.
3. Weather resistant steels, hot finished: To BS 7668.

380 Other steel sections

1. Equal flange tees: To BS EN 10055.
2. Equal and unequal angles: To BS EN 10056-1 and -2.
3. Wire, carbon steel for general engineering purposes: To BS 1052.
4. Wire and wire products, general: To BS EN 10218-2.
5. Tubes
 - 5.1. Seamless circular: To BS EN 10297-1.
 - 5.2. Seamless cold drawn: To BS EN 10305-1.
 - 5.3. Welded and cold sized square and rectangular: To BS EN 10305-5.
 - 5.4. Welded circular: To BS EN 10296-1.
 - 5.5. Welded cold drawn: To BS EN 10305-2.
 - 5.6. Welded cold sized: To BS EN 10305-3.

400 Stainless steel products

1. Chemical composition and physical properties: To BS EN 10088-1.
2. Sheet, strip and plate: To BS EN 10088-2.
3. Semi-finished products bars, rods and sections: To BS EN 10088-3.
4. Wire: To BS EN 1088-3.
5. Tubes
 - 5.1. Welded circular: To BS EN 10296-2.
 - 5.2. Seamless circular: To BS EN 10297-2.

410 Aluminium alloy products

1. Designations
 - 1.1. Designation system, chemical composition and forms: To BS EN 573-1, -2, -3 and -5.
 - 1.2. Temper designations: To BS EN 515.
2. Sheet, strip and plate: To BS EN 485-1 to -4.
3. Cold drawn rods, bars and tubes: To BS EN 754-1 and -2.
4. Extruded rods, bars, tubes and profiles: To BS EN 755-1 and -2.
5. Drawn wire: To BS EN 1301-1, -2 and -3.
6. Rivet, bolt and screw stock: To BS 1473.
7. Structural sections: To BS 1161.

420 Copper alloy products

1. Sheet, strip, plate and circles for general purposes: To BS EN 1652.
2. Sheet and strip for building purposes: To BS EN 1172.
3. Rods: To BS EN 12163.
4. Profiles and rectangular bars: To BS EN 12167.
5. Wire: To BS EN 12166.
6. Tubes: To BS EN 12449.

Fabrication

515 Fabrication generally

1. Contact between dissimilar metals in components: Avoid.

2. Finished components: Rigid and free from distortion, cracks, burrs and sharp arrises.
 - 2.1. Moving parts: Free moving without binding.
3. Corner junctions of identical sections: Mitre.

520 Cold formed work

1. Profiles: Accurate, with straight arrises.

525 Adhesive bonding

1. Preparation of surfaces of metals to receive adhesives
 - 1.1. Degrease.
 - 1.2. Abrade mechanically or chemically etch.
 - 1.3. Prime: To suit adhesive.
2. Adhesive bond: Form under pressure.

527 Welding

1. Description:
2. Welding procedures
 - 2.1. Method and standard:
 - 2.2. Welding Procedure Specification (WPS):
3. Preparation
 - 3.1. Joint preparation: Clean thoroughly.
 - 3.2. Surfaces of materials that will be self-finished and visible in the completed work: protect from weld splatter.
4. Jointing
 - 4.1. Joints: Fully bond parent and filler metal throughout with no inclusions, holes, porosity or cracks.
 - 4.2. Dissimilar metals:
 - 4.3. Strength requirements: Welds to achieve design loads.
 - 4.4. Heat straightening:
 - 4.5. Complex assemblies: Agree priority for welding members to minimize distortion caused by subsequent welds.
 - 4.6. Tack welds: Use only for temporary attachment.
 - 4.7. Jigs: Provide to support and restrain members during welding.
 - 4.8. Filler plates:
 - 4.9. Lap joints: Minimum 5 x metal thickness or 25 mm, whichever is greater.
 - 4.10. Weld terminations: Clean and sound.

530 Stainless steel fabrication

1. Guillotining or punching: Do not use for metal thicknesses greater than 10 mm.
2. Thermal cutting
 - 2.1. Carbonation in the heat affected zone: Remove, after cutting.
3. Bending
 - 3.1. Plates or bars: Cold bending radius not less than material thickness.
 - 3.2. Tubes: Cold bending radius not less than 2 x tube diameter.
4. Welding: In addition to general welding requirements:
 - 4.1. Protect adjacent surfaces from weld spatter.

4.2. Pickle all welds before post fabrication treatments.

5. Protection: Provide protection to fabricated components during transit and on site.

555 Brazing

1. Standard: To BS EN 14324.
2. Testing
 - 2.1. Destructive testing: To BS EN 12797.
 - 2.2. Nondestructive testing: To BS EN 12799.

610 Testing

1. Description:
2. Testing standard:
3. Welding records and test results: Submit copies.

Finishing

710 Finishing welded and brazed joints visible in complete work

1. Standard: To BS EN ISO 8501-3.
 - 1.1. Preparation grade:
2. Butt joints: Smooth, and flush with adjacent surfaces.
3. Fillet joints: Neat.
4. Grinding: Grind smooth where indicated on drawings.

745 Preparation for application of coatings

1. General: Complete fabrication, and drill fixing holes before applying coatings.
2. Paint, grease, flux, rust, burrs and sharp arrises: Remove.

750 Liquid organic coating for aluminium alloy components

1. Standard: To BS 4842.

760 Zinc and cadmium plating of iron and steel surfaces

1. Zinc plating: To BS EN ISO 2081.
2. Cadmium plating: To BS EN ISO 2082.

770 Chromium plating

1. Standard: To BS EN ISO 1456.

780 Galvanizing

1. Standard: To BS EN ISO 1461.
2. Preparation
 - 2.1. Vent and drain holes: Provide in accordance with BS EN ISO 14713-1 and -2. Seal after sections have been drained and cooled.
 - 2.2. Components subjected to cold working stresses: Heat treat to relieve stresses before galvanizing.
 - 2.3. Welding slag: Remove.
 - 2.4. Component cleaning: To BS EN ISO 8501-3.
 - 2.5. Grade:

790 Vitreous enamelling

1. Standard: To BS EN ISO 28722.
2. Substrate metal: Steel to BS EN 10209.

Completion

910 Documentation

1. Submit
 - 1.1. Manufacturer's maintenance instructions.
 - 1.2. Guarantees, warranties, test certificates, record schedules and log books.

920 Completion

1. Protection: Remove.
2. Cleaning and maintenance: Carry out in accordance with procedures detailed in fabricators' guarantees.

Ω End of Section

Z21 Mortars

Cement gauged mortars

110 Cement gauged mortar mixes

1. **Specification:** Proportions and additional requirements for mortar materials are specified elsewhere.

120 Sand for site made cement gauged masonry mortars

1. **Standard:** To BS EN 13139.
2. **Grading:** 0/2 (FP or MP).
 - 2.1. **Fines content** where the proportion of sand in a mortar mix is specified as a range (e.g. 1:1: 5-6):
 - 2.1.1. **Lower proportion of sand:** Use category 3 fines.
 - 2.1.2. **Higher proportion of sand:** Use category 2 fines.
3. **Sand for facework mortar:** Maintain consistent colour and texture. Obtain from one source.

131 Ready-Mixed lime:sand for cement gauged masonry mortars

1. **Standard:** To BS EN 998-2.
2. **Lime:** Nonhydraulic to BS EN 459-1.
 - 2.1. **Type:** CL 90S.
3. **Pigments for coloured mortars:** To BS EN 12878.

135 Site made lime:sand for cement gauged masonry mortars

1. **Permitted use:** Where a special colour is not required and in lieu of factory made ready-mixed material.
2. **Lime:** Nonhydraulic to BS EN 459-1.
 - 2.1. **Type:** CL 90S.
3. **Mixing:** Thoroughly mix lime with sand, in the dry state. Add water and mix again. Allow to stand, without drying out, for at least 16 hours before using.

160 Cements for mortars

1. **Cement:** To BS EN 197-1 and CE marked.
 - 1.1. **Types:** Portland cement, CEM I.
 - 1.1.1. **Portland limestone cement,** CEM II/A-L or CEM II/A-LL.
2. **Portland slag cement,** CEM II/B-S.
3. **Portland fly ash cement,** CEM II/B-V.
 - 3.1. **Strength class:** 32.5, 42.5 or 52.5.
4. **White cement:** To BS EN 197-1 and CE marked.
 - 4.1. **Type:** Portland cement, CEM I.
 - 4.2. **Strength class:** 52.5.
5. **Sulfate resisting Portland cement**
 - 5.1. **Type:** To BS EN 197-1 Sulfate resisting Portland cement, CEM I/SR and CE marked.
6. **To BS EN 197-1 fly ash cement,** CEM II/B-V and CE marked.
 - 6.1. **Strength class:** 32.5, 42.5 or 52.5.

7. Masonry cement: To BS EN 413-1 and CE marked.
 - 7.1. Class: MC 12.5.

180 Admixtures for site made cement gauged mortars

1. Air entraining (plasticizing) admixtures: To BS EN 934-3 and compatible with other mortar constituents.
2. Other admixtures: Submit proposals.
3. Prohibited admixtures: Calcium chloride, ethylene glycol and any admixture containing calcium chloride.

190 Retarded ready to use cement gauged mortar

1. Standard: To BS EN 998-2.
2. Lime for cement:lime:sand mortars: Nonhydraulic to BS EN 459-1.
 - 2.1. Type: CL 90S.
3. Pigments for coloured mortars: To BS EN 12878.
4. Time and temperature limitations: Use within limits prescribed by mortar manufacturer.
 - 4.1. Retempering: Restore workability with water only within prescribed time limits.

200 Storage of cement gauged mortar materials

1. Sands and aggregates: Keep different types/ grades in separate stockpiles on hard, clean, free-draining bases.
2. Factory made ready-mixed lime:sand/ ready to use retarded mortars: Keep in covered containers to prevent drying out or wetting.
3. Bagged cement/ hydrated lime: Store off the ground in dry conditions.

210 Making cement gauged mortars

1. Batching: By volume. Use clean and accurate gauge boxes or buckets.
 - 1.1. Mix proportions: Based on dry sand. Allow for bulking of damp sand.
2. Mixing: Mix materials thoroughly to uniform consistency, free from lumps.
 - 2.1. Mortars containing air entraining admixtures: Mix mechanically. Do not overmix.
3. Working time (maximum): Two hours at normal temperatures.
4. Contamination: Prevent intermixing with other materials.

Lime:sand mortars - Not Used

Ω End of Section

Z22 Sealants

Products

310 Joints

1. Description:
2. Manufacturer:
 - 2.1. Product reference:
3. Primer, backing strip, bond breaker: Types recommended by sealant manufacturer.
4. Fire performance
 - 4.1. Fire resistance:
 - 4.2. Reaction to fire:

Execution

610 Suitability of joints

1. Presealing checks
 - 1.1. Joint dimensions: Within limits specified for the sealant.
 - 1.2. Substrate quality: Surfaces regular, undamaged and stable.
2. Joints not fit to receive sealant:

620 Preparing joints

1. Surfaces to which sealant must adhere
 - 1.1. Remove temporary coatings, tapes, loosely adhering material, dust, oil, grease, surface water and contaminants that may affect bond.
 - 1.2. Clean using materials and methods recommended by sealant manufacturer.
2. Vulnerable surfaces adjacent to joints: Mask to prevent staining or smearing with primer or sealant.
3. Backing strip and/ or bond breaker installation: Insert into joint to correct depth, without stretching or twisting, leaving no gaps.
4. Protection: Keep joints clean and protect from damage until sealant is applied.

630 Applying sealants

1. Substrate: Dry (unless recommended otherwise) and unaffected by frost, ice or snow.
2. Environmental conditions: Do not dry or raise temperature of joints by heating.
3. Sealant application: Fill joints completely and neatly, ensuring firm adhesion to substrates.
4. Sealant profiles
 - 4.1. Butt and lap joints: Slightly concave.
 - 4.2. Fillet joints: Flat or slightly convex.
5. Protection: Protect finished joints from contamination or damage until sealant has cured.

Ω End of Section



Specification created using NBS Chorus